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CAPSTONE PROJECT REPORT

Report 3 Software Requirement Specification

Arden Enterprise Knowledge Platform

with Human in the Loop Governance

and Permission Aware Retrieval

Ho Chi Minh, October 2026

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I. Project Report

1. Record of Changes

Date	A*, M, D	In charge	Change Description
29 Sep 2026	A	Project Team	Initial SRS covering organization setup, governed documents, review, handover, access control and permission-aware AI retrieval.

Date	A*, M, D	In charge	Change Description
18 Oct 2026	M	Project Team	Aligned with code and decisions 0004–0009: multi-organization tenancy, actors, invitations, reviewer grants, Personal Workspace, contribution copy, automatic publication.
22 Oct 2026	M	Project Team	Aligned with registration form v2.0 and decision 0010: audiences replace custom roles, Arden System Admin, scope changes, acceptance criteria, six measurable NFRs, redrawn diagrams.
27 Oct 2026	M	Ngô Quang Khôi	Review feedback: added Project Report; aligned with decisions 0011–0015 (approve/reject only, Organization Admin final authority, derived responsible department, Department/Organization reviewer grants, PREPARING → indexed ACTIVE publication, Arden-AI service); detailed specification of every function with screen layouts; action-level authorization; new non-screen functions; Usability NFR; glossary, assumptions, known limitations and requirement traceability matrix.
31 Oct 2026	M	Ngô Quang Khôi	Aligned with the staging branch (commit e9e4b3a): registration, password reset with single-use code, PDF indexing and Ask Arden are Implemented; department deletion (archive) and invitations without a department added to FR-ORG-03/04, rules and entities; issues and known limitations updated; migrations 0001–0023.

A - Added M - Modified D - Deleted

2. Status Report

Reporting date: 31/10/2026, end of Iteration 1. Status is taken from the staging branch of the Arden repository (commit e9e4b3a, 31/10/2026), which is deployed to the staging environment. It contains the knowledge workflow (decisions 0011–0015), the durable indexing pipeline, PDF indexing, permission-aware Ask Arden, registration and password reset, department deletion and invitations without a department (migrations 0001–0023). The separate Arden-AI repository (main, commit cc0b4d2) holds the Python AI service. Status values in this report: Implemented = present on the staging branch; Partial = only an API or a preview; Planned = not implemented.

Feature group	Status	Evidence and scope	Next step
Accounts, sign-in and sessions	Implemented	Email/password registration with email confirmation, Google sign-in, password setup, password reset with a single-use emailed code, sign-out and cookie sessions.	Profile screen (FR-AUTH-05) is planned.
Organizations, invitations and members	Implemented	Create organization, create and delete (archive) departments, email invitations with an optional department (send, resend, revoke, accept), primary department change.	Member removal and admin promotion are disabled in the UI.
Audiences and reviewer grants	Implemented	Organization, Department and Group audiences; Department and Organization reviewer grants (decision 0015).	Keep the disposable PostgreSQL CI job required for pull requests.
Private notes and contribution copy	Implemented	Owner-only notes with versions and search; separate shared copy.	Personal PDF documents and collections are planned.

Feature group	Status	Evidence and scope	Next step
Shared drafts, submission and review	Implemented	Text and PDF drafts, reader picker, submit snapshot, approve/reject, Organization Admin final authority, contribution history.	Withdraw submission is planned.
Publication preparation and activation	Implemented	Approval → PREPARING; durable index job → Arden-AI → atomic ACTIVE; failures stay invisible. Verified locally on 27/10/2026 with a stub AI endpoint.	Text and PDF snapshots are indexed; unpublish and restore are planned.
Library and reader	Implemented	Authorized publication list and reader.	Keyword and semantic search are planned.
Ask Arden (permission-aware RAG)	Implemented	Live answers on staging in organization scope: Arden-AI retrieves only from the asker's allowed publications and Core rechecks every citation before showing it.	Run the NFR-03 evaluation on staging; Personal/Hybrid modes, rating and history are planned.
Personal/hybrid AI, summary, rating, history	Planned	—	Iteration 2–3.
Archive/unpublish and restore	Planned	—	Iteration 2.
Dashboard and analytics	Planned	—	Iteration 3, minimal level.
Arden System Admin	Partial	Bootstrap command and /api/v1/system/administrators API.	System Administration screen.

Verification performed on 27/10/2026 (before the branches were merged into staging): all migrations 0001–0020 of the working branch were applied to a clean PostgreSQL 16 database without error; the web client and API were run in local test mode with four synthetic accounts; a member created, submitted and contributed drafts, a Department reviewer approved and rejected, an Organization reviewer approved an organization-wide policy, the indexing worker activated both approved snapshots, and a member of another department could read only the organization-wide publication. The screen layouts in section 3 are screenshots of that run (synthetic data, no real personal data). On 31/10/2026 the statuses were re-checked against the staging source code by reading the code; migrations 0021–0023 were not re-run in that check.

3. Team Involvements

Contribution is summarised from the commit history of the Arden repository (all branches, merges excluded) and the Arden-AI repository on 27/10/2026. Commit counts are evidence, not a measure of effort; the share column is a proposal for the team to confirm.

Member	Role	Main contributions	Commits (Arden / Arden-AI)	Share (to confirm)
Ngô Quang Khôi	Team leader; product owner of the knowledge workflow; UI design	Product blueprint and Mainflow 1 workspace previews; Mainflow 2 governance (shared drafts, submission, review, PDF drafts, cross-actor review fix); workspace, administration and Island UI redesigns; review-authority decisions 0013–0015; SRS/SDD/test documents.	31 / 0	30%
Đặng Trường Hưng	Backend and AI engineer	Scoped audiences and replacement of custom roles (decision 0010); Google OAuth and password setup; Flow 2 publication handoff; durable Flow 3 indexing pipeline; Arden-AI service (compiler, Chroma indexing, permission-aware RAG, PDF indexing); Ask Arden citations.	21 / 10	30%
Mai Quốc Anh	DevOps and repository maintainer	Repository scaffold and standards; hosted development workflow on Coolify (remote CLI, slot fencing); CI hardening; contribution, branch and specification-continuity rules; feature index; approved brand package.	27 / 2	20%
Lâm Gia Kiệt	Frontend and authentication	Mainflow 1 implementation; login, invitation and register UI; HTML invitation email; registration and password reset; migration 0005 documentation.	11 / 0	20%

4. Issues and Suggestions

ID	Issue	Impact	Action / suggestion
I-01	The working branch, feat/PERMISSION-AWARE-RAG and the auth fixes were merged into staging on 30–31/10/2026; one UI commit of feat/ard-16-knowledge-workflow-updates (9bee5cd) is still outside staging, and main is behind staging.	The demo build may miss that UI change; main does not reflect the reviewed build.	Merge the remaining commit into staging through a pull request, then tag review-2 on staging.
I-02	Decisions 0013–0015 were accepted in working sessions; the project lead's confirmation is to be recorded in the pull request.	Reviewers may ask who approved the change of review rules.	Record the confirmation in the PR and in BM03.
I-03	Resolved: CI now runs a disposable PostgreSQL job for the authorization and migration tests.	—	Keep the job required for pull requests to staging.
I-04	Department deletion: the API declares CANNOT_DELETE_BASE_DEPARTMENT, but the repository never returns it, so the organization’s first department can also be archived.	An organization can lose its base department; its members become unassigned.	Return BASE_DEPARTMENT from deleteDepartment and add a test.
I-05	Legacy schema remains: classification, reviewer level and role tables.	Extra columns and a broader SQL review gate (decision 0015 known debt).	Schedule the contraction migration after no legacy submission is pending.

ID	Issue	Impact	Action / suggestion
I-06	UI defects found during the screenshot run: Workspace Select shows raw role codes (DEPT_DEFAULT_BOOTSTRAP, ORG_ADMIN) and a hard-coded welcome name; a reviewer who is not in a group sees "Inactive audience"; My Work is disabled for signed-in users.	Confusing for users and reviewers.	Fix the copy and enable My Work with server data.
I-07	Ask Arden sends authorized excerpts and questions to the OpenAI API.	Data leaves the customer environment; conflicts with a strictly self-hosted reading of the product.	Document the data flow, add an organization-level AI opt-in and record it in Report 4.
I-08	Many registered features are still planned (personal/hybrid AI, summary, rating, history, archive, collections, dashboard).	Risk to the Defense 1.1 scope.	Freeze the mandatory list in BM01; implement minimal versions first.

Support requested from the supervisor and the department:

- Supervisor confirmation of the scope changes in BM03 (classification removed, approve/reject only).
- An OpenAI API budget for the evaluation set and the demo.
- Access to a staging slot with a disposable PostgreSQL database for CI integration tests.

II. Software Requirement Specification

1. Product Overview

Arden is an open-source, self-hosted knowledge management and retrieval system for small and medium-sized enterprises. One installation can host several organizations. Any verified account can create an organization and becomes its first Organization Admin. Every organization is an isolated tenant: its members, departments, audiences, notes, knowledge, review records and audit events are never visible to another organization.

Arden manages knowledge as governed documents rather than project records. The required hierarchy inside an organization is Organization → Department. Each member has a private Personal Workspace and at most one primary department; a member may be unassigned after being invited without a department or after their department is deleted. Team and project values are optional tags only.

Shared knowledge follows one governed lifecycle: a member writes a shared draft (or contributes a separate copy of a private note), chooses who may read it, and submits an immutable snapshot. An eligible reviewer, or the Organization Admin, approves or rejects that exact snapshot. Approval does not publish immediately: the snapshot is prepared and indexed by the Arden-AI service, and only a complete index generation is activated atomically. Search and the Ask Arden assistant use the same active publications under the same access policy, and every AI answer cites its sources.

Arden distinguishes four actors: Member, Knowledge Reviewer, Organization Admin and Arden System Admin. Readers are defined by audiences (the whole organization, a whole department or an audience group), review authority by reviewer grants (Department or Organization scope), and

organization management by the Organization Admin capability. None of these capabilities opens another member's private notes. Arden does not provide project planning, ad-hoc per-user sharing, autonomous publication by AI or autonomous permission changes.

The context diagram shows Arden (web/desktop clients, Arden core API and the Arden-AI service) and the external entities it exchanges data with.

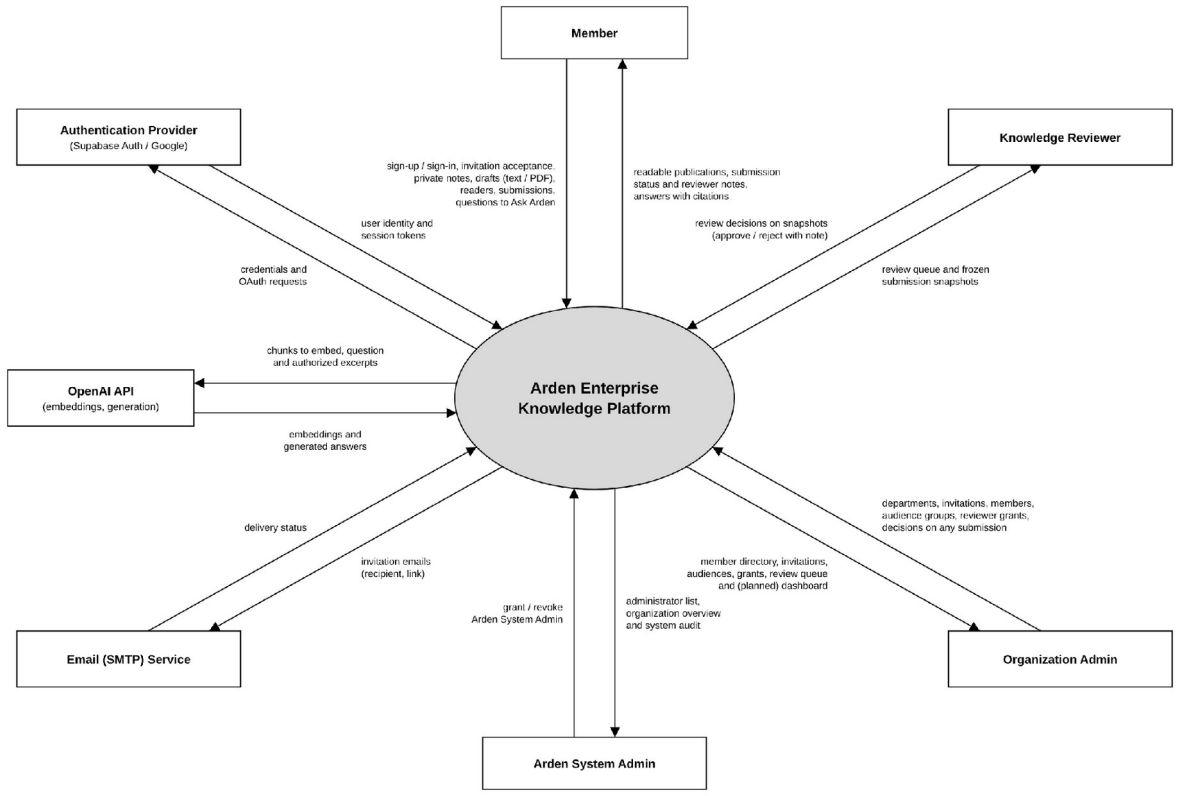


Figure 1: Context diagram

1.1 Scope and Implementation Status

Scope follows the proposed registration form version 2.0. Every functional requirement, use case and screen carries a status: Implemented (on the staging branch), Partial (API or preview only) or Planned. The detailed status by feature group and the open issues are given in Part I. Out of scope: Knowledge Handover, Knowledge Graph, external work-source connectors, notifications other than invitation email, and OCR of scanned PDFs.

1.2 Glossary

Term	Meaning
Organization	An isolated tenant inside one Arden installation.
Department / primary department	Organizational unit; each member has at most one primary department, which owns the member's shared drafts. Deleting a department archives it.
Audience	Defines who may read a publication: Organization (all active members), Department (active members of one department) or Group.
Audience group	An Organization Admin-managed list of members used only as a reader audience.

Term	Meaning
Reviewer grant	Authority to decide submissions. Department scope covers department and group-only snapshots of one department; Organization scope covers snapshots readable by the whole organization.
Private note	An owner-only item in the Personal Workspace.
Shared draft / working version	An author-owned document that becomes shared knowledge only through review; every save creates an immutable numbered version.
Semantic Document	Arden's canonical block-based document format (version 1) with stable block IDs.
Submission snapshot (release candidate)	The exact frozen version, metadata and audiences sent for review; at most one is open per draft.
Self-approval	An Organization Admin approving their own submission; recorded with selfApproved evidence.
Publication states	NONE, PREPARING (approved, being indexed), ACTIVE (visible), FAILED (index failed, invisible).
Index generation / index job	The durable record and queue item that send an approved snapshot to Arden-AI and hold its manifest.
Current publication pointer	The single active published version of a knowledge item, changed only by atomic activation.
Arden-AI	The separate Python 3.11 service that compiles, chunks, embeds, indexes (ChromaDB) and answers with OpenAI.
Allow-list	The generation IDs the asker may read, computed by Arden core before retrieval; Arden-AI may narrow but never broaden it.
RLS	PostgreSQL row-level security, used as defense in depth behind API authorization.
Idempotency-Key / If-Match	Request headers that prevent duplicate creation and lost updates.

1.3 Assumptions and Dependencies

ID	Type	Description
AD-01	Dependency	Supabase Auth provides email/password accounts, email confirmation and Google sign-in; the Arden API is the only client of Supabase.
AD-02	Dependency	A third-party SMTP service delivers invitation emails; credentials are runtime secrets.
AD-03	Dependency	PostgreSQL 16 with row-level security stores all business state; migrations 0001–0023 are applied in order.
AD-04	Dependency	The Arden-AI service (Python 3.11, FastAPI, LangChain) runs on the private network and is called only by Arden core with a service token.
AD-05	Dependency	ChromaDB (self-hosted, persistent volume) stores chunks and embeddings; it is a derived index and never an authorization source.
AD-06	Dependency	The OpenAI API provides embeddings and answer generation; only the question and authorized excerpts are sent.
AD-07	Dependency	Langfuse (self-hosted) records AI traces and cost for the Arden operations team.

ID	Type	Description
AD-08	Dependency	Deployment uses Docker Compose on Coolify (Ubuntu); the desktop client is Electron for Windows.
AD-09	Assumption	Each member has at most one primary department, set by the Organization Admin or by the invitation.
AD-10	Assumption	Demo and test data are synthetic; no real personal or partner data is used.
AD-11	Assumption	Users use a current Chrome or Edge browser on a screen at least 1280 px wide, or the Windows desktop client.

1.4 Known Limitations

ID	Limitation
KL-01	Ask Arden answers in organization scope only; Personal and Hybrid modes, answer rating and durable chat history are not implemented.
KL-02	Members without a primary department (invited without one, or whose department was deleted) cannot open Private Notes or Reviews until an Organization Admin assigns a department.
KL-03	Used password-reset codes are remembered in API memory for 24 hours, so the single-use check is lost when the API restarts.
KL-04	My Work is disabled for signed-in users (preview data only).
KL-05	Withdraw, unpublish/archive and restore have database states but no API or screen yet.
KL-06	Reviewer levels and classification columns remain only for legacy submissions.
KL-07	Removing members and promoting another Organization Admin are not available in the UI.
KL-08	Scanned PDFs (OCR) and files over 5 MB are not supported.
KL-09	The System Administration screen is not built; the API is available.

2. User Requirements

2.1 Actors

#	Actor	Description
1	Member	An active member of an organization. Writes private notes; creates text or PDF shared drafts; contributes a separate copy of a private note; chooses readers and submits snapshots; edits the working draft and resubmits after a rejection; reads published knowledge in their audiences; (Partial/Planned) asks Ask Arden, summarizes documents, rates answers and keeps personal documents and collections.
2	Knowledge Reviewer	A member holding an active reviewer grant. A Department grant covers department and group-only snapshots owned by one department; an Organization grant covers snapshots readable by the whole organization. Approves or rejects pending snapshots within the grant. Cannot review their own submission; approval does not make the reviewer a reader.

#	Actor	Description
3	Organization Admin	The final authority of an organization; the creator becomes the first Organization Admin. Manages departments, invitations, members, audience groups and reviewer grants, and may decide any pending submission, including their own (audited as self-approval). (Planned) Unpublishes and restores knowledge and views the dashboard. Gains no readership or private-note access from this capability.
4	Arden System Admin	A member of the Arden operations team who administers the installation from outside every organization (capability INSTANCE_ADMIN). The first is created by a one-time bootstrap command; later ones are granted by an active System Admin; the last active one cannot be revoked. Has no organization, content, review or AI access through this capability.

2.2 Use Cases

2.2.1 Diagrams

Members are specialised by Knowledge Reviewer and Organization Admin (generalization). The implementation status of each use case is given in section 2.2.2.

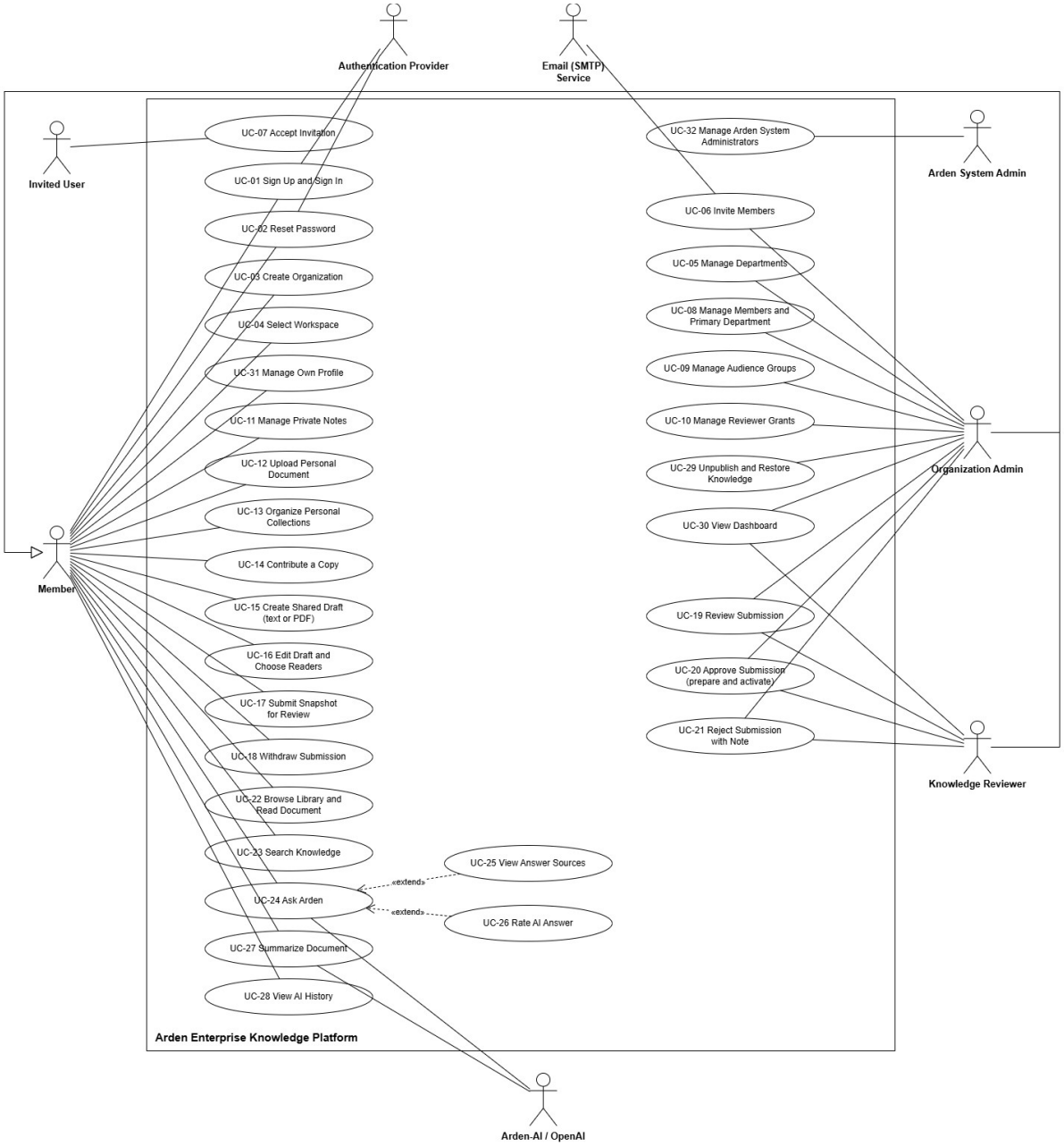


Figure 2: Use case diagram

2.2.2 Descriptions

ID	Use Case	Actors	Description	Status
UC-01	Sign Up and Sign In	All users	Create an account, sign in with email/password or Google, set a password after Google sign-in, sign out.	Implemented
UC-02	Reset Password	All users	Request a reset code and set a new password.	Implemented
UC-03	Create Organization	Member	Create an organization with its first department and become its Organization Admin.	Implemented
UC-04	Select Workspace	Member	Choose one of the user's organizations with an active membership.	Implemented

ID	Use Case	Actors	Description	Status
UC-05	Manage Departments	Organization Admin	Create or delete (archive) departments; each receives its managed Department audience.	Implemented
UC-06	Invite Members	Organization Admin	Send, resend or revoke a single-use, 7-day invitation, optionally for a department.	Implemented
UC-07	Accept Invitation	Invited user	Sign in or create an account with the invited email and accept.	Implemented
UC-08	Manage Members and Primary Department	Organization Admin	View members and move a member to another primary department.	Implemented
UC-09	Manage Audience Groups	Organization Admin	Create, rename or deactivate groups and add or remove members.	Implemented
UC-10	Manage Reviewer Grants	Organization Admin	Create or revoke Department or Organization grants with optional tags and validity.	Implemented
UC-11	Manage Private Notes	Member	Create, search, open and edit owner-only notes.	Implemented
UC-12	Upload Personal Document	Member	Upload a private PDF to the Personal Workspace.	Planned
UC-13	Organize Personal Collections	Member	Group own notes and documents in collections.	Planned
UC-14	Contribute a Copy	Member	Copy a saved private note version into a separate shared draft.	Implemented
UC-15	Create Shared Draft	Member	Start a text draft or create one from a PDF up to 5 MB.	Implemented
UC-16	Edit Draft and Choose Readers	Member	Edit the working version, choose scope, audiences and optional tags.	Implemented
UC-17	Submit Snapshot for Review	Member	Freeze the current version, metadata and audiences into a release candidate.	Implemented
UC-18	Withdraw Submission	Member; Organization Admin	Cancel a pending submission.	Planned
UC-19	Review Submission	Knowledge Reviewer; Organization Admin	Open a pending snapshot covered by a grant (or any, for the Organization Admin).	Implemented
UC-20	Approve Submission	Knowledge Reviewer; Organization Admin; System	Approve the exact snapshot; the system prepares, indexes and activates it.	Implemented
UC-21	Reject Submission	Knowledge Reviewer; Organization Admin	Reject with a mandatory note to the author.	Implemented
UC-22	Browse Library and Read Document	Member	List and read active publications in the member's audiences.	Implemented

ID	Use Case	Actors	Description	Status
UC-23	Search Knowledge	Member	Keyword and semantic search over readable knowledge.	Planned
UC-24	Ask Arden	Member; AI service	Ask a question and receive an answer grounded in readable publications.	Implemented
UC-25	View Answer Sources	Member	Extends UC-24: open the cited document and page.	Implemented
UC-26	Rate AI Answer	Member	Extends UC-24: rate 1–5 with a comment.	Planned
UC-27	Summarize Document	Member; AI service	Summarize a readable publication or own note.	Planned
UC-28	View AI History	Member	List own questions and answers.	Planned
UC-29	Unpublish and Restore Knowledge	Organization Admin	Remove an active publication from reading and AI, or restore it.	Planned
UC-30	View Dashboard	Organization Admin; Knowledge Reviewer	View counts, AI usage, feedback and quality reports.	Planned
UC-31	Manage Own Profile	Member	Change the display name.	Planned
UC-32	Manage Arden System Administrators	Arden System Admin	List, grant and revoke System Admins.	Partial (API)

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screen Flow

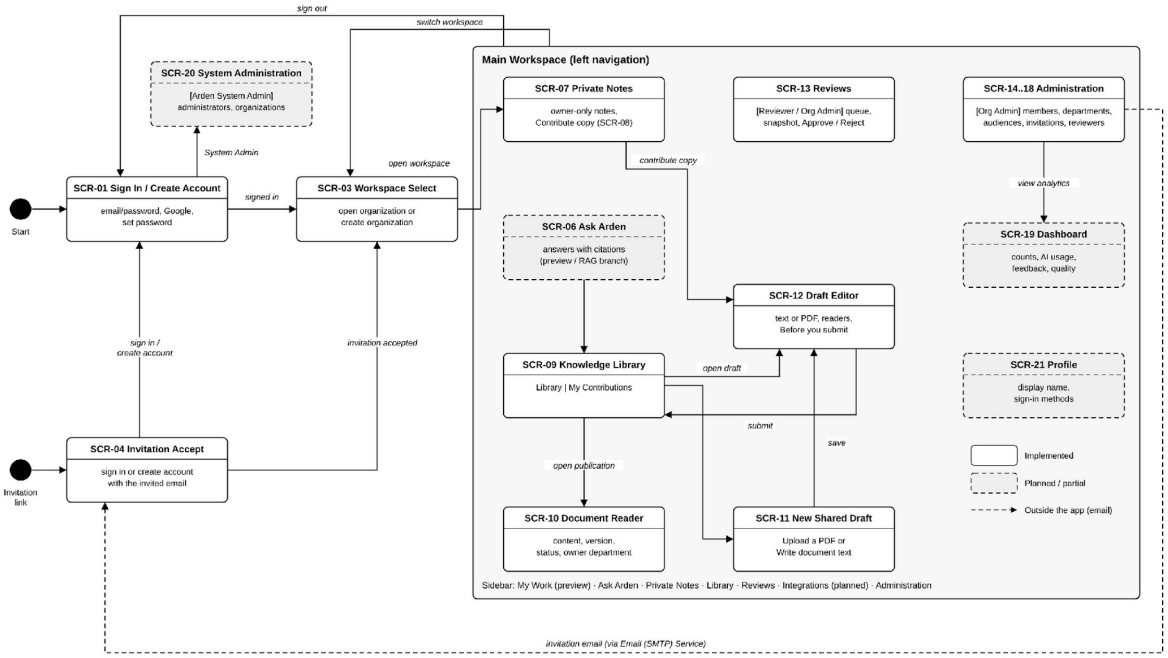


Figure 3: Screen flow

3.1.2 Screen Descriptions

ID	Screen	Description	Status
SCR-01	Sign In / Create Account	Email/password sign-in, Google sign-in, account creation and password setup.	Implemented
SCR-02	Reset Password	Request a reset code by email and set a new password.	Implemented
SCR-03	Workspace Select	List the user's organizations and create a new organization.	Implemented
SCR-04	Invitation Accept	Show the invitation and let the invited account sign in or create an account and accept.	Implemented
SCR-05	My Work	Home overview of drafts, submissions and assigned reviews.	Partial (preview)
SCR-06	Ask Arden	Question box, answers with citations and starter prompts.	Implemented
SCR-07	Private Notes	Owner-only note list with search and editor.	Implemented
SCR-08	Contribute a Copy (dialog)	Confirms the saved version that becomes a separate shared draft.	Implemented
SCR-09	Knowledge Library	Tabs Library (readable publications) and My Contributions (own drafts and their review status).	Implemented
SCR-10	Document Reader	Content, version selector, status, effective date and owner department of an active publication.	Implemented
SCR-11	New Shared Draft	Choose Upload a PDF or Write document text, enter title and text.	Implemented

ID	Screen	Description	Status
SCR-12	Draft Editor and Before You Submit	Rich-text editor, reader picker, tags, readiness checklist, Save version and Submit.	Implemented
SCR-13	Reviews	Review queue, frozen snapshot, decision panel (Approve/Reject with note).	Implemented
SCR-14	Administration – Members	Member list with filters; member detail with department, grants and audiences.	Implemented
SCR-15	Administration – Departments	Department list and creation.	Implemented
SCR-16	Administration – Audiences	Managed audiences and audience groups with members.	Implemented
SCR-17	Administration – Invitations	Invitation list by status; invite, resend and revoke.	Implemented
SCR-18	Administration – Reviewer Access	Active/inactive grants and the New grant form.	Implemented
SCR-19	Dashboard	Organization statistics, AI usage, feedback and quality lists.	Planned
SCR-20	System Administration	Arden System Admin list and organization overview.	Planned
SCR-21	Profile	Display name and sign-in information.	Planned

3.1.3 Screen Authorization

Authorization is evaluated on the server for every request and again by PostgreSQL row-level security. The table lists screens and the activities inside them. X = allowed; Own = only own items; Grant = only within an active reviewer grant; Audience = only items whose audiences include the user; — = not allowed. Knowledge Reviewer and Organization Admin also have every Member right.

Screen / activity	Member	Knowledge Reviewer	Organization Admin	Arden System Admin
Sign In / Workspace Select / Invitation Accept	X	X	X	Sign in only
Create organization	X	X	X	—
Private Notes: create, read, edit, search	Own	Own	Own	—
Contribute a copy	Own	Own	Own	—
Library: list and read publications	Audience	Audience	Audience	—
New draft (text or PDF), edit, choose readers	Own	Own	Own	—
Submit snapshot	Own	Own	Own	—
Withdraw submission (planned)	Own	Own	X	—
Reviews: view queue and snapshot	—	Grant	X	—
Approve / reject	—	Grant, not own	X (own: audited)	—
Unpublish / restore (planned)	—	—	X	—

Screen / activity	Member	Knowledge Reviewer	Organization Admin	Arden System Admin
Administration: view members and departments	—	—	X	—
Create department	—	—	X	—
Invite, resend, revoke invitation	—	—	X	—
Change primary department	—	—	X	—
Create, edit, deactivate audience group; add/remove members	—	—	X	—
Create / revoke reviewer grant	—	—	X	—
Ask Arden	Audience	Audience	Audience	—
Dashboard (planned)	—	Grant (quality lists)	X	—
System Administration: list, grant, revoke System Admin	—	—	—	X
Organization overview (planned)	—	—	—	X

3.1.4 Non-Screen Functions

ID	System function	Description
NSF-01	Session and authentication context	Signs users in through Supabase Auth behind the Arden API, keeps access and refresh tokens in HttpOnly cookies, refreshes expired access tokens on request, and derives membership, primary department, group memberships, grants and System Admin status on the server for each request. Untrusted browser origins are refused (UNTRUSTED_ORIGIN).
NSF-02	Authorization decision	Applies ownership, audience, grant and administrator rules in the API before any content, metadata, count or citation is returned; PostgreSQL RLS repeats the check under the non-bypass role arden_runtime. Unauthorized and missing resources return the same RESOURCE_NOT_AVAILABLE response.
NSF-03	Invitation email delivery	Creates a random 43-character token stored only as a SHA-256 hash, expiring after 7 days, and sends the link through SMTP. A delivery failure keeps the invitation and returns INVITATION_DELIVERY_FAILED so the admin can resend.
NSF-04	Private content storage	Stores note and draft content and PDF files in private storage behind the API with a SHA-256 content hash; integrity is checked on every read.
NSF-05	Semantic Document normalization	Converts plain-text input into Semantic Document v1 and validates stored documents (stable block IDs, at most 2,000 blocks, depth 8, 131,072 characters, safe links).
NSF-06	Idempotency and concurrency control	Requires Idempotency-Key on create, submit and decision requests and If-Match on edits; repeats return the first result and stale versions return VERSION_CONFLICT.
NSF-07	Review decision and publication preparation	In one transaction records the decision and audit event, checks the frozen snapshot hash, and on approval creates the PREPARING publication, an index generation and an index job.

ID	System function	Description
NSF-08	Index-generation background job	A Node.js worker polls every second, leases one job for 180 s, reads the snapshot, verifies its hash and calls Arden-AI POST /v1/index-generations (timeout 120 s). Retryable failures are retried up to 3 attempts; non-retryable failures mark the generation FAILED.
NSF-09	Manifest validation and atomic activation	Validates the returned manifest (identity, chunk ordinals, chunk digest) and, in one transaction, marks the generation READY, sets the publication ACTIVE and replaces the current-publication pointer. A failed or partial generation never becomes visible.
NSF-10	Permission-aware answer (Arden-AI)	Computes the READY generations the asker may read, sends the question and allow-list to Arden-AI POST /v1/answers, and rechecks every citation against current access before responding; if access changed, the answer is refused (KNOWLEDGE_ACCESS_CHANGED).
NSF-11	Audit logging	Records actor, time, object, action and outcome for organization, invitation, membership, audience, grant, submission, decision (with selfApproved), activation and System Admin events.
NSF-12	System Admin bootstrap	One-time server command system-admin:bootstrap; refuses when any System Admin exists.
NSF-13	Health checks	GET /api/health/live and /api/health/ready for Coolify; Arden-AI exposes /health/live and /health/ready.
NSF-14	Analytics aggregation	(Planned) Computes dashboard counts from knowledge, review, audit and AI records without exposing unreadable content.

3.1.5 Entity Relationship Diagram

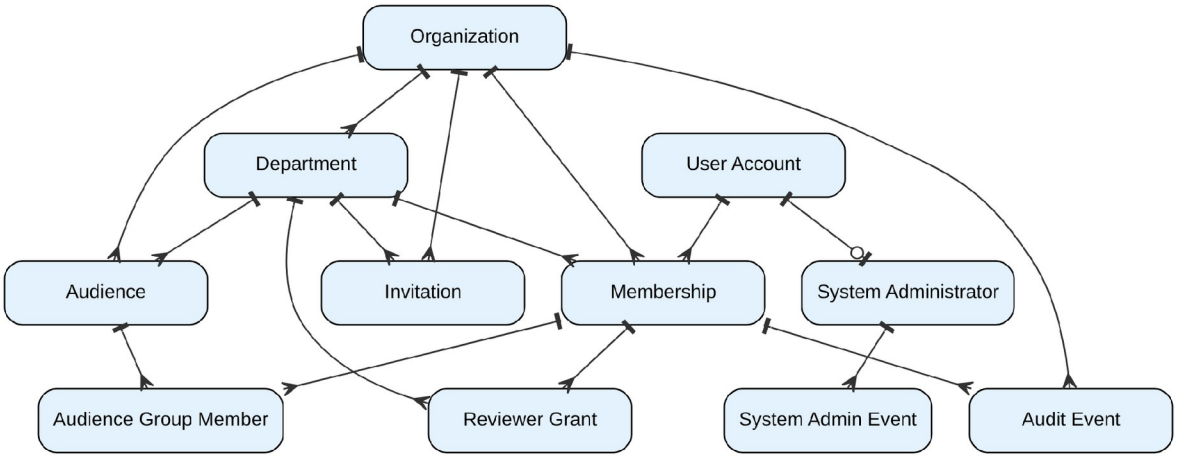


Figure 4: Logical ERD – organization, access and administration

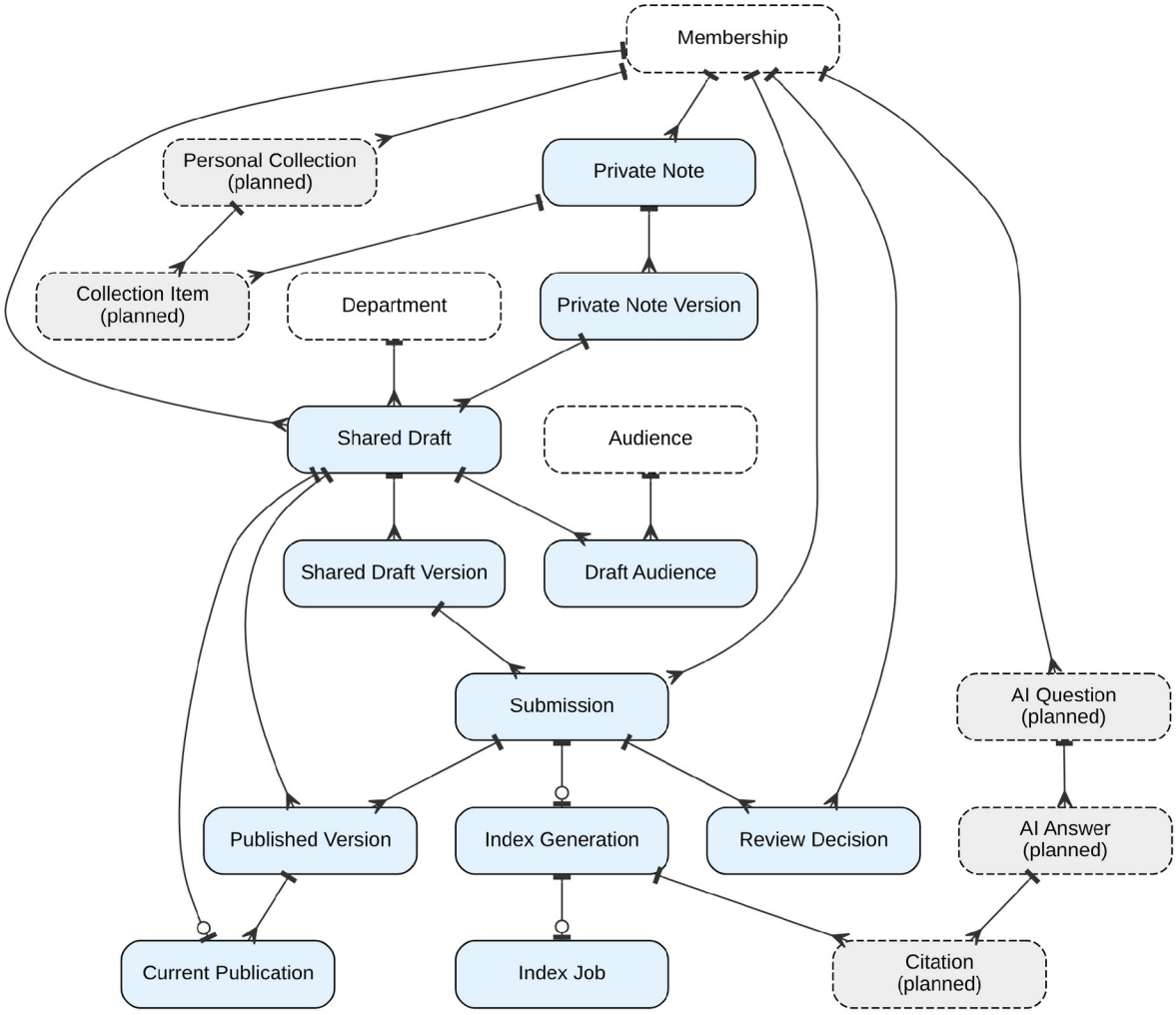


Figure 5: Logical ERD – personal workspace, knowledge lifecycle, indexing and AI

Shaded, dashed entities are planned. Legacy tables (role, permission, role_permission, access_scope, role_assignment, authorization_policy) and legacy columns (classification, reviewer level) remain only for compatibility and are not part of the logical model. Physical names are given in Report 4.

Entity Descriptions

#	Entity	Description
1	Organization	An isolated tenant.
2	Department	Unit inside an organization; has one managed Department audience.
3	User Account	An authenticated identity (Supabase Auth); may belong to several organizations.
4	Membership	A user in an organization with status (Invited, Active, Suspended, Left) and at most one primary department.
5	Invitation	Hashed single-use token for one email and an optional department; status Pending, Accepted, Revoked or Expired.
6	Audience	Organization, Department or Group reader set; Active or Inactive.
7	Audience Group Member	A member in an audience group (Active, Revoked, Expired).

#	Entity	Description
8	Reviewer Grant	Review authority with scope Department (one department) or Organization, optional tags, validity and status.
9	System Administrator / System Admin Event	Installation administrator (Active, Revoked) and the audit of its changes.
10	Private Note / Private Note Version	Owner-only item with immutable versions.
11	Personal Collection / Collection Item	(Planned) Owner-only grouping of private items.
12	Shared Draft	Author-owned shared item with title, responsible department, scope, tags and status (Active, Archived).
13	Shared Draft Version	Immutable content version (Semantic Document or PDF) with content hash.
14	Draft Audience	The audiences chosen for a shared draft.
15	Submission	Immutable snapshot with frozen metadata, audiences and review rule; status Pending Review, Approved, Rejected, Withdrawn; publication state None, Preparing, Active, Failed.
16	Review Decision	Approve or Reject, note, reviewer, time and selfApproved flag.
17	Published Version / Current Publication	The approved snapshot and the single active pointer per knowledge item.
18	Index Generation / Index Job	Durable indexing record (Pending, Processing, Ready, Failed) with manifest, and its leased job (Pending, Running, Succeeded, Failed; max 3 attempts).
19	Audit Event	Traceable record of a governed action and its outcome.
20	AI Question / Answer / Citation	(Planned in core) Question, answer status (Answered, Insufficient evidence), citations to generation, chunk and block, rating and comment.

3.2 Account and Session

Lets people create an Arden account, sign in, keep a secure session and recover access.

ID	Function	Actor	Status
FR-AUTH-01	Sign up and sign in with email and password	Anyone (unauthenticated). After sign-in the user acts as a Member of their organizations.	Implemented
FR-AUTH-02	Sign in with Google and set a password	Anyone with a Google account.	Implemented
FR-AUTH-03	Reset a forgotten password	Anyone.	Implemented
FR-AUTH-04	Keep and refresh the session	System.	Implemented
FR-AUTH-05	Manage own profile	Member	Planned

3.2.1 FR-AUTH-01 Sign up and sign in with email and password

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-01; SCR-01; NSF-01
Function trigger	The user opens Arden without a session, or clicks Sign out.
Actor and permission	Anyone (unauthenticated). After sign-in the user acts as a Member of their organizations.
Function description	Create an account with email and password, confirm the email, sign in and sign out. Arden calls Supabase Auth on the server and stores the session in HttpOnly cookies.
Preconditions	For sign-in, the account exists and its email is confirmed.
Postconditions	A session cookie pair is set; the user reaches Workspace Select or their last organization.
Main flow	1. User enters email and password and selects Sign in. 2. API forwards the credentials to Supabase Auth. 3. On success, API sets the access and refresh cookies and returns the user profile. 4. Client loads /api/v1/me and the user's organizations and opens the workspace. 5. Sign out: POST /auth/sign-out revokes the session and clears both cookies.
Alternate and error flows	• Wrong email or password → 401 INVALID_CREDENTIALS "Email or password is incorrect." (does not reveal whether the email exists). • Supabase unavailable → 503 AUTH_UNAVAILABLE. • Request from an untrusted origin → 403 UNTRUSTED_ORIGIN. • Weak password at sign-up → PASSWORD_REJECTED.
Inputs and validation	Email (valid format, ≤320); password 1–1,024 characters at sign-in; Supabase password policy at sign-up.
Data read / changed	Reads/creates the Supabase user and the Arden user_account record; no organization data is changed.
Business rules	BR-ROLE-001; CR-01.
Screen / API / job	POST /api/v1/auth/sign-up, /auth/sign-in, /auth/sign-out; GET /api/v1/me. Figure: Sign In.
Acceptance criteria	AC1: A confirmed account signs in and reaches its workspace. AC2: A wrong password returns the generic message and sets no cookie. AC3: After sign-out, any protected request with the old cookies returns 401 UNAUTHENTICATED.

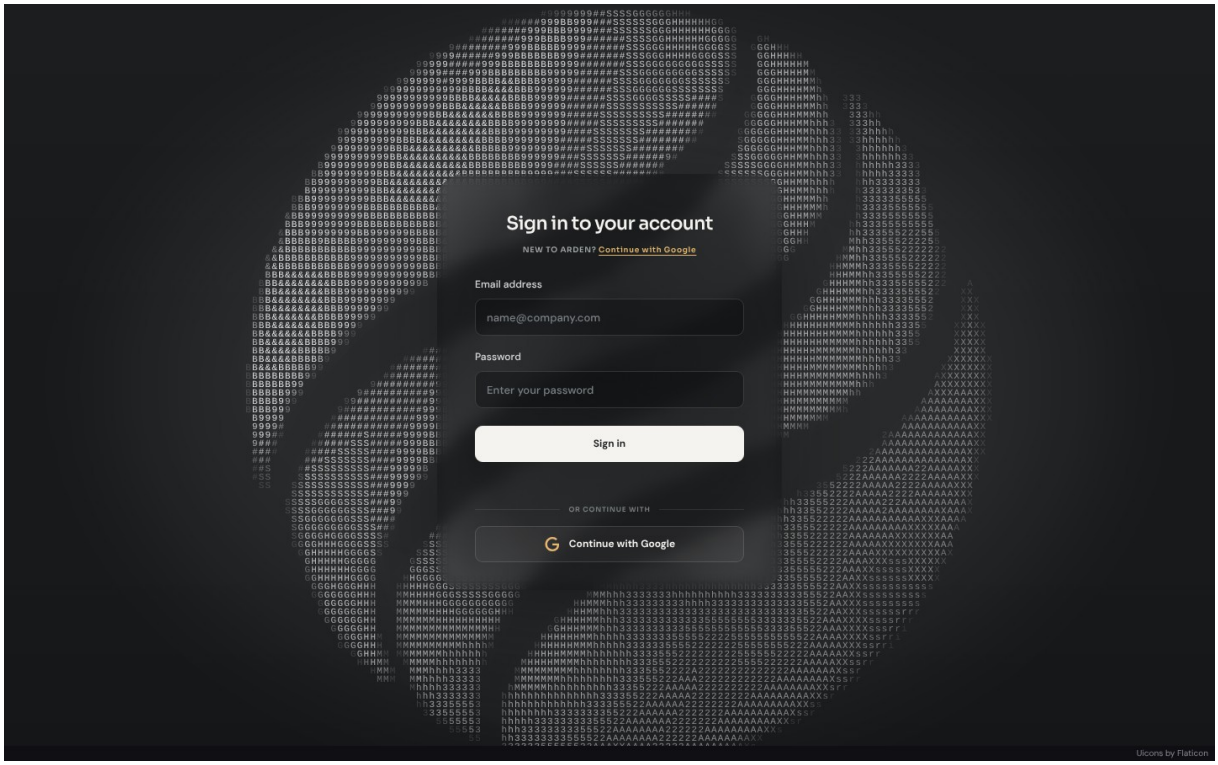


Figure 6: Sign In / Create Account (SCR-01)

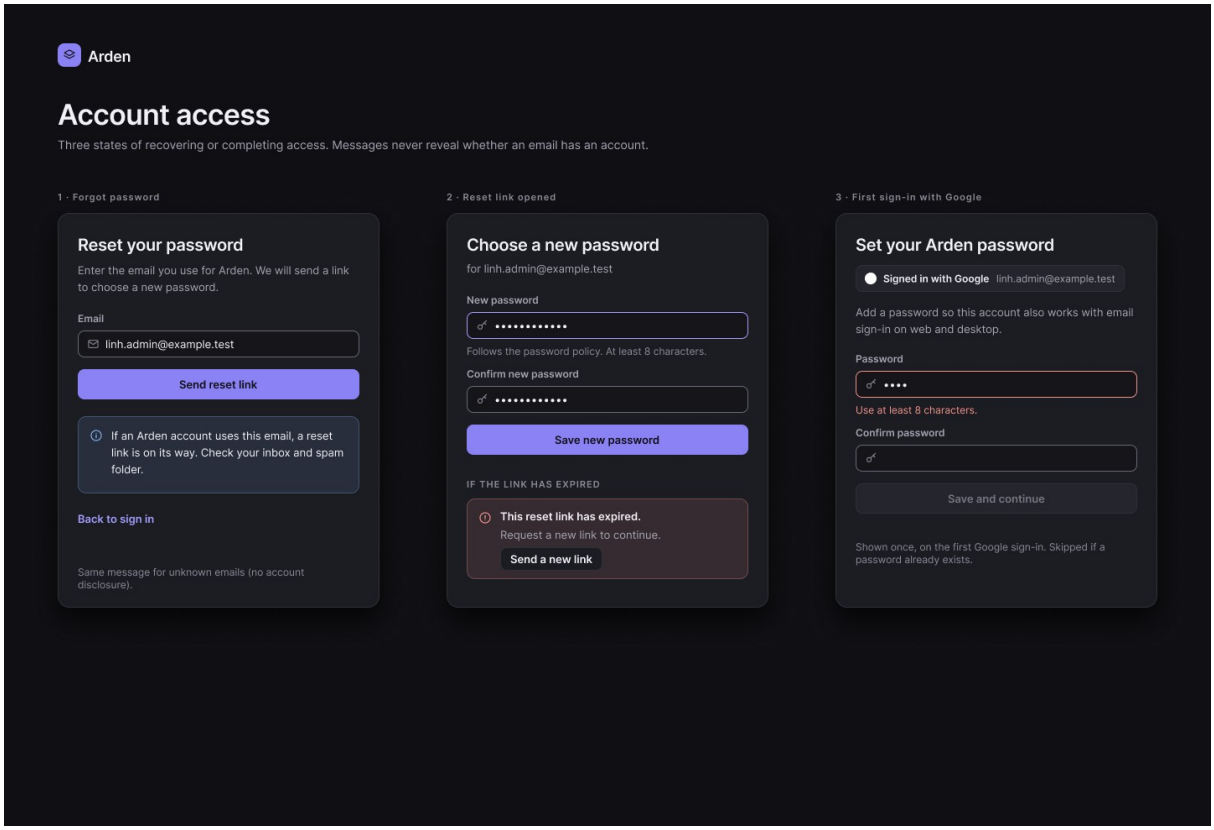
3.2.2 FR-AUTH-02 Sign in with Google and set a password

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-01; SCR-01
Function trigger	The user selects Continue with Google.
Actor and permission	Anyone with a Google account.
Function description	OAuth sign-in through Supabase (PKCE). On first Google sign-in the user must set an Arden password so the account also works with email sign-in; the desktop client completes the flow in its main process.
Preconditions	Google sign-in is configured for the installation.
Postconditions	Authenticated session; password initialized.
Main flow	1. Client calls GET /auth/google/start (web or desktop). 2. User consents at Google; Supabase redirects to /auth/google/callback. 3. API exchanges the code, sets cookies and redirects to the web app (or to /auth/google/desktop-complete). 4. If no password exists, the Set password screen requires one (POST /auth/password).
Alternate and error flows	• Callback without a valid code or flow ID → sign-in refused. • Password already set → PASSWORD_ALREADY_INITIALIZED. • Password shorter than 8 characters → validation error.
Inputs and validation	New password 8–1,024 characters.

Item	Specification
Data read / changed	Supabase identity; user_account.
Business rules	BR-ROLE-001.
Screen / API / job	GET /auth/google/start, /auth/google/callback, /auth/google/desktop-complete; POST /auth/password.
Acceptance criteria	AC1: A first Google sign-in cannot reach the workspace before a password is set. AC2: The desktop client never exposes tokens to the renderer.

3.2.3 FR-AUTH-03 Reset a forgotten password

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-02; SCR-02
Function trigger	The user selects Forgot password on the sign-in screen.
Actor and permission	Anyone.
Function description	Send a reset code by email and let the user set a new password; each code can be used once.
Preconditions	The email belongs to an account.
Postconditions	The new password works; old sessions are refused.
Main flow	1. User enters email (POST /auth/forgot-password). 2. Supabase sends a reset code to the email. 3. User enters the code and a new password (POST /auth/reset-password); a code that was already used is refused.
Alternate and error flows	• Unknown email → the same neutral confirmation (no account disclosure). • Expired link → user must request a new one.
Inputs and validation	Email; new password per Supabase policy.
Data read / changed	Supabase user.
Business rules	CR-06.
Screen / API / job	POST /auth/forgot-password, /auth/reset-password, /auth/verify.
Acceptance criteria	AC1: The response is identical for known and unknown emails. AC2: After reset, the old password is rejected. AC3: A reset code that was already used is refused.



Figma screen R3-02: Password reset and Google sign-in

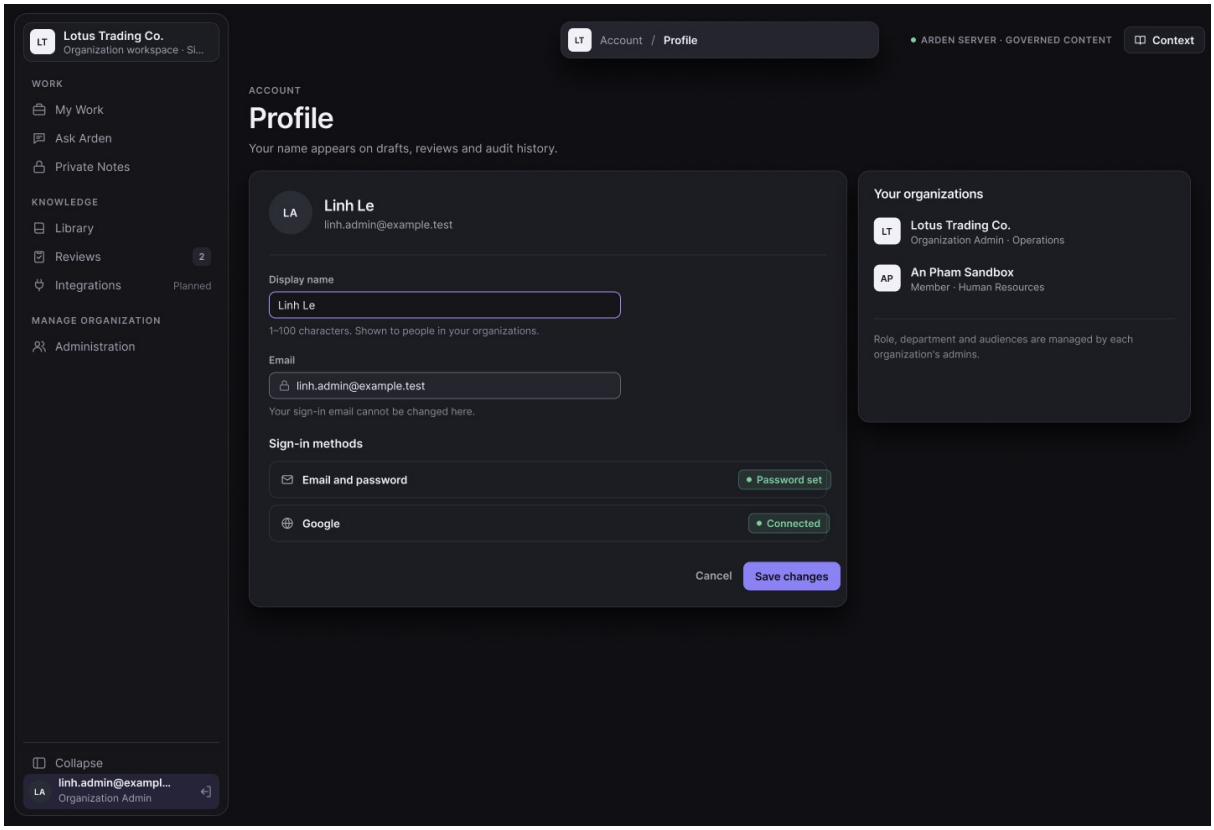
3.2.4 FR-AUTH-04 Keep and refresh the session

Item	Specification
Status	Implemented
Related UC / screen / NSF	NSF-01
Function trigger	Any API request with session cookies.
Actor and permission	System.
Function description	Resolve the user from the access token; when it has expired, refresh it with the refresh token and rotate both cookies.
Preconditions	Cookies present.
Postconditions	Valid session or 401.
Main flow	1. API reads the access and refresh cookies. 2. API validates the access token with Supabase. 3. If expired, API calls refreshSession, verifies the new token and sets rotated cookies.
Alternate and error flows	• Invalid or revoked refresh token → 401 UNAUTHENTICATED; the client returns to Sign In.
Inputs and validation	Cookies only.
Data read / changed	None.
Business rules	CR-01.

Item	Specification
Screen / API / job	Cookie names arden_access / arden_refresh (__Host- prefix in production).
Acceptance criteria	AC1: An expired access token with a valid refresh token is renewed without signing in again. AC2: A revoked session cannot be refreshed.

3.2.5 FR-AUTH-05 Manage own profile

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-31; SCR-21
Function trigger	The user opens Profile from the sidebar.
Actor and permission	Member
Function description	View email and sign-in methods and change the display name.
Preconditions	Signed in.
Postconditions	Display name updated everywhere it is shown.
Main flow	1. User edits the display name (1–100 characters) and saves. 2. API updates the user account and returns the profile.
Alternate and error flows	• Empty or too long name → validation message MSG-001.
Inputs and validation	Display name 1–100 characters.
Data read / changed	user_account.full_name.
Business rules	—
Screen / API / job	Planned screen SCR-21.
Acceptance criteria	AC1: The new name appears in member lists and review history. AC2: A user cannot change another user's profile.



3.3 Organization and Membership

Creates isolated organizations and controls who joins them and in which department.

ID	Function	Actor	Status
FR-ORG-01	Create an organization	Any signed-in user with a verified email.	Implemented
FR-ORG-02	Select a workspace	Member	Implemented
FR-ORG-03	Manage departments	Organization Admin	Implemented
FR-ORG-04	Invite members	Organization Admin with a verified email	Implemented
FR-ORG-05	Accept an invitation	Invited user	Implemented
FR-ORG-06	Manage members and primary department	Organization Admin	Implemented (removal and admin promotion disabled)

3.3.1 FR-ORG-01 Create an organization

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-03; SCR-03
Function trigger	The user selects Create organization on Workspace Select.

Item	Specification
Actor and permission	Any signed-in user with a verified email.
Function description	Create a new isolated organization with its first department. The creator becomes the first Organization Admin; the Organization audience and the first Department audience are created automatically.
Preconditions	Signed in, verified email.
Postconditions	Organization, department, membership (Organization Admin, primary department), managed audiences and audit events exist.
Main flow	1. User enters the organization name and the primary department name. 2. API validates and creates all records in one transaction. 3. Client opens the new organization.
Alternate and error flows	• Empty name → 400 INVALID_ORGANIZATION. • Conflicting details → 409 ORGANIZATION_CONFLICT.
Inputs and validation	Organization name 1–120; department name 1–80 characters.
Data read / changed	Creates organization, department, organization_membership, department_membership, audiences, audit_event.
Business rules	BR-ORG-001, BR-ORG-002, BR-ORG-005.
Screen / API / job	POST /api/v1/organizations. Figure: Workspace Select.
Acceptance criteria	AC1: The creator can open Administration of the new organization. AC2: Another user without membership receives RESOURCE_NOT_AVAILABLE for every route of it.

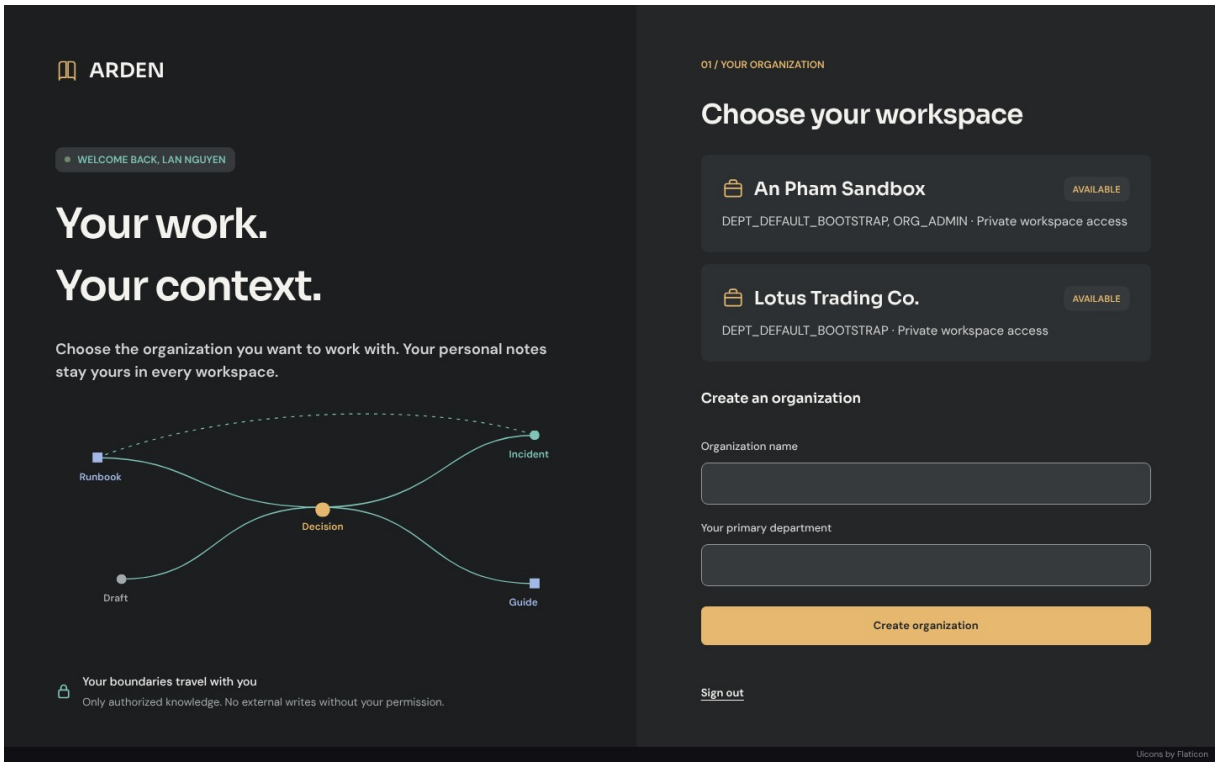


Figure 7: Workspace Select with Create organization (SCR-03)

3.3.2 FR-ORG-02 Select a workspace

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-04; SCR-03
Function trigger	After sign-in when the user has zero or several organizations, or via Switch workspace.
Actor and permission	Member
Function description	List the user's organizations and open one.
Preconditions	Signed in.
Postconditions	The chosen organization becomes the context of every request.
Main flow	1. Client lists organizations from /api/v1/me. 2. User selects one marked Available.
Alternate and error flows	• Suspended or Left membership → organization not enterable.
Inputs and validation	Organization ID.
Data read / changed	Reads memberships.
Business rules	BR-MEM-001.
Screen / API / job	GET /api/v1/me.
Acceptance criteria	AC1: Only the user's own organizations are listed. AC2: A Suspended membership cannot open the organization.

3.3.3 FR-ORG-03 Manage departments

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-05; SCR-15
Function trigger	Administration → Departments → New department.
Actor and permission	Organization Admin
Function description	Create or delete a department. Arden creates its managed "Everyone in <department>" audience; deleting a department archives it instead of removing it.
Preconditions	Actor is Organization Admin of the organization.
Postconditions	Department and Department audience exist; after deletion the department is Archived, its audience, access scope and reviewer grants are inactive, and its members have no primary department until reassigned.

Item	Specification
Main flow	1. Admin enters a name and selects Create department. 2. API creates the department and its audience; the list shows members, primary members and reviewers per department. 3. Delete: Admin confirms; API archives the department, ends its department memberships and default-role assignments, clears it from pending invitations, deactivates its audience, access scope and reviewer grants, and records DEPARTMENT_DELETED.
Alternate and error flows	• Empty name → "Enter a valid department name." • Non-admin → RESOURCE_NOT_AVAILABLE.
Inputs and validation	Name 1–80 characters.
Data read / changed	department, audience (DEPARTMENT_ALL), department_membership, role_assignment, organization_invitation, access_scope, reviewer_grant, audit_event.
Business rules	BR-ORG-003, BR-ORG-005.
Screen / API / job	POST /api/v1/organizations/{organizationId}/departments; POST …/departments/{departmentId}/delete; GET /api/v1/organizations/{organizationId}/administration.
Acceptance criteria	AC1: The new department appears as a primary-department and reader option. AC2: A Member calling the API is denied. AC3: After deletion the department is Archived, its members can no longer read its Department publications on the next request, and they appear as unassigned.

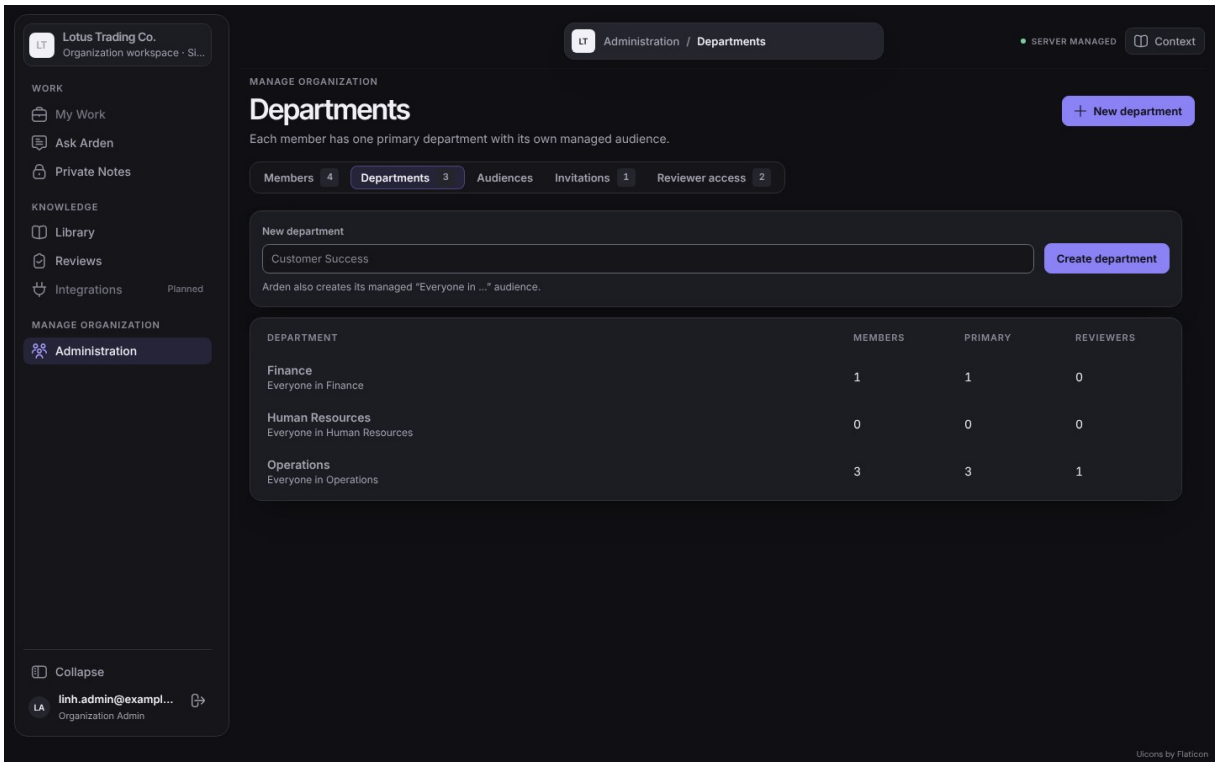


Figure 8: Administration – Departments (SCR-15)

3.3.4 FR-ORG-04 Invite members

Item	Specification
Status	Implemented

Item	Specification
Related UC / screen / NSF	UC-06; SCR-17; NSF-03
Function trigger	Administration → Invitations → Send invitation, or Resend / Revoke on a row.
Actor and permission	Organization Admin with a verified email
Function description	Invite a person by email, optionally into a primary department. The link is single-use and expires after 7 days.
Preconditions	No pending invitation exists for the same email in the organization.
Postconditions	Invitation Pending; email sent.
Main flow	1. Admin enters the work email and, optionally, a primary department. 2. API stores the hashed token and expiry, then sends the email. 3. The Invitations tab lists the invitation with status, department and expiry. 4. Resend creates a new link; Revoke makes the old link unusable.
Alternate and error flows	• Pending invitation exists → 409 INVITATION_CONFLICT. • Admin email not verified → 403 VERIFIED_EMAIL_REQUIRED. • SMTP failure → 502 INVITATION_DELIVERY_FAILED (invitation kept; can be resent). • SMTP not configured → 503 INVITATIONS_UNAVAILABLE.
Inputs and validation	Email (valid, ≤320); department ID of an active department.
Data read / changed	organization_invitation (token_hash, status, expires_at), audit_event.
Business rules	BR-MEM-002, BR-MEM-003.
Screen / API / job	POST /api/v1/organizations/{organizationId}/invitations; POST …/invitations/{id}/resend; POST …/invitations/{id}/revoke.
Acceptance criteria	AC1: The recipient receives a link valid for 7 days (MSG-017). AC2: A second invitation to the same pending email is refused. AC3: A revoked link no longer works. AC4: An invitation without a department creates an Active membership with no primary department when accepted.

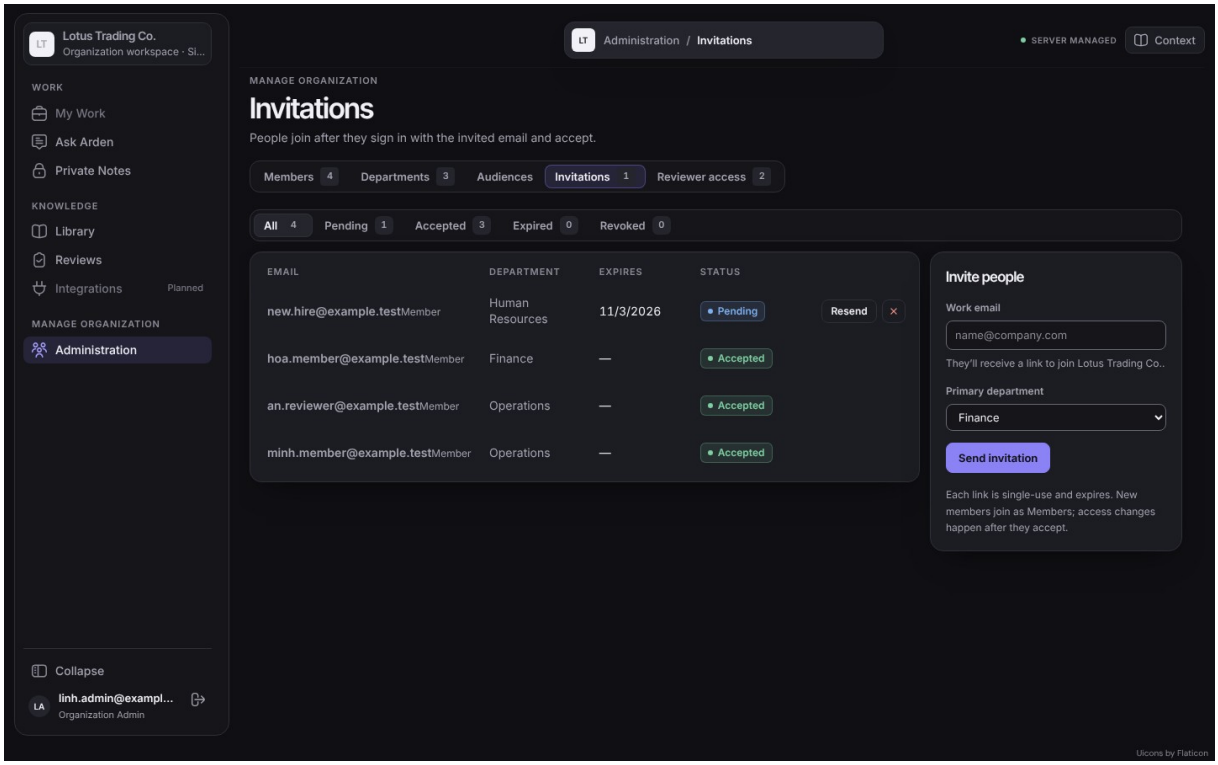


Figure 9: Administration – Invitations with the invite form (SCR-17)

3.3.5 FR-ORG-05 Accept an invitation

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-07; SCR-04
Function trigger	The invited person opens the email link /invite/{token}.
Actor and permission	Invited user
Function description	Sign in or create an account with the invited email and explicitly accept to become a Member of the invited department.
Preconditions	Invitation Pending and not expired.
Postconditions	Active membership with the invited primary department, or with no department if none was invited; invitation Accepted.
Main flow	1. Page shows Continue to Arden with Sign in / Create account. 2. User authenticates with the invited email and selects Sign in and accept. 3. API verifies the token hash, expiry and verified email match, then creates the membership.
Alternate and error flows	• Expired, revoked, already used or other-account invitation → INVITATION_NOT_AVAILABLE (MSG-016), nothing created.
Inputs and validation	Token: 43 URL-safe characters.
Data read / changed	organization_invitation, organization_membership, department_membership, audit_event.
Business rules	BR-MEM-001.

Item	Specification
Screen / API / job	POST /api/v1/invitations/accept.
Acceptance criteria	AC1: Accepting with the invited verified email creates exactly one Active membership. AC2: Another account cannot accept the link.

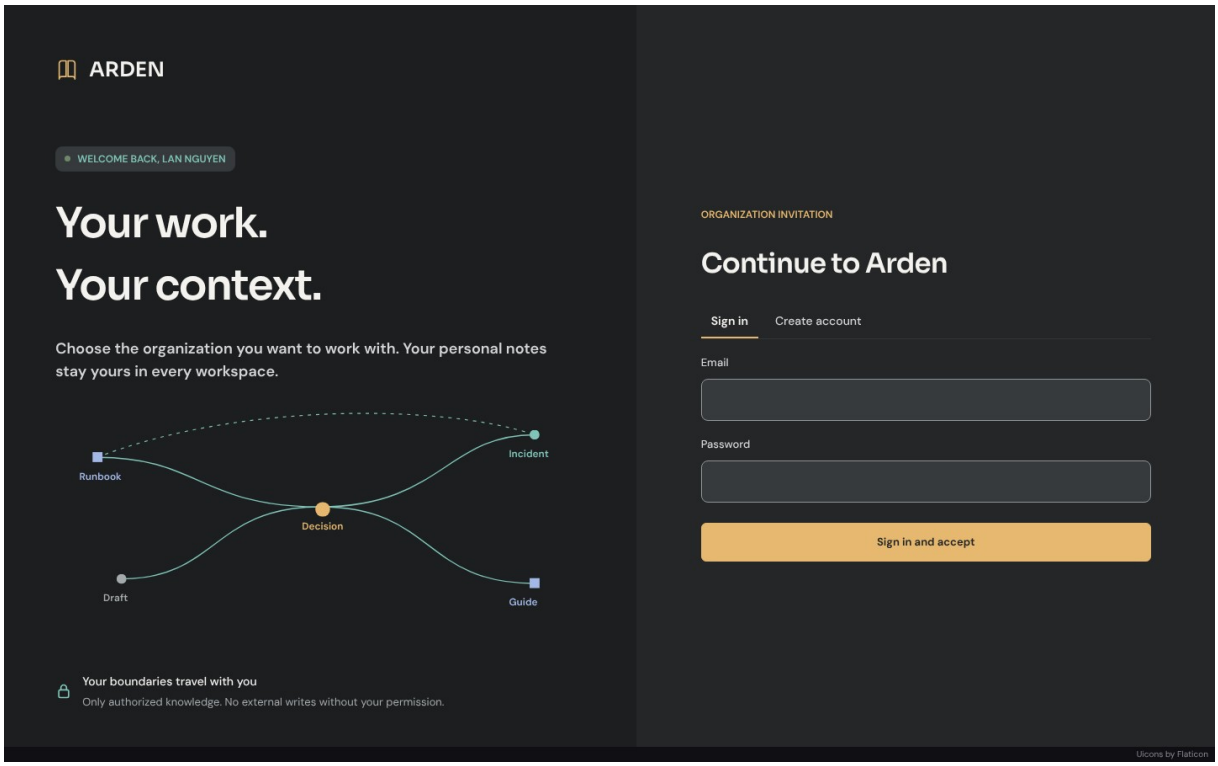


Figure 10: Invitation Accept (SCR-04)

3.3.6 FR-ORG-06 Manage members and primary department

Item	Specification
Status	Implemented (removal and admin promotion disabled)
Related UC / screen / NSF	UC-08; SCR-14
Function trigger	Administration → Members → Manage.
Actor and permission	Organization Admin
Function description	View members with department, access, grants and audiences; move a member to another primary department.
Preconditions	Target is an Active member.
Postconditions	New primary department; Department-audience readership and the responsible department of new drafts follow it.
Main flow	1. Admin filters or searches members and opens Manage. 2. Admin selects Move and a department. 3. API changes the primary department and records audit.

Item	Specification
Alternate and error flows	• Unknown member or department → RESOURCE_NOT_AVAILABLE. • "Remove from organization" and "Make Org Admin" are shown disabled.
Inputs and validation	Department ID.
Data read / changed	department_membership, audit_event.
Business rules	BR-ORG-005, BR-MEM-004, BR-ACC-006.
Screen / API / job	GET /api/v1/organizations/{organizationId}/administration; POST /api/v1/organizations/{organizationId}/members/{membershipId}/department.
Acceptance criteria	AC1: After the move the member reads Department publications of the new department only, on the next request. AC2: Pending submissions keep their frozen responsible department.

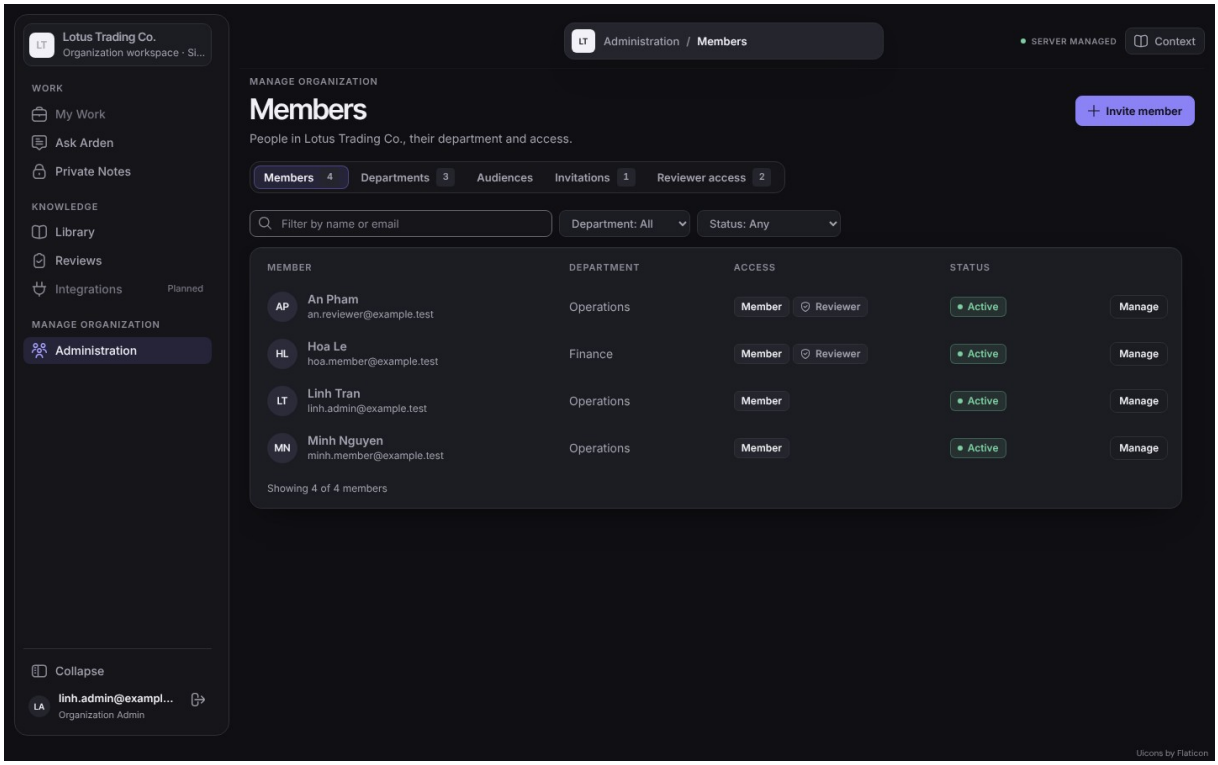


Figure 11: Administration – Members (SCR-14)

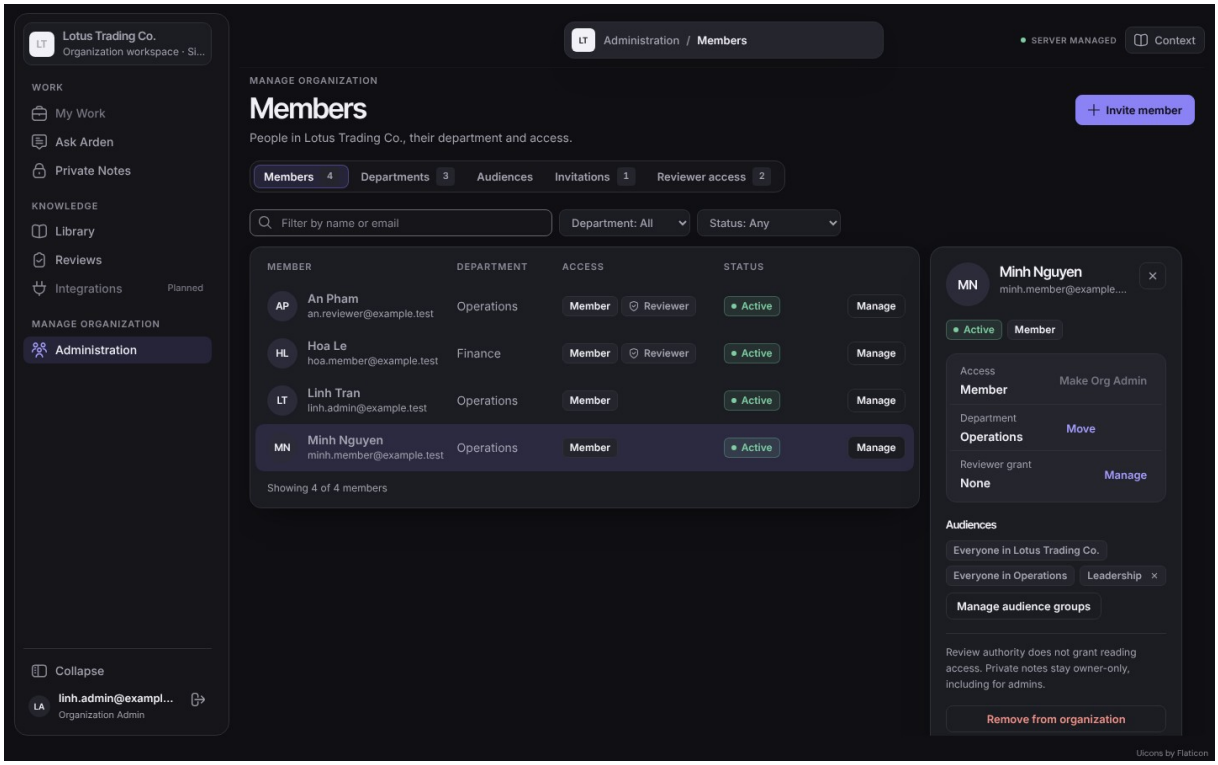


Figure 12: Member detail

3.4 Audiences, Reviewer Grants and System Administration

Defines who may read, who may review, and who administers the installation.

ID	Function	Actor	Status
FR-ACC-01	Manage audience groups	Organization Admin	Implemented
FR-ACC-02	Manage reviewer grants	Organization Admin	Implemented
FR-SYS-01	Bootstrap the first Arden System Admin	Server operator	Implemented
FR-SYS-02	Manage Arden System Admins	Arden System Admin	Partial (API only)
FR-SYS-03	View the organization overview	Arden System Admin	Planned

3.4.1 FR-ACC-01 Manage audience groups

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-09; SCR-16
Function trigger	Administration → Audiences → New audience group, or select a group.
Actor and permission	Organization Admin
Function description	Create, rename (with version check) or deactivate groups and add or remove active members. Managed Organization and Department audiences are shown but cannot be edited.

Item	Specification
Preconditions	Actor is Organization Admin.
Postconditions	Group membership changes take effect on the next request.
Main flow	1. Admin creates a group with name and description. 2. Admin selects a group, chooses an active member and selects Add, or removes a member. 3. Deactivate requires the current version.
Alternate and error flows	• Stale version → 428/409 "Reload the audience group before saving." • Member or admin of another organization → non-enumerating denial.
Inputs and validation	Name 1–80; description ≤300 characters; membership ID of an active member.
Data read / changed	audience (GROUP), audience_group_membership, audit_event.
Business rules	BR-ACC-003, BR-ACC-004, BR-ACC-006.
Screen / API / job	GET/POST /api/v1/organizations/{organizationId}/audience-groups; PATCH …/{id}; POST …/{id}/members; POST …/members/{id}/remove; POST …/{id}/deactivate.
Acceptance criteria	AC1: A member added to a group reads publications shared with it on the next request. AC2: After removal the member is denied on the next request.

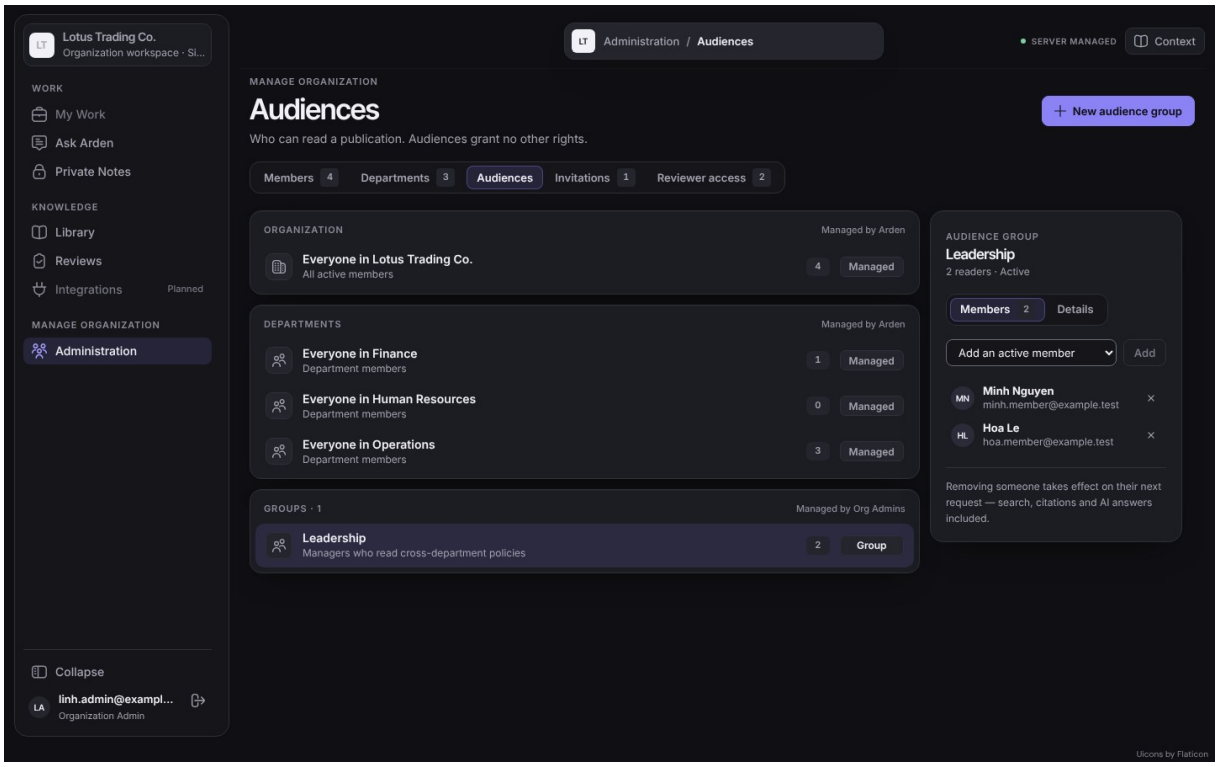


Figure 13: Administration – Audiences (SCR-16)

3.4.2 FR-ACC-02 Manage reviewer grants

Item	Specification
Status	Implemented

Item	Specification
Related UC / screen / NSF	UC-10; SCR-18
Function trigger	Administration → Reviewer access → New grant, or Revoke.
Actor and permission	Organization Admin
Function description	Grant review authority with scope One department (department and group-only snapshots owned by that department) or Whole organization (snapshots readable by everyone). Optional team and project tags and an expiry narrow the grant.
Preconditions	Reviewer is an active member.
Postconditions	Grant Active; queues and decisions recheck it on every request.
Main flow	1. Admin chooses reviewer, scope, department (department scope only), optional tags and expiry. 2. The form previews what the grant covers. 3. Create grant; Revoke deactivates it.
Alternate and error flows	• Department scope without department, Organization scope with a department, or past expiry → "The reviewer grant is invalid." • The authoring screen warns when an organization-wide audience is chosen and no Organization grant exists.
Inputs and validation	Reviewer membership ID; scope DEPARTMENT/ORGANIZATION; tags 1–100 characters; expiry in the future.
Data read / changed	reviewer_grant (scope, department_id, tags, valid_until, status), audit_event.
Business rules	BR-GOV-002, BR-GOV-003.
Screen / API / job	GET/POST /api/v1/organizations/{organizationId}/knowledge/reviewer-grants; POST …/reviewer-grants/{id}/deactivate.
Acceptance criteria	AC1: A Department grant for Operations sees Operations department and group-only snapshots only. AC2: An Organization grant sees only snapshots that include the Organization audience. AC3: After revoke the reviewer's queue is empty.

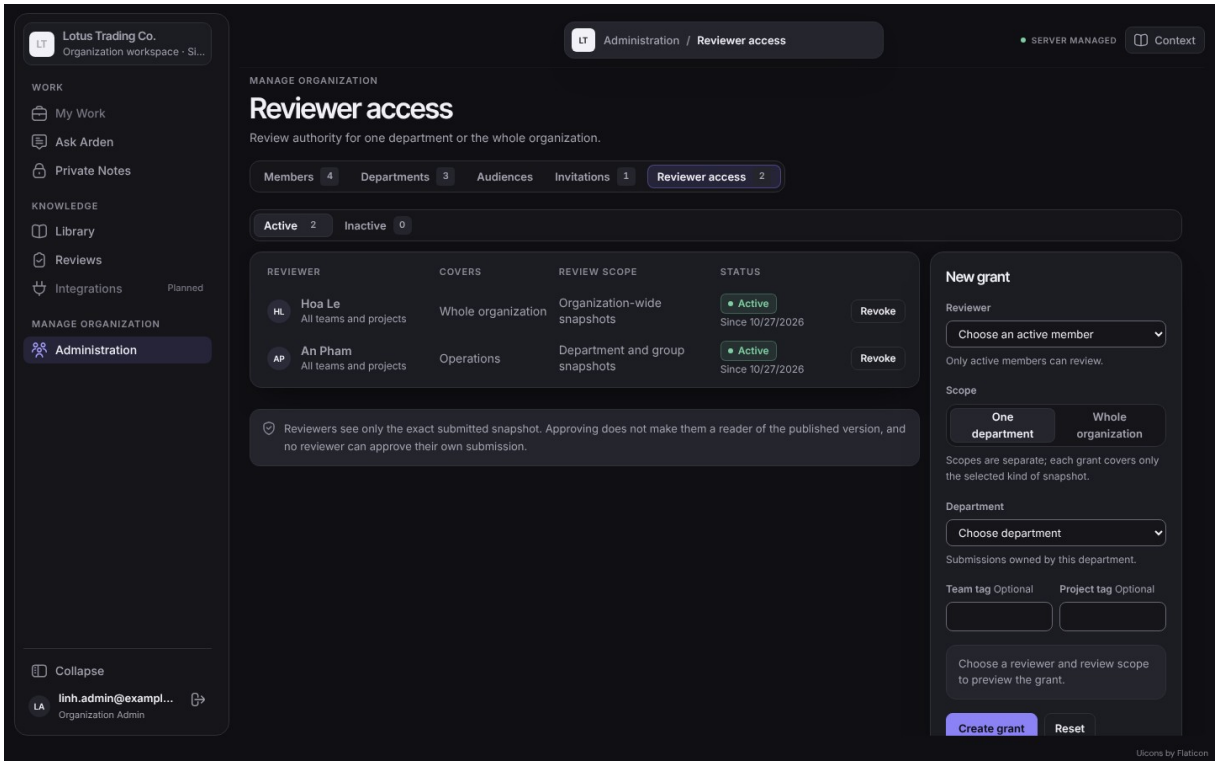


Figure 14: Administration – Reviewer access (SCR-18)

3.4.3 FR-SYS-01 Bootstrap the first Arden System Admin

Item	Specification
Status	Implemented
Related UC / screen / NSF	NSF-12
Function trigger	An operator runs pnpm system-admin:bootstrap on the server.
Actor and permission	Server operator
Function description	Create the first installation administrator for an existing active user account.
Preconditions	Migration 0017 applied; no System Admin exists.
Postconditions	One Active System Admin.
Main flow	1. Operator sets the migration database URL and the target user ID. 2. Command locks, checks preconditions and inserts the administrator.
Alternate and error flows	• A System Admin already exists → command fails, nothing changes. • Unknown or inactive account → fails.
Inputs and validation	User ID (UUID).
Data read / changed	system_administrator.
Business rules	BR-ROLE-003.
Screen / API / job	Command-line only; no HTTP route.

Item	Specification
Acceptance criteria	AC1: A second run fails. AC2: No HTTP route can create the first System Admin.

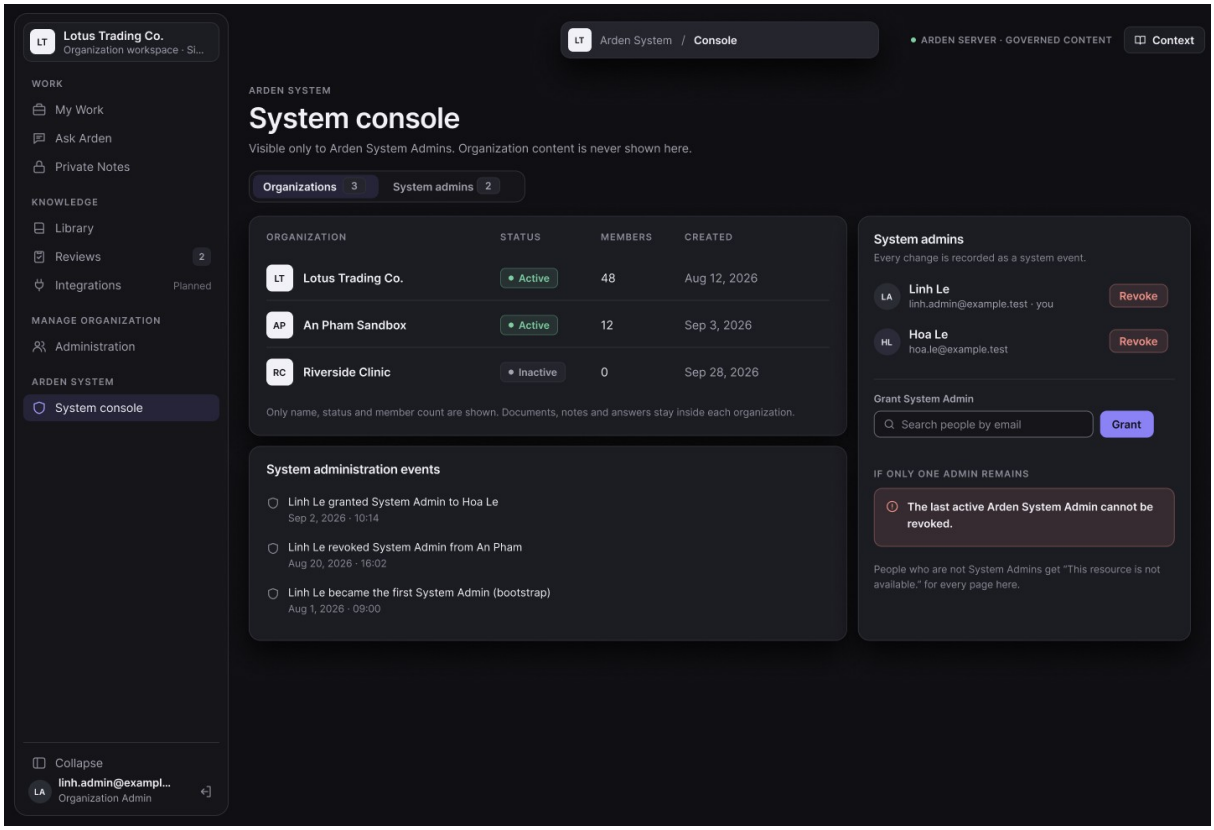
3.4.4 FR-SYS-02 Manage Arden System Admins

Item	Specification
Status	Partial (API only)
Related UC / screen / NSF	UC-32; SCR-20
Function trigger	System Administration screen (planned) or API call.
Actor and permission	Arden System Admin
Function description	List, grant and revoke System Admins; every change is recorded as a system administration event.
Preconditions	Actor is an Active System Admin.
Postconditions	Administrator list changed.
Main flow	1. GET /api/v1/system/administrators. 2. POST /api/v1/system/administrators {userId}. 3. POST /api/v1/system/administrators/{userId}/revoke.
Alternate and error flows	• Non-administrator → 404 "The resource is not available." • Revoking the last active System Admin → 409 SYSTEM_ADMINISTRATOR_LAST_ACTIVE (MSG-020).
Inputs and validation	User ID.
Data read / changed	system_administrator, system_administration_event.
Business rules	BR-ROLE-003.
Screen / API / job	/api/v1/system/administrators.
Acceptance criteria	AC1: Organization Admins and Members cannot enumerate or change System Admins. AC2: The last active System Admin cannot be revoked. AC3: A System Admin without membership cannot open any organization.

3.4.5 FR-SYS-03 View the organization overview

Item	Specification
Status	Planned
Related UC / screen / NSF	SCR-20
Function trigger	System Administration → Organizations.
Actor and permission	Arden System Admin
Function description	List organizations with status and member count only.

Item	Specification
Preconditions	Active System Admin.
Postconditions	None.
Main flow	1. Open the overview.
Alternate and error flows	• Non-administrator → not available.
Inputs and validation	—
Data read / changed	Reads organization and membership counts.
Business rules	BR-ROLE-003.
Screen / API / job	Planned API under /api/v1/system.
Acceptance criteria	AC1: No document title, content or private item is returned.



Figma screen R3-11: System console and organization overview

3.5 Personal Knowledge

Keeps each member's own notes private and lets them share a separate copy through governance.

ID	Function	Actor	Status
FR-PKM-01	Manage private notes	Member (owner)	Implemented
FR-PKM-02	Contribute a copy	Member (owner)	Implemented
FR-PKM-03	Upload a personal document	Member	Planned
FR-PKM-04	Organize personal collections	Member	Planned

3.5.1 FR-PKM-01 Manage private notes

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-11; SCR-07
Function trigger	Sidebar → Private Notes.
Actor and permission	Member (owner)
Function description	Create, list, search, open and edit owner-only notes. Each save creates a new version.
Preconditions	Active membership.
Postconditions	New note version saved.
Main flow	1. Member selects + to create a note, or selects a note. 2. Member edits title and text and selects Save. 3. Search filters the member's notes (Ctrl K).
Alternate and error flows	• Stale version → VERSION_CONFLICT "This note changed. Reload it before saving again." (MSG-018). • Missing If-Match → PRECONDITION_REQUIRED. • Another user opening the note → RESOURCE_NOT_AVAILABLE.
Inputs and validation	Title ≤800; body ≤100,000 bytes.
Data read / changed	draft_document, draft_version (private workspace), private storage.
Business rules	BR-ACC-001, BR-PKM-001.
Screen / API / job	GET/POST /api/v1/organizations/{organizationId}/private-workspace/drafts; GET/PATCH …/drafts/{draftId}.
Acceptance criteria	AC1: Another member, the Organization Admin and the System Admin cannot open the note. AC2: Each save increases the version by one.

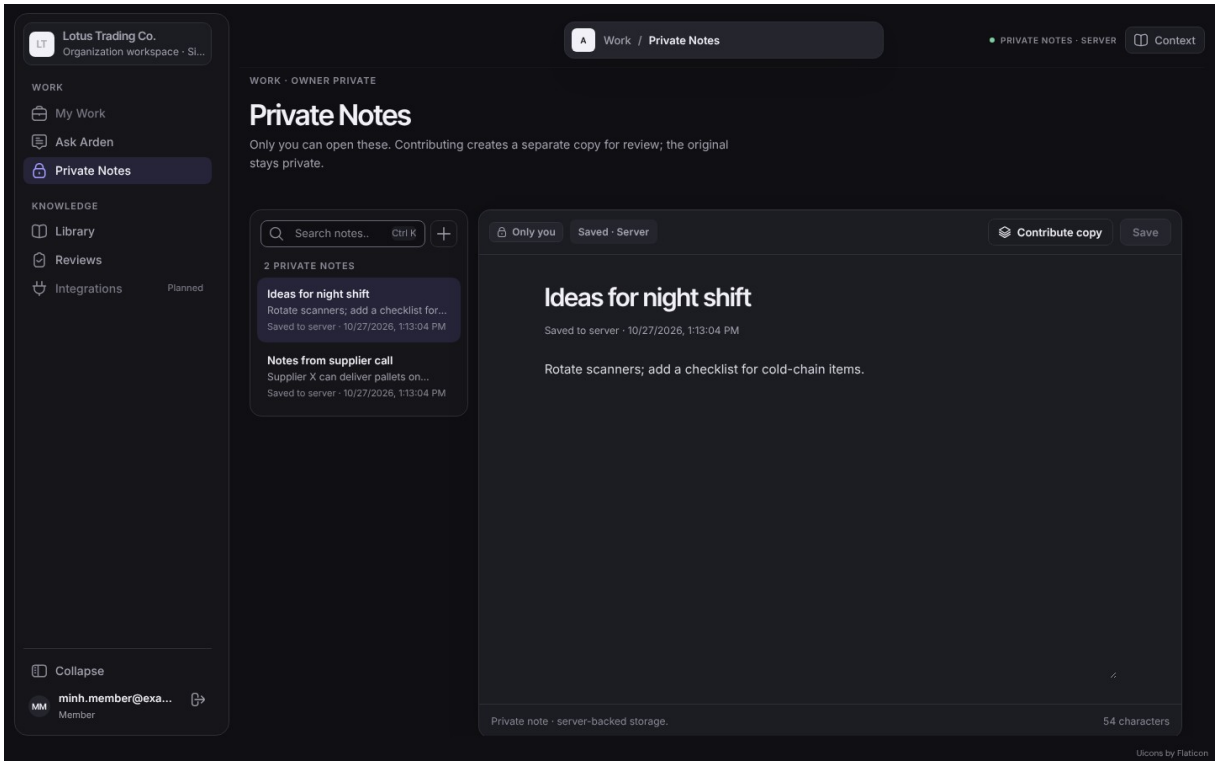


Figure 15: Private Notes (SCR-07)

3.5.2 FR-PKM-02 Contribute a copy

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-14; SCR-08
Function trigger	Private Notes → Contribute copy.
Actor and permission	Member (owner)
Function description	Create a separate shared draft from the exact saved version of a private note. Nothing is submitted, approved or published; readers and reviewers are chosen later in the draft.
Preconditions	The note version is saved and owned by the actor.
Postconditions	New shared draft (version 1) authored by the actor; the private original is unchanged.
Main flow	1. Member selects Contribute copy. 2. Dialog shows the saved version to copy. 3. Member selects Create shared draft. 4. API copies the version with an Idempotency-Key.
Alternate and error flows	• Unsaved or unsupported note format → SOURCE_NOT_SUPPORTED. • Repeated request with the same key → the first draft is returned.
Inputs and validation	Source private version ID; Idempotency-Key.
Data read / changed	shared_draft, shared_draft_version, audit_event.

Item	Specification
Business rules	BR-ACC-001, BR-DOC-003.
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/drafts {sourcePrivateVersionId}.
Acceptance criteria	AC1: The new draft has the same content hash as the copied version. AC2: Later edits of either side do not affect the other.

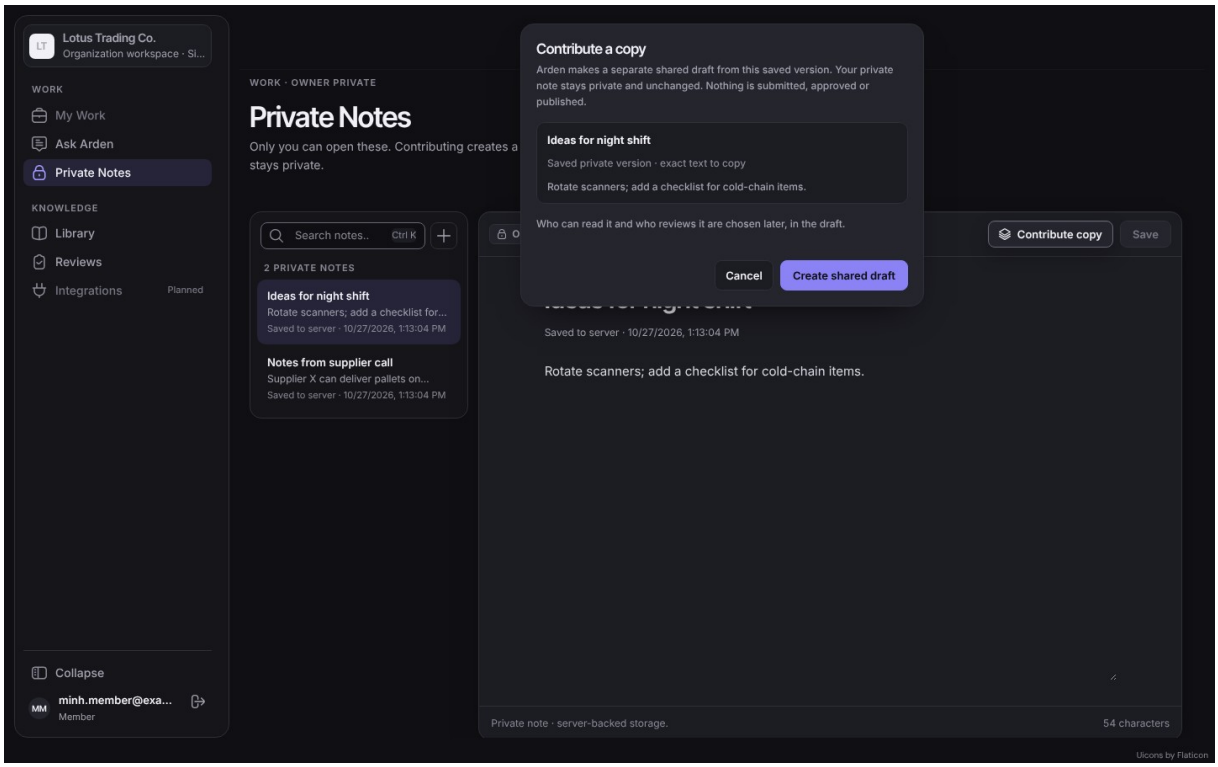


Figure 16: Contribute a copy dialog (SCR-08)

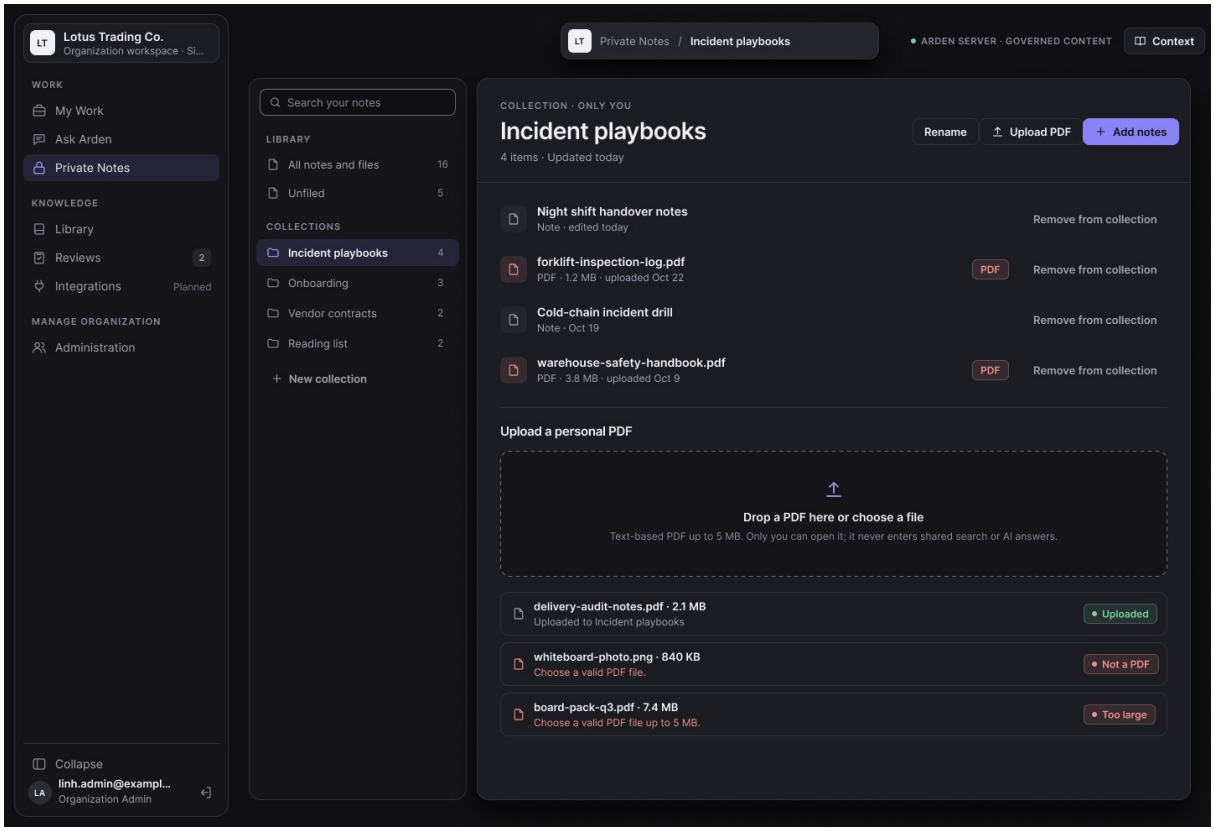
3.5.3 FR-PKM-03 Upload a personal document

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-12; SCR-07
Function trigger	Private Notes → Upload PDF.
Actor and permission	Member
Function description	Store a text-based PDF (≤5 MB) as an owner-only item.
Preconditions	Active membership.
Postconditions	Private item stored.
Main flow	1. Member chooses a PDF and confirms.
Alternate and error flows	• Non-PDF → MSG-013; >5 MB → MSG-014.

Item	Specification
Inputs and validation	PDF ≤5 MB.
Data read / changed	Private item and version.
Business rules	BR-DOC-002, BR-PKM-001.
Screen / API / job	Planned.
Acceptance criteria	AC1: Only the owner can open or download the document.

3.5.4 FR-PKM-04 Organize personal collections

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-13; SCR-07
Function trigger	Private Notes → Collections.
Actor and permission	Member
Function description	Create, rename and delete collections and add or remove own items.
Preconditions	Active membership.
Postconditions	Collection changed; items unaffected by deletion.
Main flow	1. Member creates a collection and adds items.
Alternate and error flows	• Adding another member's item → refused.
Inputs and validation	Name 1–80.
Data read / changed	Personal collection, collection item.
Business rules	BR-PKM-001.
Screen / API / job	Planned.
Acceptance criteria	AC1: Collections are visible only to the owner. AC2: Deleting a collection keeps its items.



Figma screen R3-13: Upload personal document and organize collections

3.6 Shared Drafts and Submission

Lets members author shared knowledge, choose its readers and freeze an exact snapshot for review.

ID	Function	Actor	Status
FR-DOC-01	Create a shared draft from text	Member	Implemented
FR-DOC-02	Create a shared draft from a PDF	Member	Implemented
FR-DOC-03	Edit the draft and choose readers	Author	Implemented
FR-DOC-04	Submit a snapshot for review	Author	Implemented
FR-DOC-05	Track contributions and history	Author	Implemented
FR-DOC-06	Withdraw a pending submission	Author; Organization Admin	Planned

3.6.1 FR-DOC-01 Create a shared draft from text

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-15; SCR-11
Function trigger	Library → New draft → Write document text.

Item	Specification
Actor and permission	Member
Function description	Create an author-owned draft. Plain text is normalized to Semantic Document v1.
Preconditions	Active membership.
Postconditions	Draft version 1 exists in My Contributions with status "Draft · metadata incomplete".
Main flow	1. Member enters a title and text. 2. Client sends POST …/drafts/new with an Idempotency-Key. 3. API normalizes and stores the content and returns the draft.
Alternate and error flows	• Empty title or invalid text → INVALID_REQUEST / INVALID_DOCUMENT. • Same Idempotency-Key with different content → IDEMPOTENCY_CONFLICT.
Inputs and validation	Title 1–160 characters in the UI (server ≤200 code points); body ≤100,000 bytes.
Data read / changed	shared_draft, shared_draft_version, storage, audit_event.
Business rules	BR-DOC-001.
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/drafts/new.
Acceptance criteria	AC1: A valid request creates exactly one draft at version 1. AC2: Repeating it with the same key creates no second draft.

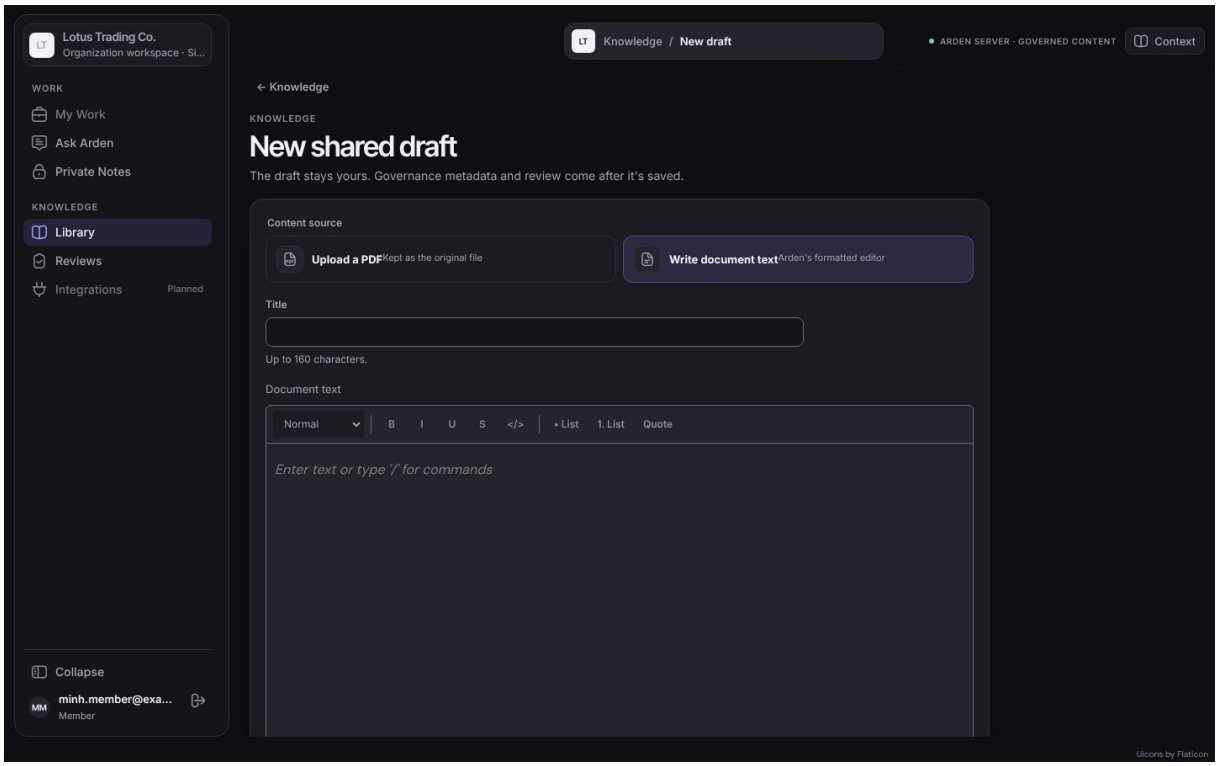
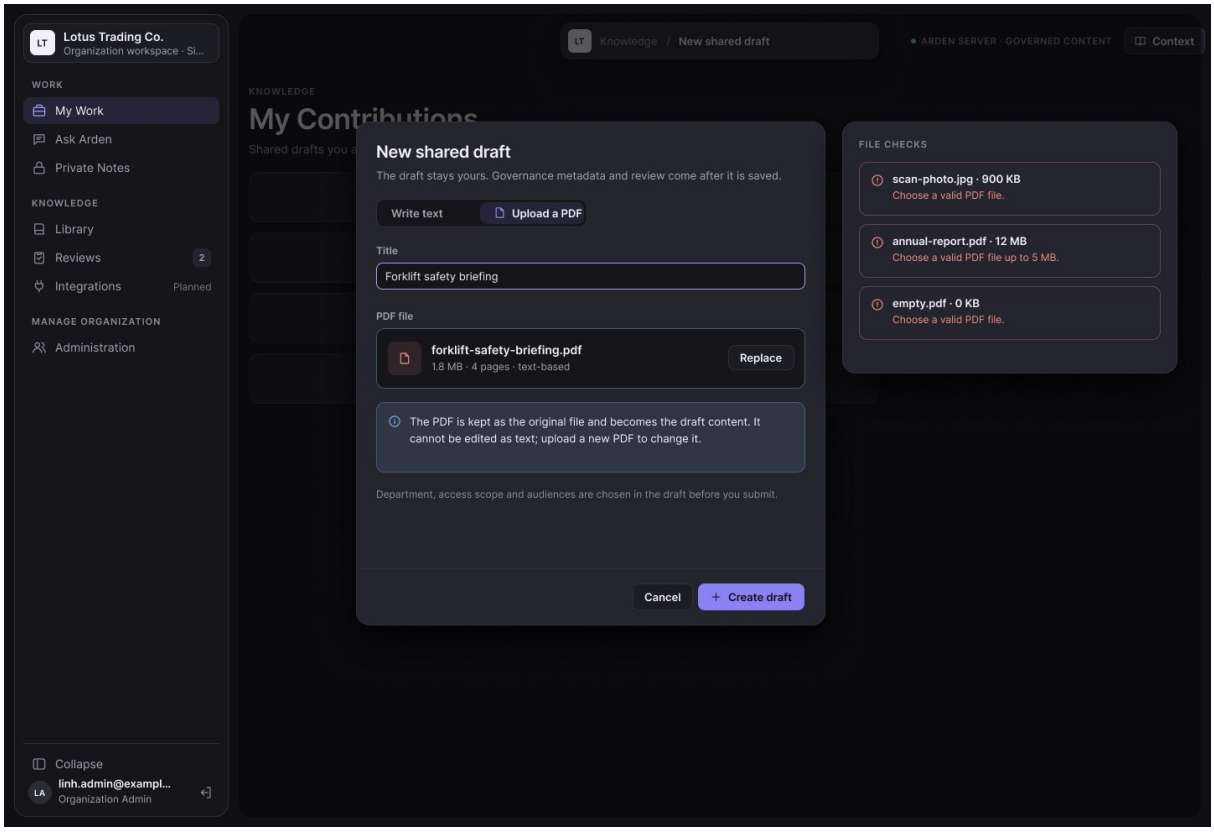


Figure 17: New shared draft (SCR-11)

3.6.2 FR-DOC-02 Create a shared draft from a PDF

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-15; SCR-11
Function trigger	Library → Create from PDF, or New draft → Upload a PDF.
Actor and permission	Member
Function description	Upload a PDF kept as the original file; the PDF is the draft content and cannot be edited as text.
Preconditions	Active membership.
Postconditions	PDF draft version 1.
Main flow	1. Member chooses the file and title. 2. Client sends the file (base64) with an Idempotency-Key. 3. API checks type and size and stores the file.
Alternate and error flows	• Not a PDF → INVALID_PDF (MSG-013). • Larger than 5 MB → INVALID_PDF / REQUEST_TOO_LARGE (MSG-014). • Text edit of a PDF draft → "Uploaded PDFs cannot be edited as text."
Inputs and validation	PDF only, non-empty, ≤5 MB; title as FR-DOC-01.
Data read / changed	shared_draft, shared_draft_version (application/pdf), storage.
Business rules	BR-DOC-002.
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/drafts/pdf.
Acceptance criteria	AC1: A 2 MB PDF becomes a draft; a 6 MB PDF and a .docx are rejected with the correct message.



Figma screen R3-15: Create a shared draft from PDF

3.6.3 FR-DOC-03 Edit the draft and choose readers

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-16; SCR-12
Function trigger	My Contributions → select a draft → Open draft.
Actor and permission	Author
Function description	Edit the working version and choose who can read it: the Organization audience, the author's Department audience and/or audience groups the author belongs to. The responsible department is not chosen: it is the author's primary department ("Owned by you · reviewed by <department>"). The working draft stays editable while a snapshot is under review.
Preconditions	Actor is the author.
Postconditions	New immutable version with metadata.
Main flow	1. Author edits text (rich-text toolbar: heading, bold, italic, lists, quote, code). 2. Author types in Who can read it and picks audiences; optional team/project tags. 3. Readiness checklist shows Title, Your department, Readers. 4. Save version (If-Match) creates the next version.

Item	Specification
Alternate and error flows	• Stale version → VERSION_CONFLICT (MSG-018). • Audience not readable by the author, Organization audience with Department scope, or another department → INVALID_METADATA. • Author without primary department can save but not submit.
Inputs and validation	Audience IDs 1–100 (unique); scope ORGANIZATION/DEPARTMENT; tags ≤100 characters.
Data read / changed	shared_draft, shared_draft_version, shared_draft_audience.
Business rules	BR-ACC-002, BR-ACC-004, BR-ACC-005, BR-GOV-001.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/options; PATCH /api/v1/organizations/{organizationId}/knowledge/drafts/{draftId}.
Acceptance criteria	AC1: The reader list offers only audiences the author can read. AC2: A Department publication shared with a group is readable only by group members of that department. AC3: Editing during review does not change the submitted snapshot.

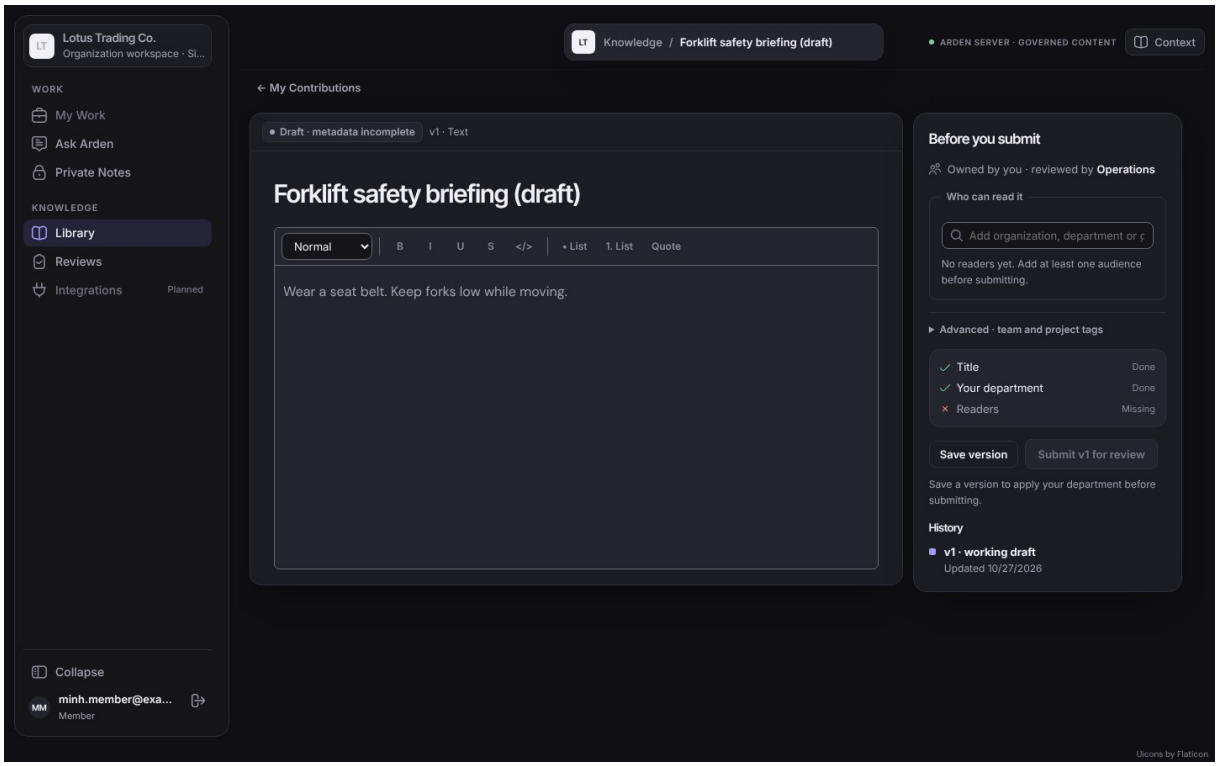


Figure 18: Draft editor with Before you submit (SCR-12)

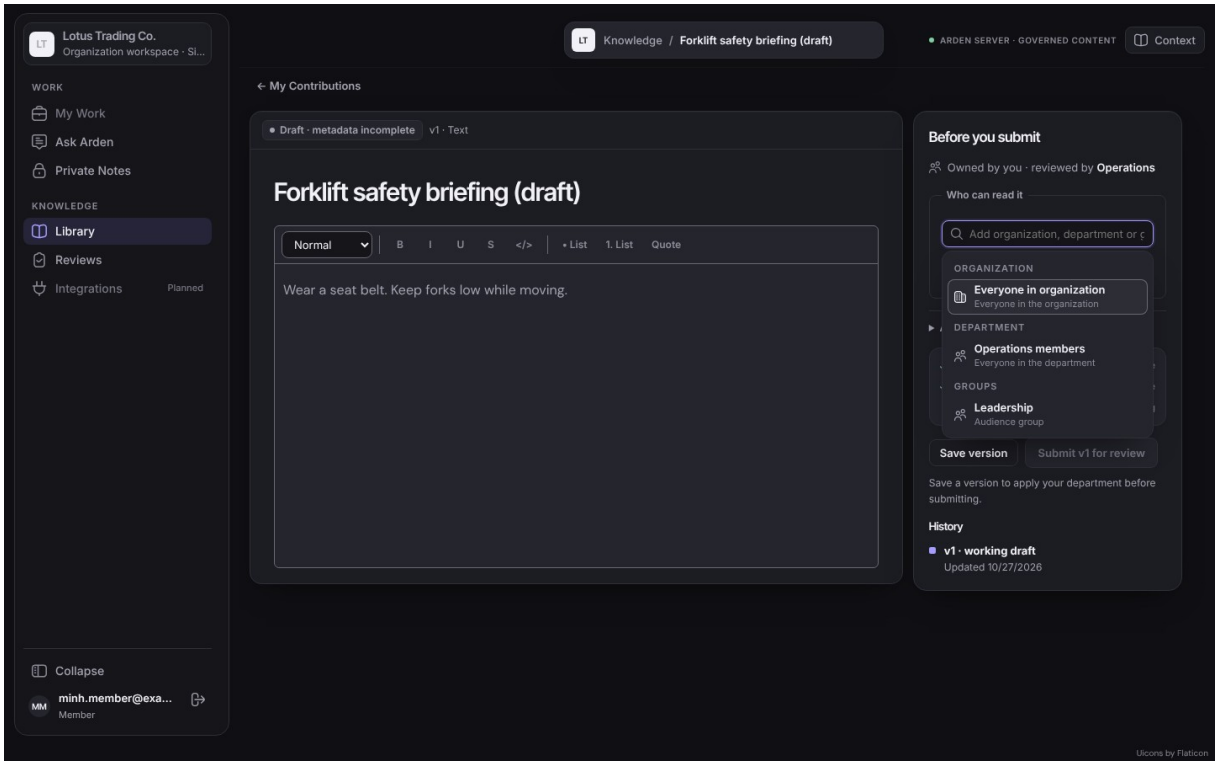


Figure 19: Reader picker

3.6.4 FR-DOC-04 Submit a snapshot for review

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-17; SCR-12
Function trigger	Draft editor → Submit vN for review.
Actor and permission	Author
Function description	Freeze the exact current version, title, responsible department, scope, tags, audiences, review rule and content hash into an immutable submission.
Preconditions	All checklist items done; no other open submission for the draft; the stored department equals the author's current primary department.
Postconditions	Submission Pending Review; appears in eligible review queues.
Main flow	1. Author selects Submit (If-Match, Idempotency-Key). 2. API checks readiness and creates the snapshot. 3. My Contributions shows Awaiting review.
Alternate and error flows	• Missing department, scope or audience → DRAFT_NOT_READY (MSG-004). • Open candidate exists or stale version → SUBMISSION_CONFLICT.
Inputs and validation	Draft ID, version (If-Match).
Data read / changed	shared_submission, audit_event.

Item	Specification
Business rules	BR-GOV-001, BR-GOV-005, BR-LIFE-001.
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/drafts/{draftId}/submit.
Acceptance criteria	AC1: A second concurrent submit of the same draft gets a conflict; exactly one candidate is open. AC2: The submission keeps its hash even if the draft is edited later.

3.6.5 FR-DOC-05 Track contributions and history

Item	Specification
Status	Implemented
Related UC / screen / NSF	SCR-09
Function trigger	Library → My Contributions.
Actor and permission	Author
Function description	List own drafts with version, readers, department, status (Draft · metadata incomplete, Awaiting review, Rejected, Published) and the reviewer note; open the history of versions and submissions.
Preconditions	Signed in.
Postconditions	None.
Main flow	1. Open My Contributions. 2. Select a row to see details, reviewer note and history.
Alternate and error flows	• No drafts → empty state.
Inputs and validation	—
Data read / changed	Reads shared_draft, shared_submission, review_decision.
Business rules	BR-LIFE-001.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/drafts; GET …/knowledge/submissions.
Acceptance criteria	AC1: A rejected draft shows the reviewer note. AC2: Versions and history entries cannot be edited.

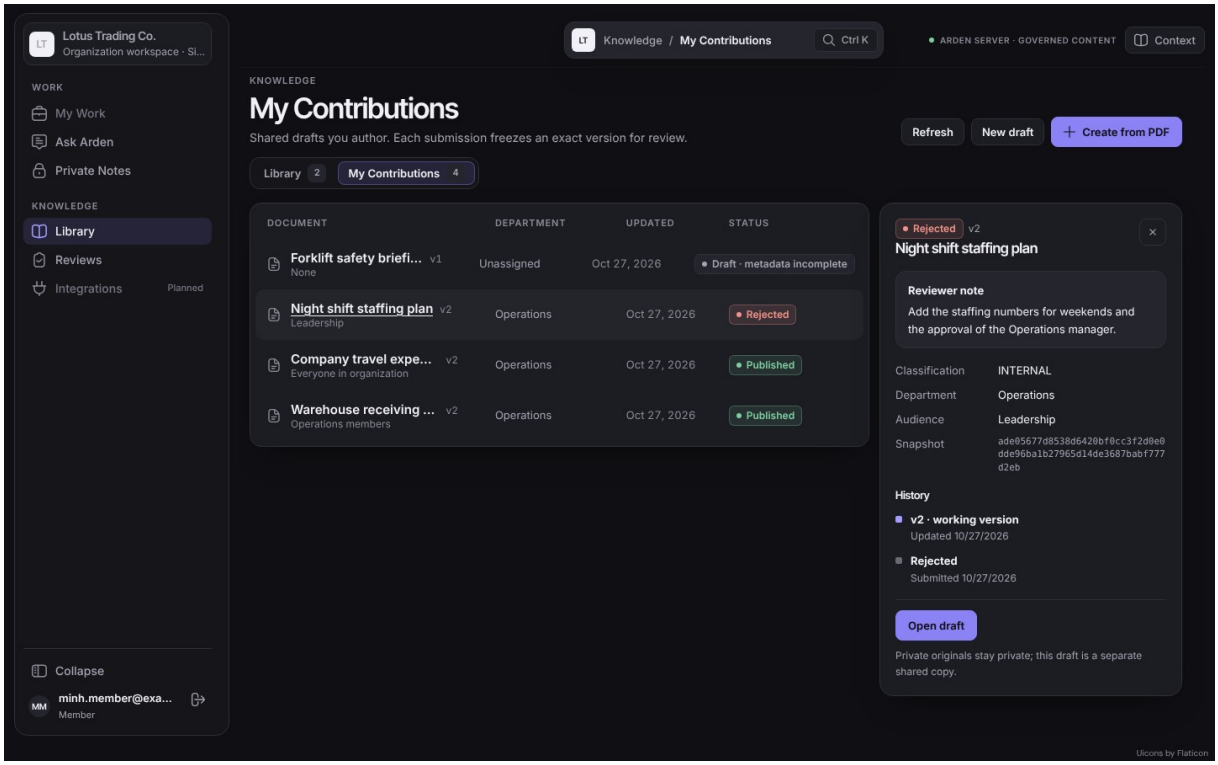
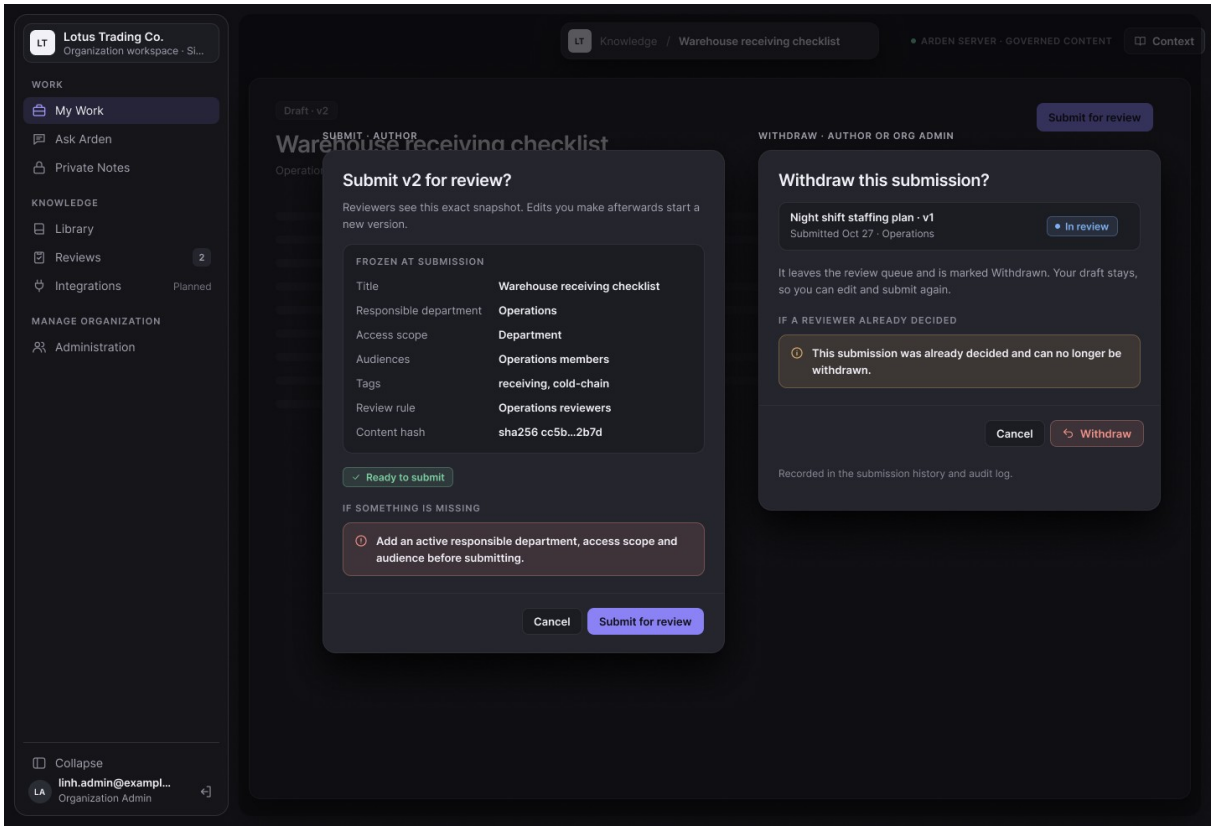


Figure 20: My Contributions with a rejected draft (SCR-09)

3.6.6 FR-DOC-06 Withdraw a pending submission

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-18
Function trigger	My Contributions → Withdraw.
Actor and permission	Author; Organization Admin
Function description	Cancel a pending submission (status Withdrawn).
Preconditions	Submission Pending Review.
Postconditions	Submission Withdrawn; removed from queues.
Main flow	1. Select Withdraw and confirm.
Alternate and error flows	• Already decided → conflict.
Inputs and validation	Submission ID.
Data read / changed	shared_submission, audit_event.
Business rules	BR-GOV-005.
Screen / API / job	Planned API.
Acceptance criteria	AC1: A withdrawn submission disappears from every review queue.



Figma screen R3-17: Submit and withdraw a pending submission

3.7 Review and Publication

Lets authorized people decide the exact snapshot and makes approved knowledge visible only after it is indexed.

ID	Function	Actor	Status
FR-REV-01	Open the review queue	Knowledge Reviewer; Organization Admin	Implemented
FR-REV-02	Inspect a submission	Knowledge Reviewer; Organization Admin	Implemented
FR-REV-03	Approve a submission	Knowledge Reviewer (grant, not own); Organization Admin (any, own audited)	Implemented
FR-REV-04	Reject a submission	Knowledge Reviewer; Organization Admin	Implemented
FR-PUB-01	Index and activate an approved snapshot	System (indexing worker, Arden-AI)	Implemented (text and PDF)
FR-PUB-02	Unpublish and restore knowledge	Organization Admin	Planned

3.7.1 FR-REV-01 Open the review queue

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-19; SCR-13
Function trigger	Sidebar → Reviews.
Actor and permission	Knowledge Reviewer; Organization Admin
Function description	List pending submissions the actor may decide: for an ordinary reviewer those covered by a current grant (Department grant: the responsible department's department/group-only snapshots; Organization grant: snapshots that include the Organization audience), never their own; for the Organization Admin every pending submission. Metadata only.
Preconditions	Active membership.
Postconditions	None.
Main flow	1. Open Reviews. 2. Queue shows title, department and date; the count is shown in the sidebar.
Alternate and error flows	• No eligible submissions → "Queue is clear". • Snapshot whose audience is no longer enforceable is marked approval-blocked.
Inputs and validation	—
Data read / changed	Reads shared_submission, reviewer_grant.
Business rules	BR-GOV-002, BR-GOV-003.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/reviews.
Acceptance criteria	AC1: A Department reviewer for Operations does not see an organization-wide snapshot; an Organization reviewer does. AC2: The reviewer's own submissions never appear.

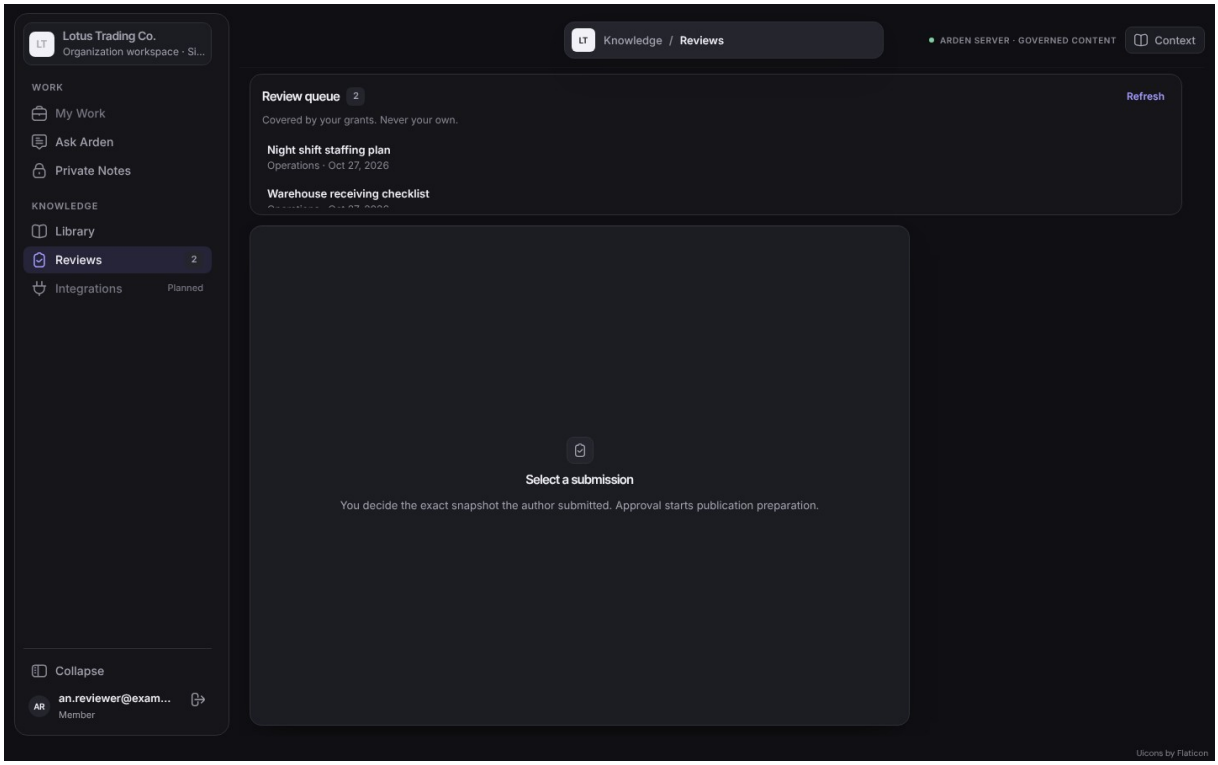


Figure 21: Review queue (SCR-13)

3.7.2 FR-REV-02 Inspect a submission

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-19; SCR-13
Function trigger	Select a submission in the queue.
Actor and permission	Knowledge Reviewer; Organization Admin
Function description	Show the exact frozen content and facts: required reviewer, department, access scope, audiences, tags and snapshot hash.
Preconditions	Actor eligible.
Postconditions	None.
Main flow	1. API reauthorizes, loads the immutable snapshot and reauthorizes again. 2. Decision panel shows frozen facts and the note field.
Alternate and error flows	• Grant revoked meanwhile → RESOURCE_NOT_AVAILABLE.
Inputs and validation	—
Data read / changed	Reads shared_submission and stored content.
Business rules	BR-GOV-006.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/reviews/{submissionId}.

Item	Specification
Acceptance criteria	AC1: The displayed content hash equals the submitted hash.

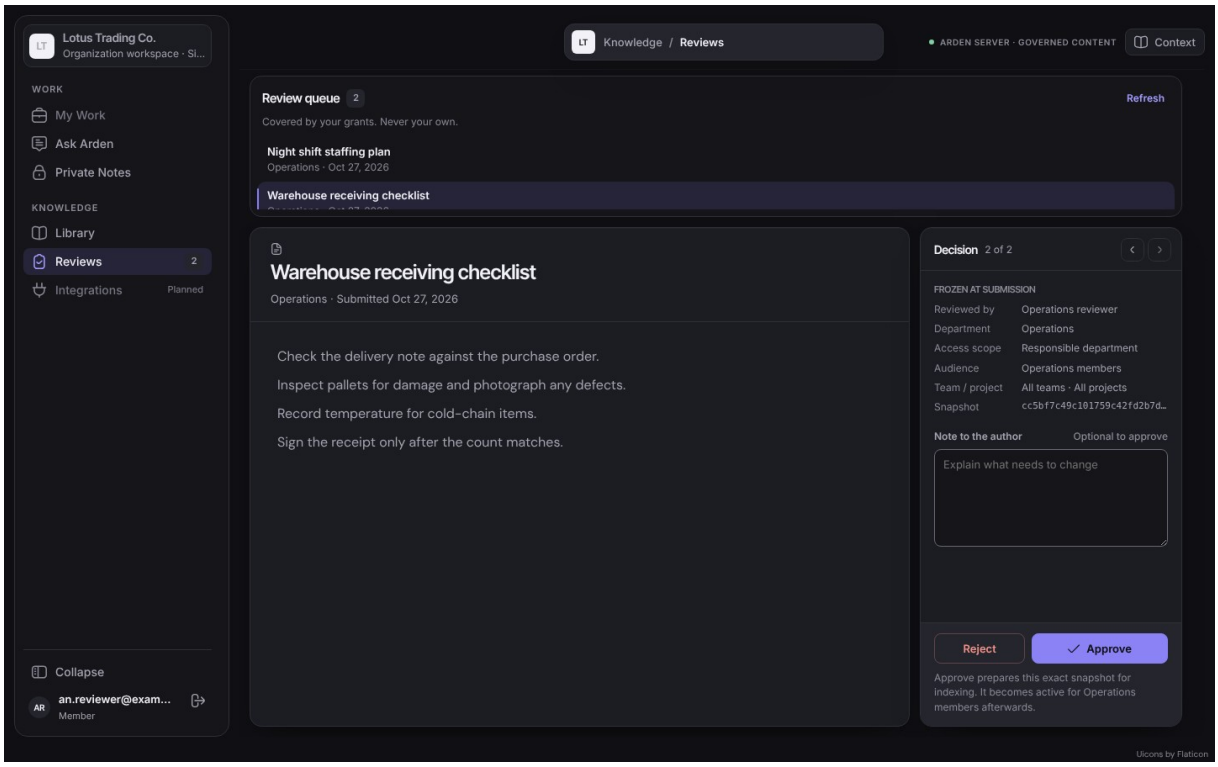


Figure 22: Submission detail and decision panel

3.7.3 FR-REV-03 Approve a submission

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-20; SCR-13; NSF-07
Function trigger	Decision panel → Approve → Approve for preparation.
Actor and permission	Knowledge Reviewer (grant, not own); Organization Admin (any, own audited)
Function description	Approve the exact snapshot. The server rechecks authority and audiences, records the decision and moves the snapshot to PREPARING. It is not visible until indexing finishes (FR-PUB-01).
Preconditions	Submission Pending Review; actor eligible; audiences enforceable.
Postconditions	Submission Approved, publication PREPARING, index job queued; toast "Approved. The exact reviewed snapshot is preparing for indexing; it is not active yet."
Main flow	1. Reviewer selects Approve; a confirmation explains that indexing follows. 2. Client sends the action, expected snapshot version ID and content hash with an Idempotency-Key. 3. API locks the submission, rechecks and commits decision, audit, PREPARING state and index job.

Item	Specification
Alternate and error flows	• Snapshot changed → REVIEW_CONFLICT. • Audience not enforceable or author no longer eligible → AUDIENCE_NOT_ENFORCEABLE (MSG-019). • Ordinary reviewer on own submission → not available.
Inputs and validation	Action APPROVE; expectedSnapshotVersionId; expectedContentHash; optional note ≤2,000.
Data read / changed	review_decision, shared_submission, knowledge_index_generation, knowledge_index_job, audit_event.
Business rules	BR-GOV-002…008, BR-LIFE-002.
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/reviews/{submissionId}/decisions.
Acceptance criteria	AC1: After approval the item is not in the Library until activation. AC2: An Organization Admin self-approval is recorded with selfApproved = true. AC3: A reviewer outside the audiences still cannot read the published item.

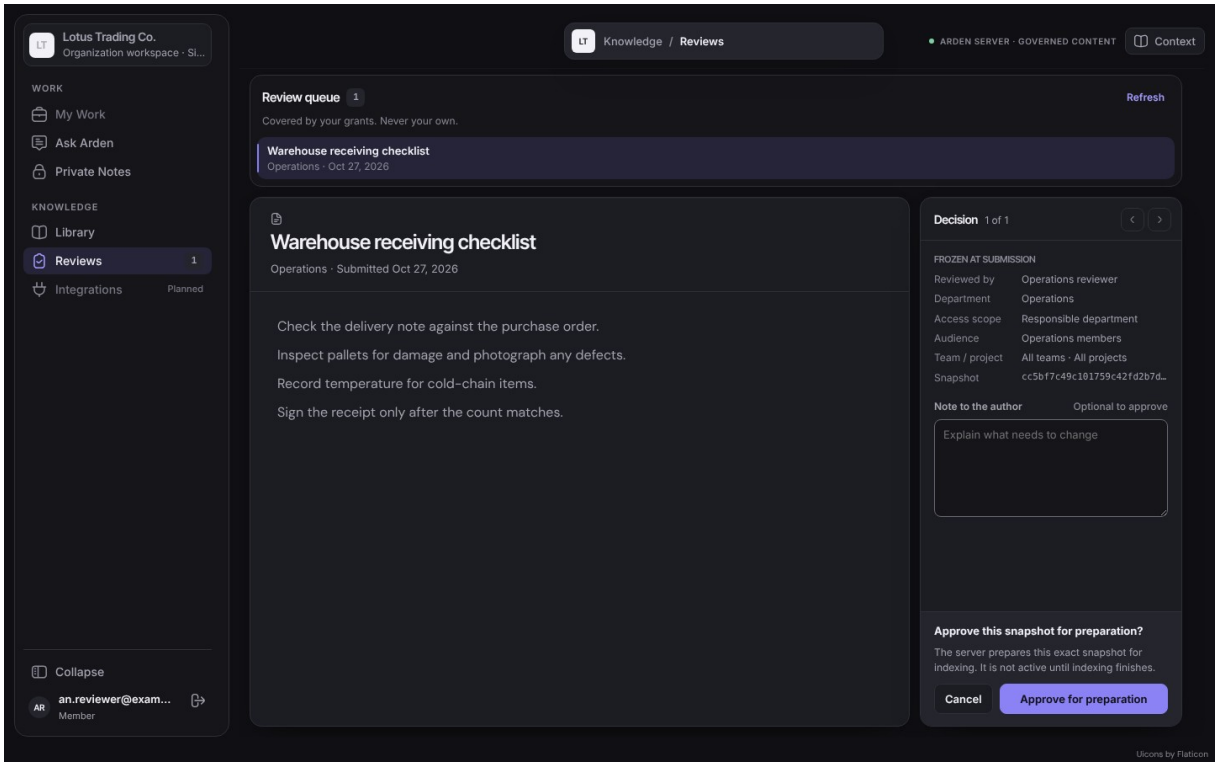


Figure 23: Approve confirmation

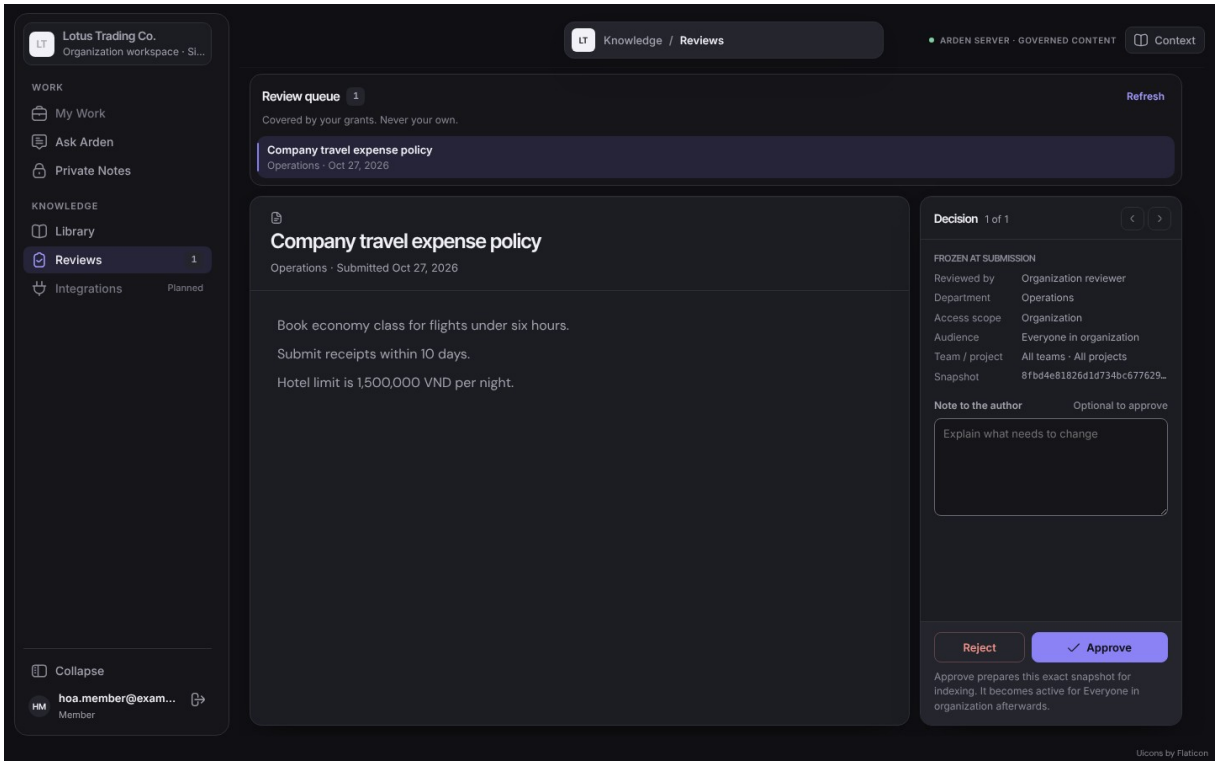


Figure 24: Organization-wide snapshot in an Organization reviewer's queue

3.7.4 FR-REV-04 Reject a submission

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-21; SCR-13
Function trigger	Decision panel → enter Note to the author → Reject → Reject with note.
Actor and permission	Knowledge Reviewer; Organization Admin
Function description	Reject the snapshot with a mandatory note. The candidate closes; the author keeps editing the working draft and may submit a new snapshot. There is no separate request-revision decision.
Preconditions	Submission Pending Review; actor eligible.
Postconditions	Submission Rejected; note visible to the author; nothing published.
Main flow	1. Reviewer writes the note and selects Reject. 2. Confirmation "Reject this snapshot?" 3. API records decision and audit.
Alternate and error flows	• Empty note → client message "Explain why this snapshot is rejected." and server 400 COMMENT_REQUIRED (MSG-015).
Inputs and validation	Action REJECT; note 1–2,000 characters; expected version and hash.
Data read / changed	review_decision, shared_submission, audit_event.
Business rules	BR-GOV-004.

Item	Specification
Screen / API / job	POST /api/v1/organizations/{organizationId}/knowledge/reviews/{submissionId}/decisions.
Acceptance criteria	AC1: Reject without a note is refused. AC2: The rejected content never appears in the Library, search or AI answers.

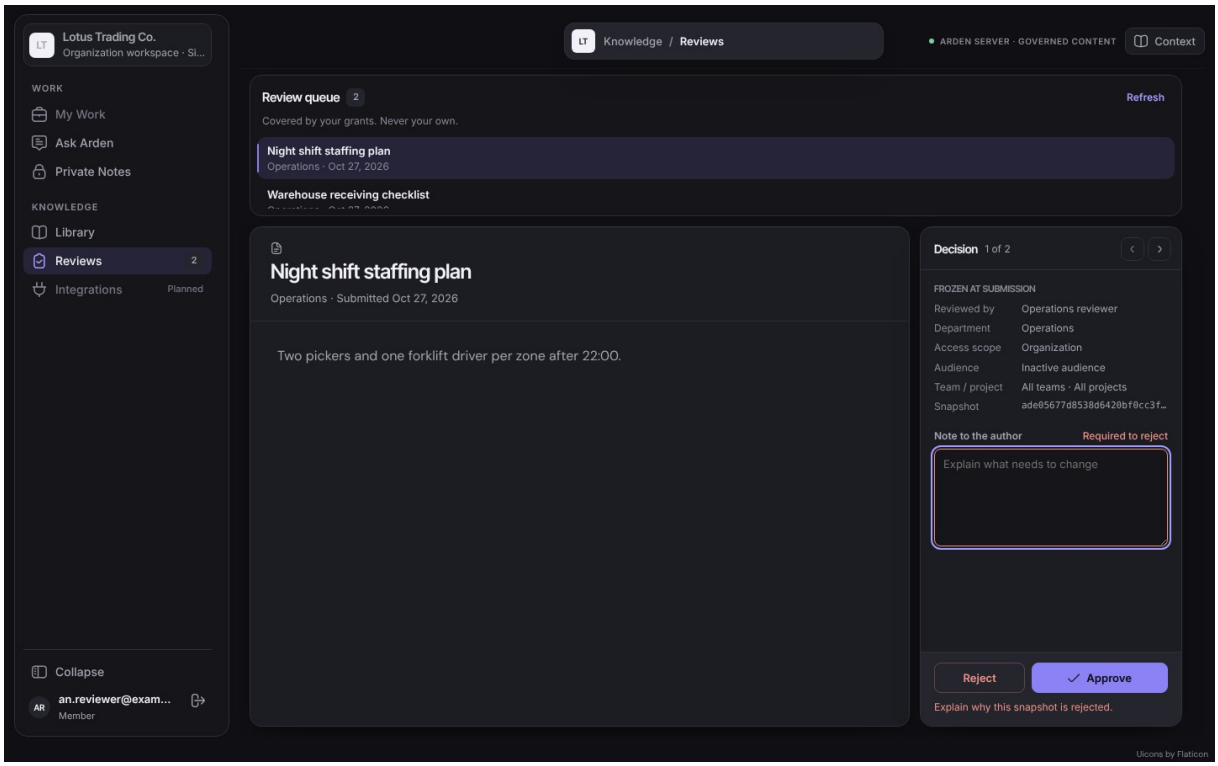


Figure 25: Reject without a note is blocked

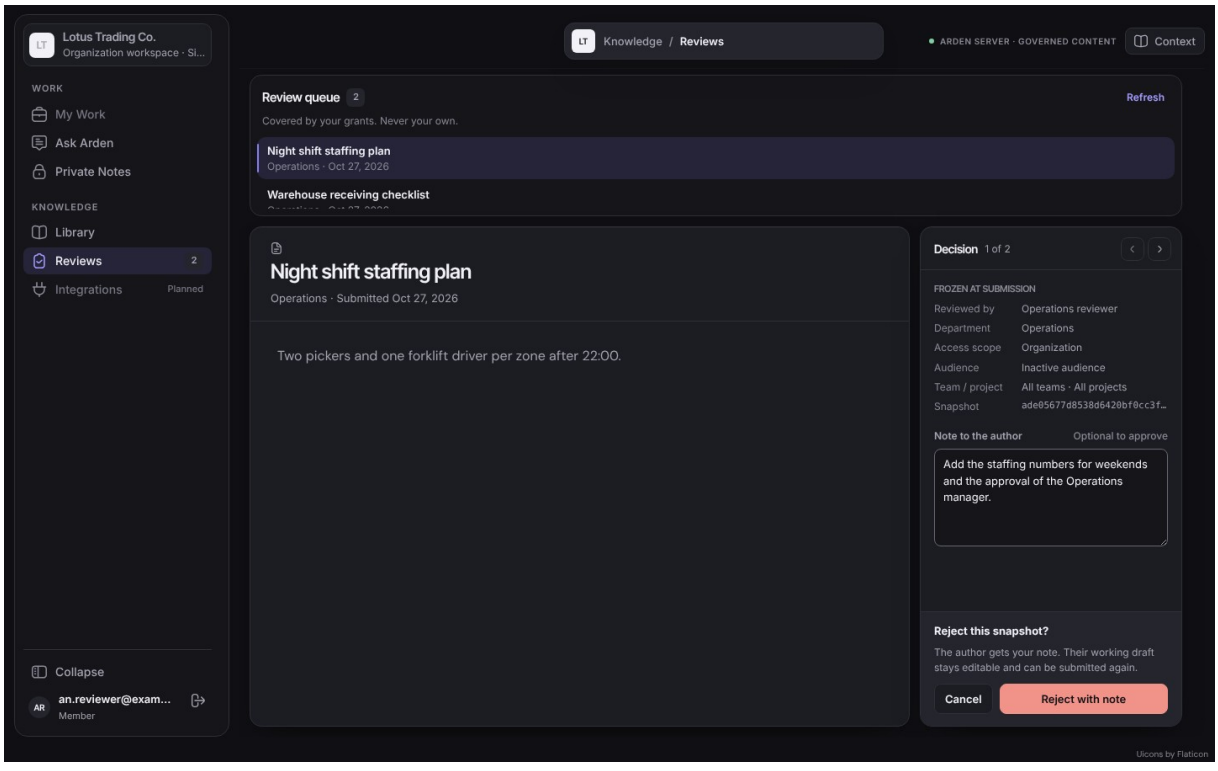
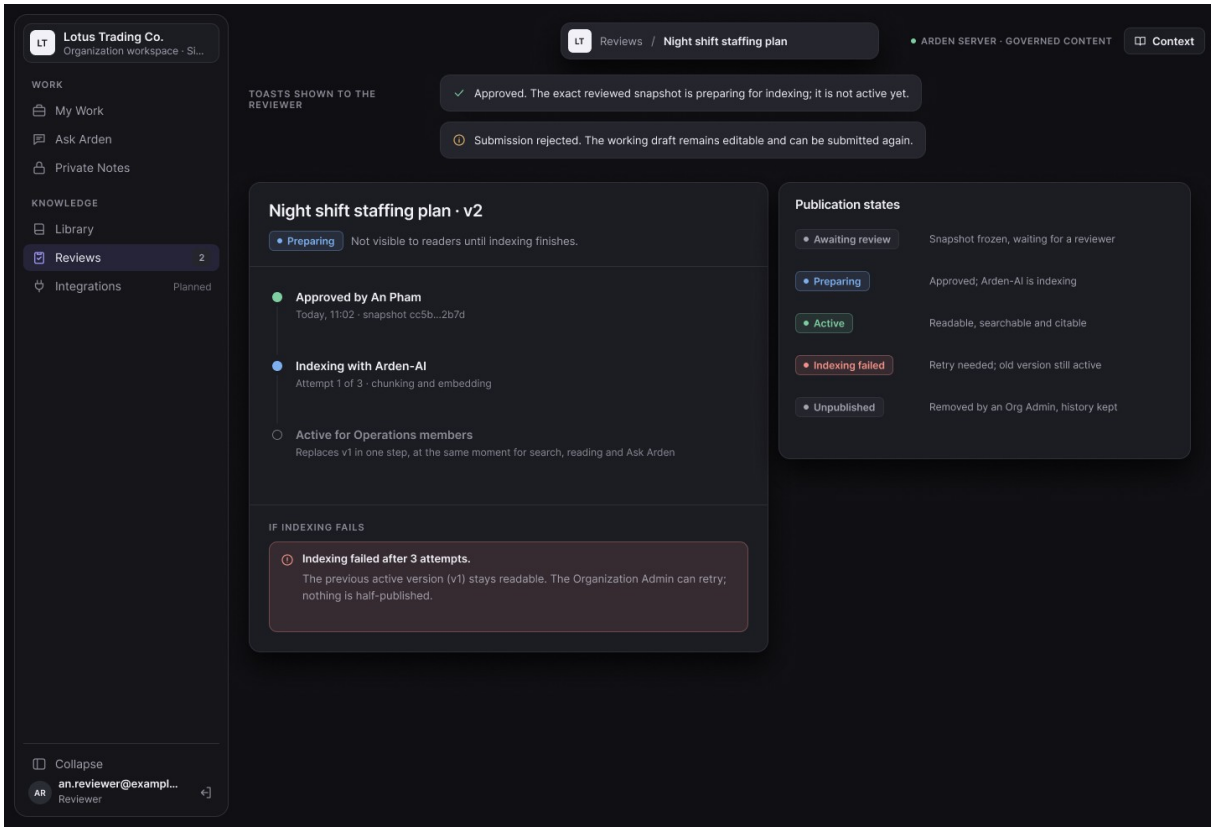


Figure 26: Reject confirmation

3.7.5 FR-PUB-01 Index and activate an approved snapshot

Item	Specification
Status	Implemented (text and PDF)
Related UC / screen / NSF	NSF-08, NSF-09
Function trigger	Background worker poll (every 1 s) finds a pending index job.
Actor and permission	System (indexing worker, Arden-AI)
Function description	Send the approved snapshot to Arden-AI, validate the returned manifest and atomically activate the publication.
Preconditions	Index job Pending.
Postconditions	Generation READY, publication ACTIVE, current pointer replaced; or generation FAILED with the previous active version unchanged.
Main flow	1. Worker leases the job for 180 s and reads the snapshot (hash verified). 2. Worker calls Arden-AI POST /v1/index-generations with a service token. 3. Arden-AI compiles, chunks, embeds and writes the candidate generation to ChromaDB, then returns the manifest. 4. Worker validates the manifest and activates in one transaction.
Alternate and error flows	• AI unavailable, timeout, HTTP 408/429/5xx → retried up to 3 attempts. • Invalid manifest or integrity failure → FAILED, not retried. • Arden-AI fails → the job is retried; after 3 attempts the generation is FAILED and the previous version stays readable.
Inputs and validation	Snapshot content, generation ID, job ID.
Data read / changed	knowledge_index_job, knowledge_index_generation, published_knowledge_version, knowledge_current_publication.
Business rules	BR-LIFE-002, BR-AI-003.
Screen / API / job	Background job; Arden-AI /v1/index-generations.
Acceptance criteria	AC1: A partially indexed generation never becomes visible. AC2: After activation the item appears in the Library of its audiences.

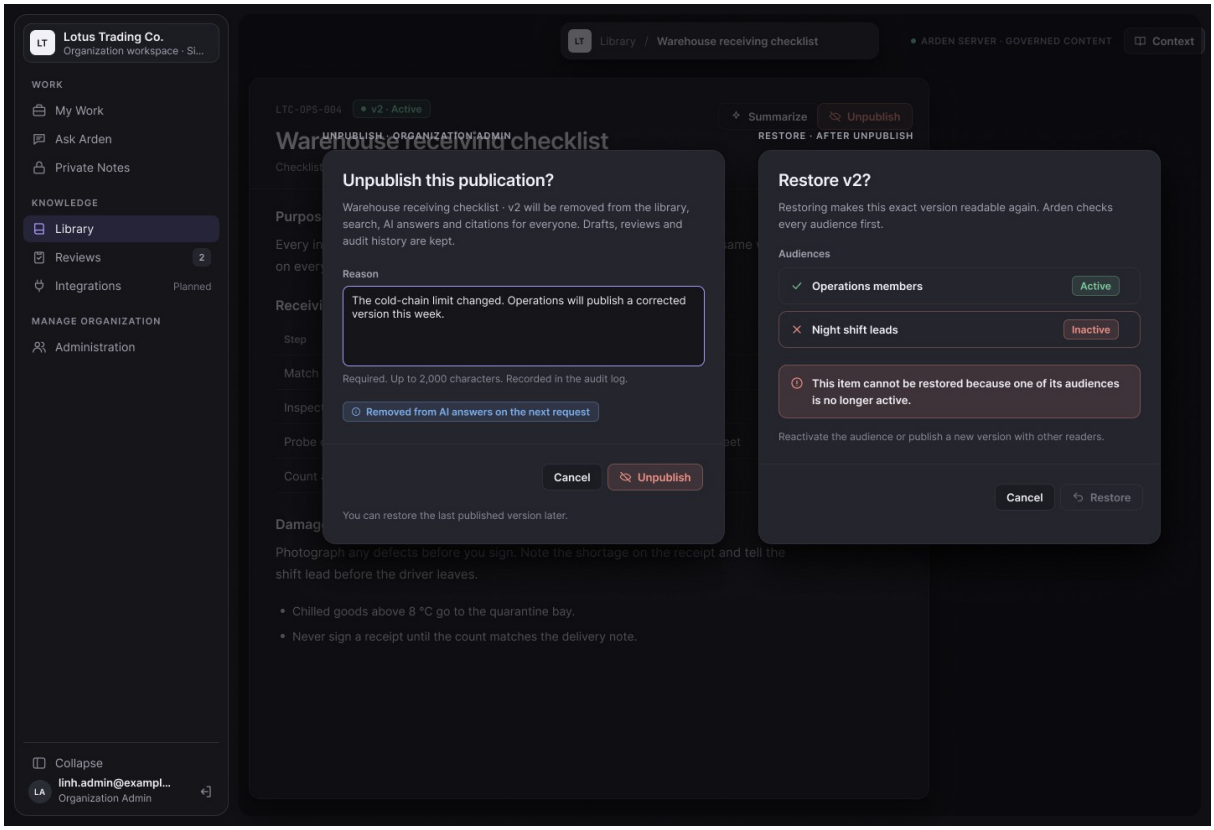


Figma screen R3-22: Preparing an approved snapshot for activation

3.7.6 FR-PUB-02 Unpublish and restore knowledge

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-29
Function trigger	Document reader → Unpublish / Restore.
Actor and permission	Organization Admin
Function description	Remove an active publication from reading, search, AI and citations with a reason, or restore the last published version after rechecking its audiences. History is kept.
Preconditions	Publication Active (unpublish) or archived (restore).
Postconditions	Pointer removed or re-created; audit.
Main flow	1. Admin selects Unpublish, enters a reason, confirms.
Alternate and error flows	• Missing reason → MSG-015; audience inactive → restore blocked (MSG-021).
Inputs and validation	Reason ≤2,000.
Data read / changed	shared_draft.status, knowledge_current_publication, audit_event.

Item	Specification
Business rules	BR-LIFE-004.
Screen / API / job	Planned API.
Acceptance criteria	AC1: After unpublish the item disappears for every reader on the next request. AC2: Restore makes it visible again to its audiences only.



Figma screen R3-23: Unpublish and restore knowledge

3.8 Library and Search

Shows only the knowledge each member may read.

ID	Function	Actor	Status
FR-LIB-01	Browse the library	Member	Implemented
FR-LIB-02	Read a published document	Member	Implemented
FR-LIB-03	Search and filter knowledge	Member	Planned

3.8.1 FR-LIB-01 Browse the library

Item	Specification
Status	Implemented

Item	Specification
Related UC / screen / NSF	UC-22; SCR-09
Function trigger	Sidebar → Library.
Actor and permission	Member
Function description	List active publications whose audiences include the member, with owner department, version label, effective date and status.
Preconditions	Active membership.
Postconditions	None.
Main flow	1. Open Library. 2. The header shows the number of readable publications.
Alternate and error flows	• No readable publication → "No accessible publication".
Inputs and validation	—
Data read / changed	Reads published_knowledge_version, knowledge_current_publication, audiences.
Business rules	BR-ACC-003, BR-LIFE-003.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/published.
Acceptance criteria	AC1: A Finance member sees the organization-wide policy but not an Operations-only checklist. AC2: Approval or grant never adds readership.

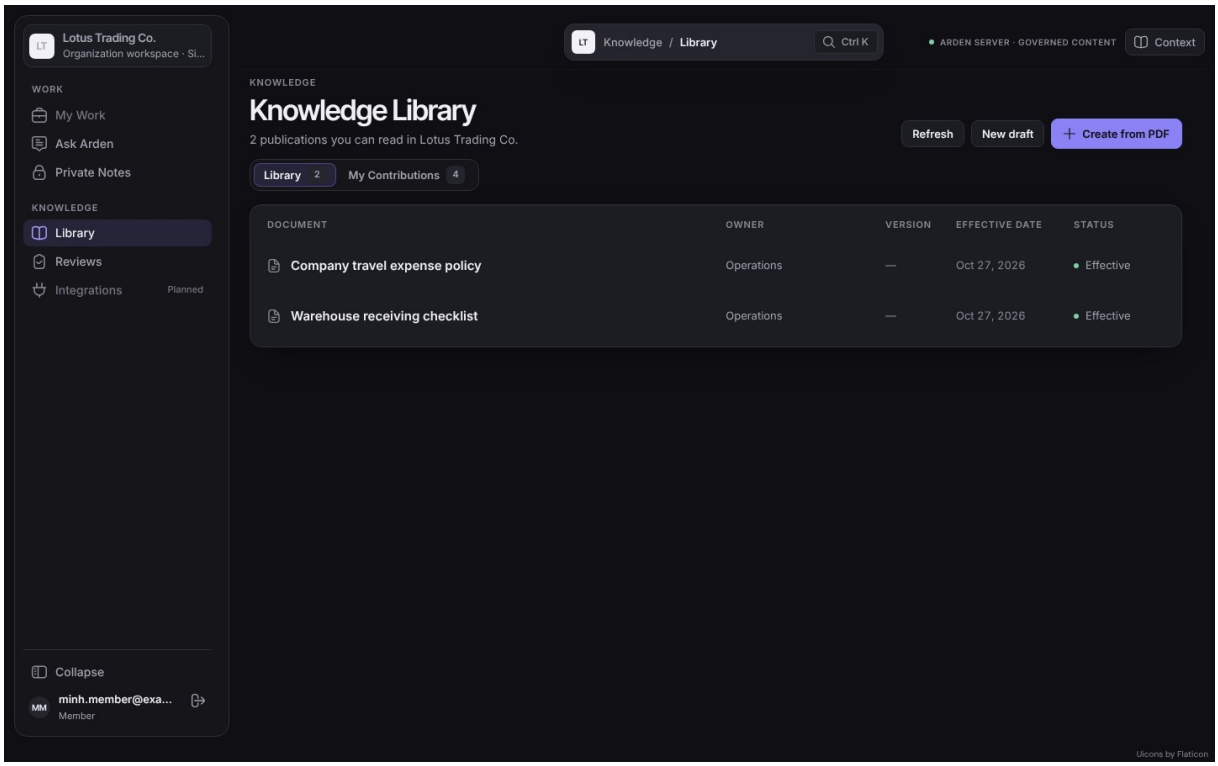


Figure 27: Knowledge Library (Operations member)

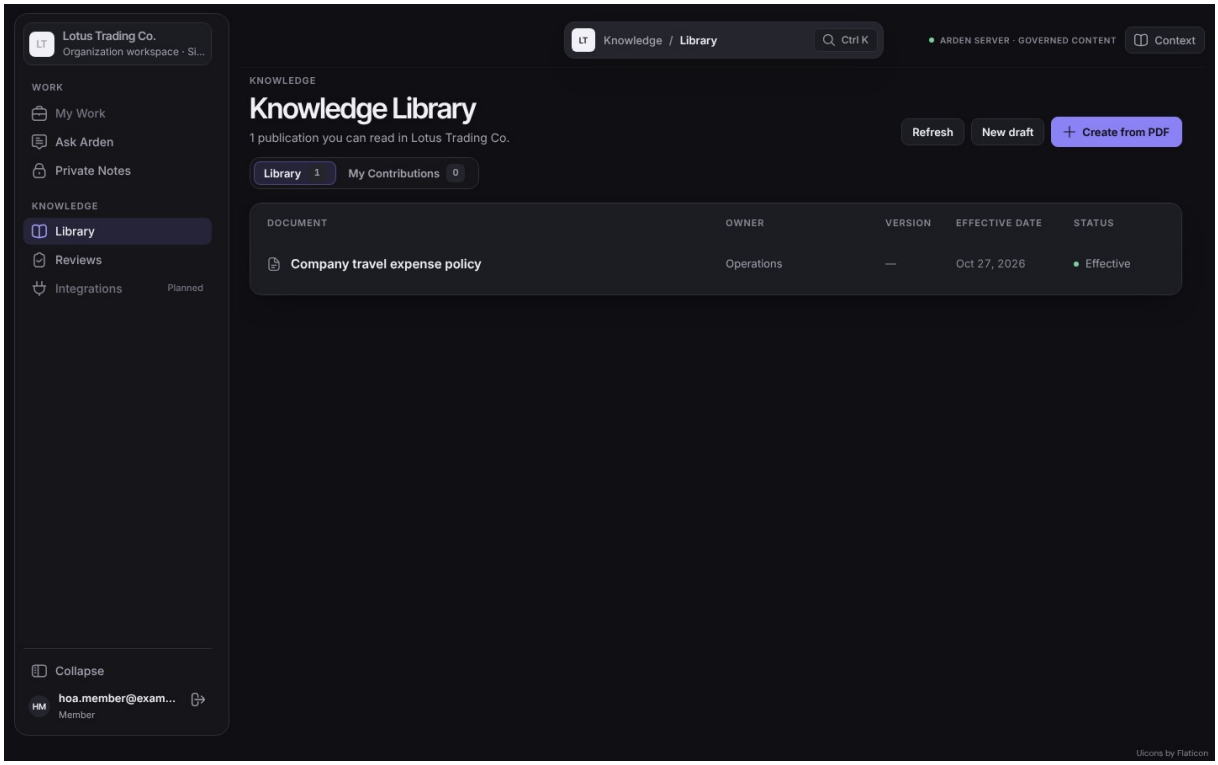


Figure 28: Knowledge Library (Finance member)

3.8.2 FR-LIB-02 Read a published document

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-22; SCR-10
Function trigger	Select a publication in the Library or a citation.
Actor and permission	Member
Function description	Show content (Semantic Document or PDF viewer), version selector, status, effective date, owner department and summary.
Preconditions	Actor may read the publication.
Postconditions	None.
Main flow	1. API reads the publication, rechecks authorization and returns it.
Alternate and error flows	• Unauthorized or unknown ID → identical RESOURCE_NOT_AVAILABLE (MSG-010).
Inputs and validation	—
Data read / changed	Reads publication and stored content.
Business rules	BR-ACC-003.
Screen / API / job	GET /api/v1/organizations/{organizationId}/knowledge/published/{publishedVersionId}.

Item	Specification
Acceptance criteria	AC1: A direct link to a document of another department returns MSG-010.

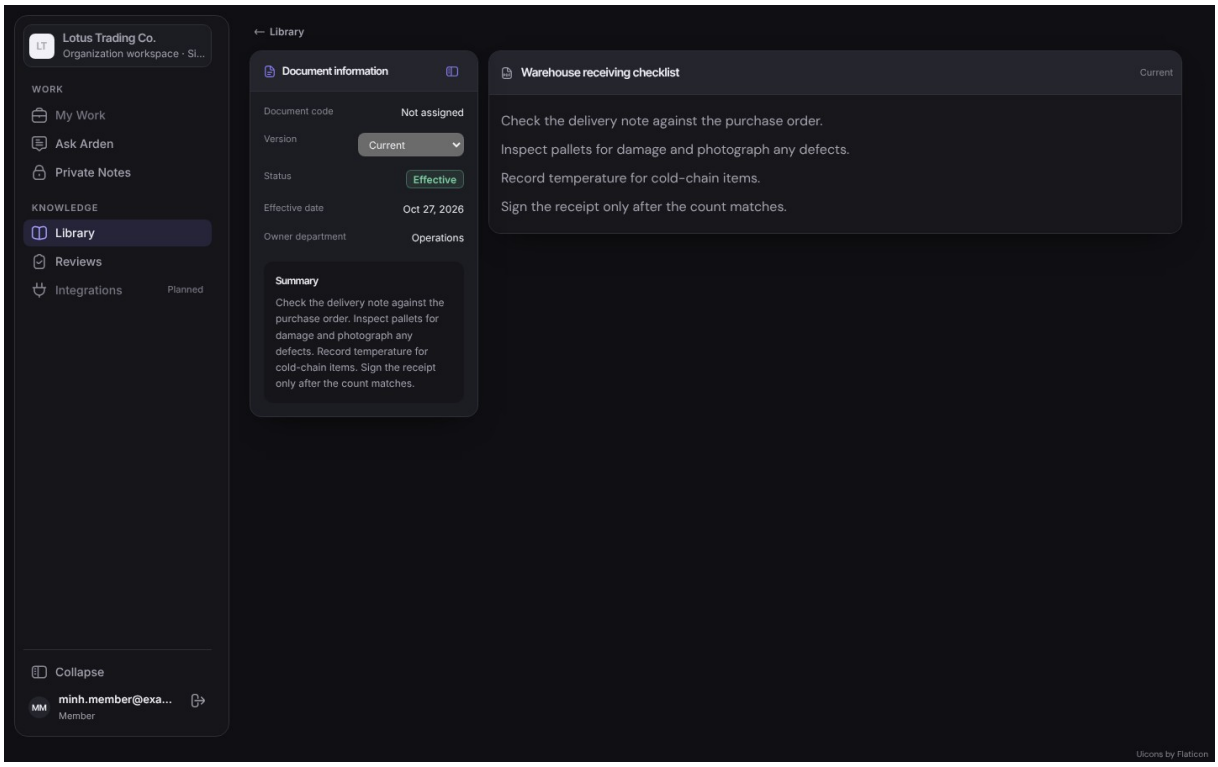
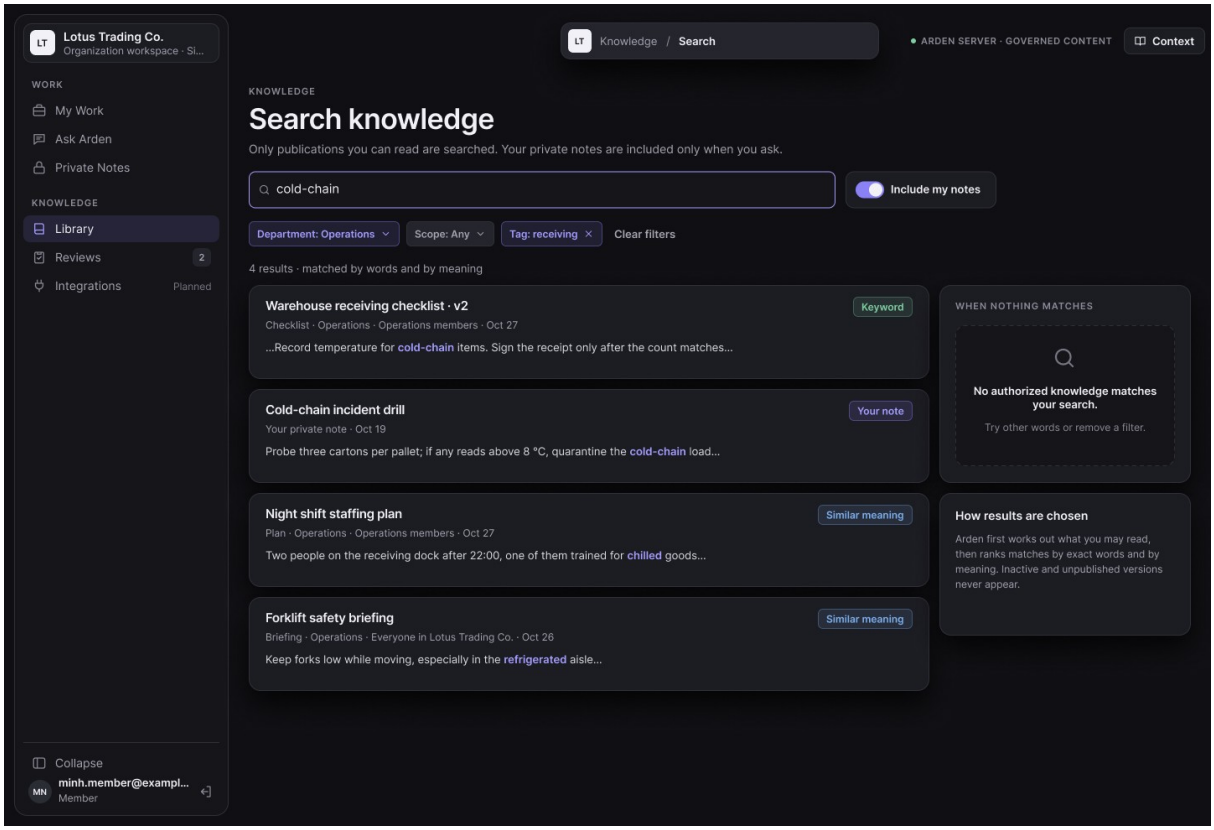


Figure 29: Document reader (SCR-10)

3.8.3 FR-LIB-03 Search and filter knowledge

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-23
Function trigger	Library search field (Ctrl K) or filters.
Actor and permission	Member
Function description	Keyword (PostgreSQL full-text) and semantic (ChromaDB) search over readable active publications, and optionally the member's own notes; filters by department, scope and tag.
Preconditions	Active membership.
Postconditions	None.
Main flow	1. Member types a query. 2. Core computes the readable allow-list and returns ranked results with snippets.
Alternate and error flows	• No result → MSG-011.
Inputs and validation	Query 1–500 characters.
Data read / changed	Reads publications, chunks.

Item	Specification
Business rules	BR-AI-001.
Screen / API / job	Planned API.
Acceptance criteria	AC1: A query matching only an unreadable document returns no result and no count. AC2: Unpublished or failed items never appear.



Figma screen R3-26: Search and filter knowledge

3.9 Ask Arden (AI Assistant)

Answers questions from knowledge the member may read, always with sources.

ID	Function	Actor	Status
FR-AI-01	Ask a question	Member	Implemented (organization scope)
FR-AI-02	Open answer sources	Member	Implemented
FR-AI-03	Personal and hybrid modes	Member	Planned
FR-AI-04	Summarize a document	Member	Planned
FR-AI-05	Rate an answer	Member	Planned
FR-AI-06	View AI history	Member	Planned

3.9.1 FR-AI-01 Ask a question

Item	Specification
Status	Implemented (organization scope)
Related UC / screen / NSF	UC-24; SCR-06; NSF-10
Function trigger	Sidebar → Ask Arden → type a question.
Actor and permission	Member
Function description	Answer from active publications the member may read. Core computes the allow-list of READY generations, Arden-AI retrieves from ChromaDB within that list and generates with OpenAI, and Core rechecks every citation before showing it.
Preconditions	Active membership; Arden-AI configured.
Postconditions	Answer with status Answered (≥1 citation) or Insufficient evidence.
Main flow	1. Member enters a question (≤4,000 characters). 2. Core lists authorized sources; if none, returns Insufficient evidence. 3. Core calls Arden-AI /v1/answers with the question and allow-list. 4. Core rechecks each citation (same publication, version and hash) and returns answer and citations.
Alternate and error flows	• Empty question → INVALID_REQUEST. • AI not configured or failing → 503 AI_UNAVAILABLE "Ask Arden is temporarily unavailable." • Access changed during generation → 409 KNOWLEDGE_ACCESS_CHANGED "Please ask again."
Inputs and validation	Question 1–4,000 characters.
Data read / changed	Reads knowledge_index_generation, publications; no answer is stored yet.
Business rules	BR-AI-001…006.
Screen / API / job	POST /api/v1/organizations/{organizationId}/ai/answers; Arden-AI POST /v1/answers.
Acceptance criteria	AC1: Every Answered response has at least one citation to a readable publication. AC2: A question answerable only from an unreadable document returns Insufficient evidence.

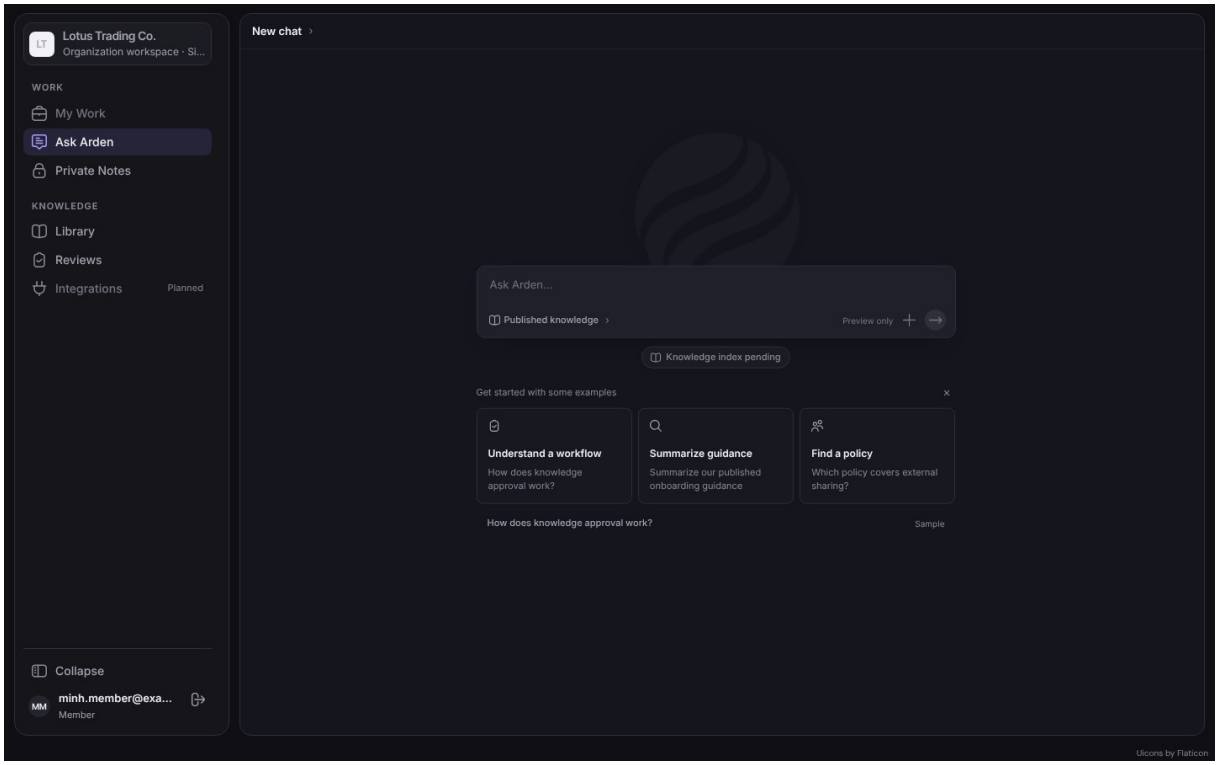


Figure 30: Ask Arden (screenshot of the 27/10 preview build; the staging build returns live answers)

3.9.2 FR-AI-02 Open answer sources

Item	Specification
Status	Implemented
Related UC / screen / NSF	UC-25
Function trigger	Select a citation chip.
Actor and permission	Member
Function description	Open the cited publication at the cited block or PDF page.
Preconditions	Citation returned by FR-AI-01.
Postconditions	Reader opens.
Main flow	1. Client opens the reader with version and block/page.
Alternate and error flows	• Source no longer readable → MSG-010.
Inputs and validation	—
Data read / changed	Reads publication.
Business rules	BR-AI-002.
Screen / API / job	Document reader deep link.
Acceptance criteria	AC1: A citation opens the exact version and page.

3.9.3 FR-AI-03 Personal and hybrid modes

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-24
Function trigger	Ask Arden → mode selector.
Actor and permission	Member
Function description	Answer from the member's own private notes (Personal) or from both private notes and readable publications (Hybrid).
Preconditions	—
Postconditions	—
Main flow	1. Member selects the mode and asks.
Alternate and error flows	• —
Inputs and validation	Mode.
Data read / changed	Private note chunks (owner only).
Business rules	BR-PKM-001.
Screen / API / job	Planned.
Acceptance criteria	AC1: Another member's private note is never retrieved.

3.9.4 FR-AI-04 Summarize a document

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-27
Function trigger	Document reader → Summarize.
Actor and permission	Member
Function description	Generate a summary of a readable publication or own note, citing the version used.
Preconditions	Actor may read the document.
Postconditions	Summary shown; not stored as knowledge.
Main flow	1. Member selects Summarize.
Alternate and error flows	• Unreadable document → MSG-010.
Inputs and validation	Document ID.

Item	Specification
Data read / changed	—
Business rules	BR-AI-005.
Screen / API / job	Planned.
Acceptance criteria	AC1: The summary names the document and version it was generated from.

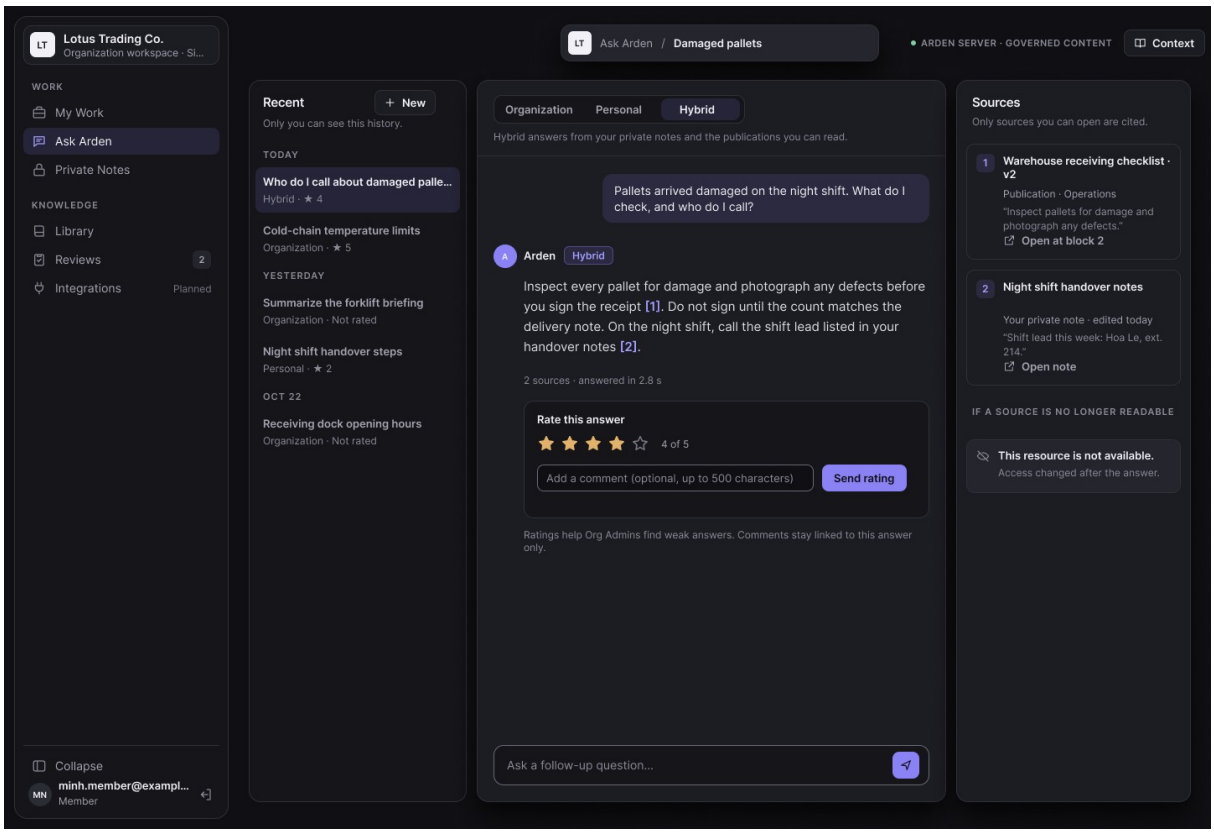
3.9.5 FR-AI-05 Rate an answer

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-26
Function trigger	Answer → rating control.
Actor and permission	Member
Function description	Rate 1–5 with an optional comment (≤500).
Preconditions	Answer exists.
Postconditions	Feedback stored; permissions unchanged.
Main flow	1. Member selects a rating and optional comment.
Alternate and error flows	• Out-of-range rating → validation error.
Inputs and validation	Rating 1–5; comment ≤500.
Data read / changed	AI answer feedback.
Business rules	BR-AI-005.
Screen / API / job	Planned.
Acceptance criteria	AC1: One rating per user and answer, updatable.

3.9.6 FR-AI-06 View AI history

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-28
Function trigger	Ask Arden → Recent.
Actor and permission	Member
Function description	List own questions, answers, citations and ratings, newest first.
Preconditions	Signed in.

Item	Specification
Postconditions	None.
Main flow	1. Open Recent.
Alternate and error flows	• —
Inputs and validation	—
Data read / changed	AI question/answer records.
Business rules	BR-PKM-001.
Screen / API / job	Planned.
Acceptance criteria	AC1: Nobody else, including administrators, can read the history. AC2: Citations opened from history are rechecked.



Figma screen R3-27: Ask Arden modes, sources, rating and history

3.10 Dashboard and Analytics

Gives administrators and reviewers minimal, permission-safe statistics.

ID	Function	Actor	Status
FR-DASH-01	Organization statistics	Organization Admin	Planned
FR-DASH-02	Feedback and quality lists	Organization Admin; Knowledge Reviewer	Planned

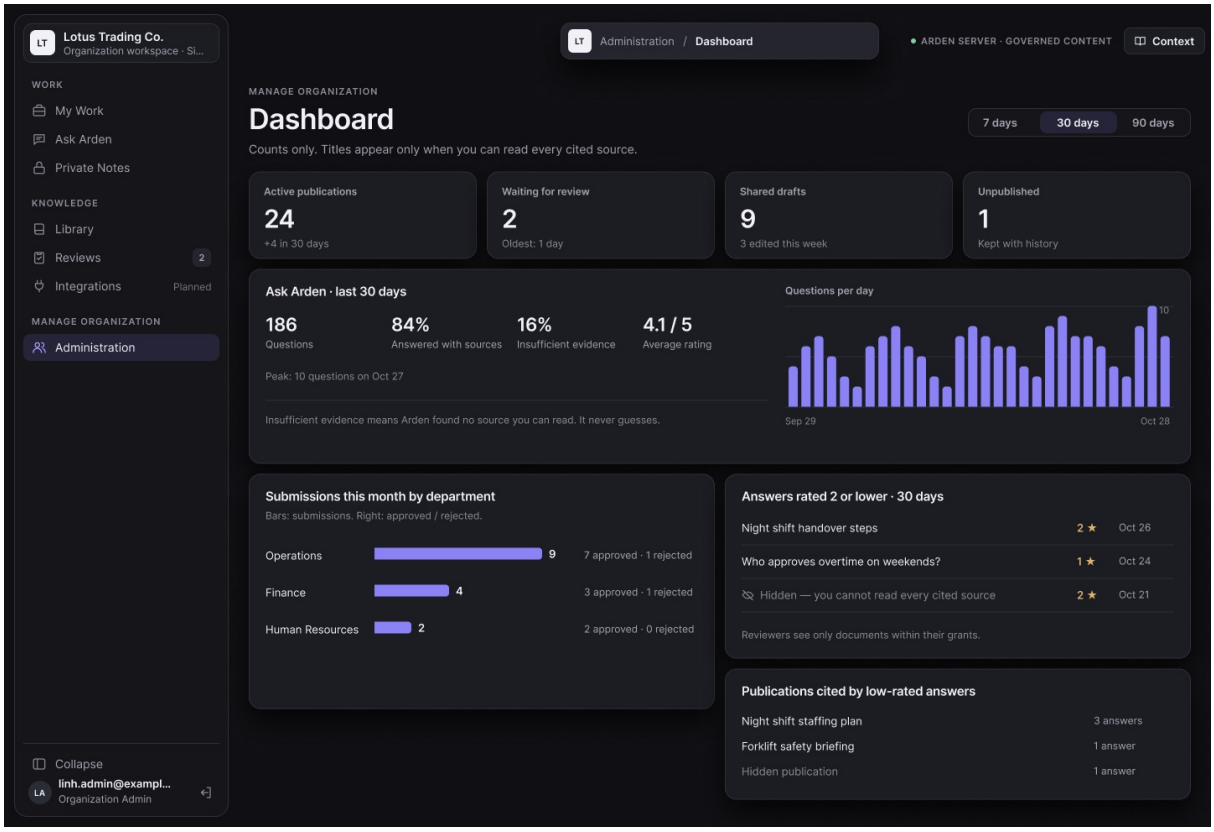
3.10.1 FR-DASH-01 Organization statistics

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-30; SCR-19
Function trigger	Sidebar → Dashboard.
Actor and permission	Organization Admin
Function description	Counts of knowledge by state (draft, pending, active, unpublished) and department; submissions and decisions per department per month; questions per day, answered vs insufficient evidence and average rating for the last 30 days.
Preconditions	Organization Admin.
Postconditions	None.
Main flow	1. Open Dashboard; choose the period.
Alternate and error flows	• No data → zero values.
Inputs and validation	Period.
Data read / changed	Aggregates only.
Business rules	BR-DASH-001.
Screen / API / job	Planned API.
Acceptance criteria	AC1: Numbers equal database counts on the seed dataset. AC2: No title or content is shown.

3.10.2 FR-DASH-02 Feedback and quality lists

Item	Specification
Status	Planned
Related UC / screen / NSF	UC-30; SCR-19
Function trigger	Dashboard → Feedback / Quality.
Actor and permission	Organization Admin; Knowledge Reviewer

Item	Specification
Function description	List rated answers and publications cited by answers rated ≤2 in the last 30 days. Text and titles are shown only when the viewer can read every cited source; reviewers see only documents within their grants.
Preconditions	Role as stated.
Postconditions	None.
Main flow	1. Open the list.
Alternate and error flows	• Unreadable source → shown as Hidden.
Inputs and validation	—
Data read / changed	Aggregates over AI feedback.
Business rules	BR-DASH-001.
Screen / API / job	Planned API.
Acceptance criteria	AC1: A viewer without access to a cited source sees Hidden. AC2: A document cited by three low-rated answers shows the count 3.



Figma screen R3-28: Dashboard and quality analytics

4. Non-Functional Requirements

4.1 External Interfaces

ID	Interface	Requirement
IF-01	Web user interface	Responsive React application. Signed-in workspace uses the Island design: left navigation (My Work, Ask Arden, Private Notes, Library, Reviews, Integrations (planned), Administration for Organization Admins) and a central work area; sign-in and onboarding use the Living Archive design.
IF-02	Windows desktop client	Electron application sharing the same UI; calls only allowlisted API routes and keeps tokens in the main process.
IF-03	Authentication provider	Supabase Auth (email/password, email confirmation, Google OAuth PKCE), called only by the Arden API.
IF-04	Email (SMTP)	Configurable SMTP relay for HTML invitation emails; all six SMTP settings are runtime secrets.
IF-05	Database	PostgreSQL 17 (pgvector image) with versioned SQL migrations 0001–0023 and row-level security under the non-bypass role arden_runtime; indexing uses the role arden_indexer.
IF-06	Arden-AI service	Private HTTP API (FastAPI, Python 3.11): POST /v1/index-generations and POST /v1/answers with a bearer service token; versioned JSON contracts (contractVersion 1).
IF-07	ChromaDB	Self-hosted vector index on the private network with a persistent volume; written and queried only by Arden-AI.
IF-08	OpenAI API	Embeddings and answer generation called only by Arden-AI; receives the question and authorized excerpts only.
IF-09	Langfuse	Self-hosted AI tracing and cost monitoring for the Arden operations team.

4.2 Quality Attributes

Each non-functional requirement is stated with its metric, threshold and measurement method so that it can be verified. Rules that are not measurable quality attributes are listed as business rules or common requirements in section 5.

4.2.1 Usability

ID	Requirement
NFR-01	A first-time user completes the main tasks without help outside the product, on web and desktop, with mouse or keyboard.
Metric	(a) Task success of first-time users for four tasks: create and submit a draft, approve a submission, invite a member, find and open a published document; (b) share of main screens without overlap or horizontal scrolling at 1280, 1024 and 390 px; (c) share of actions reachable by keyboard; (d) share of error messages that name the field and the corrective action; (e) behaviour differences between web and desktop.
Threshold	(a) at least 4 of 5 participants finish each task within 5 minutes; (b) 100%; (c) 100%; (d) 100%; (e) 0.
Method	Moderated usability test with 5 students outside the team; responsive and keyboard checklist (system test ST-24); desktop parity scenarios (ST-22).

4.2.2 Security and Privacy

ID	Requirement
NFR-02	No user can read, count or receive in AI context any item outside their authorization (other organization, other department, group not joined, another member's private item, inactive publication). Organization Admin and Arden System Admin have no content override.
Metric	Unauthorized disclosures in the permission test matrix covering list, detail, search, AI retrieval, citations and dashboard.
Threshold	0 disclosures; 100% of denial test cases pass.
Method	Automated API and PostgreSQL row-level-security tests in CI on every pull request; two-department demo script.

4.2.3 AI Answer Quality

ID	Requirement
NFR-03	Ask Arden answers only from authorized sources and cites them.
Metric	(a) Share of Answered responses with at least one valid citation; (b) share of answers containing information that exists only in documents the asker cannot read; (c) share of unanswerable questions returning Insufficient evidence.
Threshold	(a) 100%; (b) 0%; (c) at least 90%.
Method	Evaluation script with at least 30 fixed questions on the synthetic demo corpus, at least 10 answerable only from documents the asker cannot read; run before each release; results traced in Langfuse.

4.2.4 Performance

ID	Requirement
NFR-04	Common operations respond quickly.
Metric	95th percentile response time at the API; time from approval to ACTIVE.
Threshold	Ask Arden answer ≤ 5 s; library and document reads ≤ 500 ms; other API requests ≤ 500 ms; approval to ACTIVE ≤ 60 s for a document under 50 pages; with 10 concurrent users and up to 1,000 published documents.
Method	k6 load-test script on the staging environment with the seed corpus; worker logs for activation time.

4.2.5 Reliability

ID	Requirement
NFR-05	Governance actions are atomic and never duplicated, and a failed index never becomes visible.
Metric	Partial or duplicate drafts, submissions, decisions or publications after repeated requests with the same Idempotency-Key, after concurrent submit, and after forced failures of Arden-AI; failed or partial generations that become ACTIVE.
Threshold	0 in every case; the previous active version remains readable during failures.
Method	Automated integration tests that repeat requests, run concurrent submissions and inject AI failures (timeout, HTTP 500, invalid manifest).

4.2.6 Auditability

ID	Requirement
NFR-06	Every governance and security action is audited without leaking content.
Metric	Share of action types (invitation, membership change, audience group change, grant change, submission, decision incl. selfApproved, activation, System Admin change) that create an audit event with actor, time, object, action and outcome; events or logs containing content, tokens or secrets.
Threshold	100% of action types; 0 events or logs with content or secrets.
Method	Automated tests asserting one audit row per action type; log review on staging before each release.

4.2.7 Deployability and Maintainability

ID	Requirement
NFR-07	The system can be built, tested and deployed from the repositories.
Metric	(a) CI result of typecheck, unit and integration tests for Arden and Arden-AI; (b) time to deploy a clean staging environment (core, worker, Arden-AI, ChromaDB), apply all migrations and load seed data.
Threshold	(a) 100% pass on staging and on tag review-2, with at least 50% of test cases automated; (b) ≤ 30 minutes without manual SQL.
Method	GitHub Actions (Ubuntu and Windows); scripted Docker Compose deployment on Coolify, timed and recorded in the runbook.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-ORG-001	One installation may host several organizations; each organization is an isolated tenant.
BR-ORG-002	Any verified account may create an organization and becomes its first Organization Admin.
BR-ORG-003	The hierarchy is Organization → Department. A member may temporarily have no department; deleting a department archives it, unassigns its members, clears it from pending invitations and deactivates its Department audience and reviewer grants.
BR-ORG-004	Team and project are optional tags only and never create hierarchy or grant access.
BR-ORG-005	Each department has exactly one managed Department audience. Each member has at most one primary department, which decides Department-audience membership and the responsible department of the member's drafts.
BR-MEM-001	A membership is created only after authentication, verified-email match and explicit acceptance of a valid invitation.
BR-MEM-002	At most one pending invitation exists per email in an organization. An invitation link is single-use, expires after 7 days and its token is stored only as a hash.
BR-MEM-003	Only an Organization Admin with a verified email may invite or resend.

ID	Rule Definition
BR-MEM-004	The last active Organization Admin of an organization cannot be removed or demoted.
BR-ROLE-001	The actors are Member, Knowledge Reviewer, Organization Admin and Arden System Admin; contributing knowledge is a Member capability.
BR-ROLE-002	The Organization Admin manages the organization and may decide any pending submission, including their own (recorded as self-approval). This capability gives no readership and no private-note access.
BR-ROLE-003	Arden System Admin is an installation capability outside every organization; it gives no membership, content, review or AI access. The first is created only by the bootstrap command; the last active one cannot be revoked.
BR-ACC-001	A private item is readable only by its owner. Contribution creates a separate shared copy; the original never becomes shared.
BR-ACC-002	A shared draft needs a target scope, 1–100 audiences and the author's active primary department before submission.
BR-ACC-003	A member may read an active publication when at least one of its audiences includes them; a Department publication also requires active membership in its department. Empty, inactive or unknown audiences grant nothing.
BR-ACC-004	The Organization audience is valid only with Organization scope and a Department audience only with Department scope of the same department; groups are valid with either scope.
BR-ACC-005	An author may target only audiences they can currently read; hidden groups are not listed.
BR-ACC-006	Membership, department and group changes take effect on the next request for reading, search, AI and citations.
BR-GOV-001	The responsible department is the author's primary department; it is not chosen and is frozen at submission.
BR-GOV-002	Reviewer grants have disjoint scopes: a snapshot whose audiences include the Organization audience needs an Organization grant; any other snapshot needs a Department grant for its responsible department. Grants also respect tags, validity and status. Reviewer levels and classifications apply only to legacy submissions.
BR-GOV-003	An ordinary reviewer cannot see or decide their own submission.
BR-GOV-004	A decision is Approve or Reject; Reject requires a note (≤2,000 characters). There is no request-revision decision; the author edits the working draft and submits a new snapshot.
BR-GOV-005	A draft has at most one open submission (release candidate).
BR-GOV-006	A decision must quote the exact snapshot version ID and content hash; a changed snapshot is refused.
BR-GOV-007	Approval is blocked when an audience is no longer enforceable or the author can no longer target it; rejection stays possible.
BR-GOV-008	One eligible decision is final; there is no second approval step and no human publisher.
BR-LIFE-001	Saved versions and submissions are immutable; the working draft stays editable while a snapshot is under review.
BR-LIFE-002	Approval moves the snapshot to PREPARING. Only a complete index generation followed by atomic activation makes it ACTIVE; a FAILED generation stays invisible.
BR-LIFE-003	The current active version stays readable until a newer version is activated or the item is unpublished.
BR-LIFE-004	Unpublish and restore (planned) are Organization Admin actions that keep all versions, decisions and audit events.

ID	Rule Definition
BR-DOC-001	Semantic Document v1 is the only stored editable format: at most 2,000 blocks, nesting depth 8, 131,072 characters, links only http, https or mailto.
BR-DOC-002	Uploaded files must be text-based PDFs of at most 5 MB; uploaded PDFs cannot be edited as text.
BR-DOC-003	A shared draft has at most one source: a private-note version or a published version.
BR-PKM-001	Personal documents, collections and AI history are owner-only and never used in another member's search or answers.
BR-AI-001	Arden core computes the allow-list of active generations the asker may read before any retrieval; Arden-AI may narrow but never broaden it.
BR-AI-002	Arden core rechecks every citation after generation; if access changed, the answer is refused.
BR-AI-003	ChromaDB is a derived index, never an authorization source.
BR-AI-004	An Answered response has at least one citation; without authorized evidence the status is Insufficient evidence.
BR-AI-005	AI never approves, publishes, unpublishes or changes audiences, grants or permissions.
BR-AI-006	Only the question and authorized excerpts are sent to the AI provider; provider credentials stay on the server.
BR-DASH-001	Dashboards show aggregates; titles and text appear only when the viewer may read the source.
BR-DEM-001	The demo uses at least two departments and one audience group and shows that unauthorized members cannot see department, group or private content through list, detail, search or AI.

5.2 Common Requirements

ID	Common Requirement
CR-01	Every mutating action validates the current membership, group memberships, grants, ownership and System Admin status on the server.
CR-02	Inputs are validated on client and server; server validation is authoritative and uses strict JSON schemas (unknown fields rejected).
CR-03	List, detail, count, citation and dashboard views use the same access decision.
CR-04	Unauthorized and missing resources return the same RESOURCE_NOT_AVAILABLE response.
CR-05	Create, submit and decision requests require an Idempotency-Key; edits require If-Match.
CR-06	User-facing errors state the corrective action without revealing restricted data or internals; every error carries a request ID.
CR-07	Governance state changes record actor, time, object, prior and new state and outcome.
CR-08	Responses with protected data are sent with Cache-Control: no-store.
CR-09	Lists and AI answers distinguish private items, drafts, pending, active and failed knowledge.
CR-10	Automated tests and the demo include cross-organization, cross-department, group, private-item, admin-override and self-review attempts.

5.3 Application Messages List

Code	Type	Context	Content
MSG-001	Inline	Required field missing	This field is required.
MSG-002	Inline	Audience does not match scope	The draft metadata is invalid.
MSG-003	Inline	Author has no primary department	You have no primary department yet. Ask an Org Admin to assign one before submitting.
MSG-004	Inline	Submission lacks governance data	Add an active responsible department, access scope and audience before submitting.
MSG-005	Toast	Version saved	Draft v{n} saved on the server.
MSG-006	Toast	Snapshot submitted	Version {n} submitted for review (Awaiting review).
MSG-007	Toast	Approved	Approved. The exact reviewed snapshot is preparing for indexing; it is not active yet.
MSG-008	Toast	Rejected	Submission rejected. The working draft remains editable and can be submitted again.
MSG-009	Inline	Approve confirmation	The server prepares this exact snapshot for indexing. It is not active until indexing finishes.
MSG-010	Inline	Unauthorized or missing item	This resource is not available.
MSG-011	Inline	No search results (planned)	No authorized knowledge matches your search.
MSG-012	Inline	AI insufficient evidence	I could not find any indexed published knowledge you are allowed to read.
MSG-013	Inline	Wrong file type	Choose a valid PDF file.
MSG-014	Inline	File too large	Choose a valid PDF file up to 5 MB.
MSG-015	Inline	Reject note missing	Explain why this snapshot is rejected.
MSG-016	Inline	Invitation not usable	This invitation is not available for this account.
MSG-017	Toast	Invitation sent	Invitation email sent to {email}.
MSG-018	Inline	Version conflict	This draft changed. Reload it before saving again.
MSG-019	Inline	Approval blocked	This review decision cannot be applied.
MSG-020	Inline	Last System Admin	The last active Arden System Admin cannot be revoked.
MSG-021	Inline	Restore blocked (planned)	This item cannot be restored because one of its audiences is no longer active.
MSG-022	Inline	Wrong credentials	Email or password is incorrect.
MSG-023	Inline	Invitation delivery failed	The invitation was saved but its email was not delivered. Retry sending it from organization administration.
MSG-024	Inline	AI unavailable	Ask Arden is temporarily unavailable.
MSG-025	Inline	Access changed during answer	Knowledge access changed while the answer was being prepared. Please ask again.
MSG-026	Inline	Pending invitation exists	A pending invitation already exists for this email.

5.4 Other Requirements

ID	Requirement
OR-01	Arden remains open-source and self-hosted; the only external services are those listed in section 1.3.
OR-02	Sharing data between organizations is out of scope.
OR-03	Project-management functions, Knowledge Handover, Knowledge Graph and external work-source integrations are out of scope (registration form v2.0).
OR-04	Governance may be described as ISO-aligned; certification is not claimed.
OR-05	The demo dataset is synthetic and contains Organization, Department and group publications and private notes across at least two departments.
OR-06	Supported clients are the web application and the Windows desktop application; full offline operation is out of scope.
OR-07	OCR of scanned PDFs is out of scope.

5.5 Requirement Traceability Matrix

FR	Use case	Screen / API / job	Main entities	Tests
FR-AUTH-01	UC-01	SCR-01; /auth/sign-in, sign-up, sign-out	User Account	app.test.ts (auth routes); ST-01, ST-02
FR-AUTH-02	UC-01	SCR-01; /auth/google/*, /auth/password	User Account	app.test.ts (Google flow); ST-03
FR-AUTH-03	UC-02	SCR-02; /auth/forgot-password, reset-password	User Account	app.test.ts (reset routes); signInPage.test.ts
FR-AUTH-04	—	NSF-01	User Account	provider tests; ST-01
FR-AUTH-05	UC-31	SCR-21	User Account	Planned
FR-ORG-01	UC-03	SCR-03; POST /organizations	Organization, Department, Membership, Audience	app.test.ts; repository integration; ST-04
FR-ORG-02	UC-04	SCR-03; /me	Membership	ST-04
FR-ORG-03	UC-05	SCR-15; /departments	Department, Audience	app.test.ts; ST-05
FR-ORG-04	UC-06	SCR-17; /invitations	Invitation	app.test.ts; smtp-mailer.test.ts; ST-06
FR-ORG-05	UC-07	SCR-04; /invitations/accept	Invitation, Membership	app.test.ts; ST-06, ST-07
FR-ORG-06	UC-08	SCR-14; /members/{id}/department	Membership	app.test.ts; ST-21
FR-ACC-01	UC-09	SCR-16; /audience-groups	Audience, Audience Group Member	app.test.ts (audience groups); ST-14, ST-21
FR-ACC-02	UC-10	SCR-18; /knowledge/reviewer-grants	Reviewer Grant	policy.test.ts; review-queue.test.ts; ST-15
FR-SYS-01	—	system-admin:bootstrap	System Administrator	ST-23

FR	Use case	Screen / API / job	Main entities	Tests
FR-SYS-02	UC-32	/system/administrators	System Administrator, System Admin Event	app.test.ts (system admin); ST-23
FR-SYS-03	—	SCR-20	Organization	Planned
FR-PKM-01	UC-11	SCR-07; /private-workspace/drafts	Private Note, Version	app.test.ts; content-storage.test.ts; ST-08, ST-09
FR-PKM-02	UC-14	SCR-08; POST /knowledge/drafts	Shared Draft	app.test.ts; ST-12
FR-PKM-03	UC-12	SCR-07	Private Note	Planned
FR-PKM-04	UC-13	SCR-07	Personal Collection	Planned
FR-DOC-01	UC-15	SCR-11; /knowledge/drafts/new	Shared Draft, Version	app.test.ts; semantic-document.test.ts; ST-10
FR-DOC-02	UC-15	SCR-11; /knowledge/drafts/pdf	Shared Draft, Version	app.test.ts; ST-11
FR-DOC-03	UC-16	SCR-12; PATCH /knowledge/drafts/{id}; /knowledge/options	Shared Draft, Draft Audience	policy.test.ts; readerSelection.test.ts; ST-10
FR-DOC-04	UC-17	SCR-12; /knowledge/drafts/{id}/submit	Submission	app.test.ts; policy.test.ts; ST-10, ST-13
FR-DOC-05	—	SCR-09; /knowledge/drafts, /knowledge/submissions	Shared Draft, Submission	knowledgeStatus.test.ts; ST-18
FR-DOC-06	UC-18	—	Submission	Planned
FR-REV-01	UC-19	SCR-13; /knowledge/reviews	Submission, Reviewer Grant	review-queue.test.ts; ST-16
FR-REV-02	UC-19	SCR-13; /knowledge/reviews/{id}	Submission	app.test.ts; ST-16
FR-REV-03	UC-20	SCR-13; /decisions	Review Decision, Index Generation	review-decision.test.ts; ST-17; AT-02
FR-REV-04	UC-21	SCR-13; /decisions	Review Decision	review-decision.test.ts; ST-18, ST-19
FR-PUB-01	UC-20	NSF-08, NSF-09; Arden-AI /v1/index-generations	Index Generation, Index Job, Published Version	worker.test.ts; contracts.test.ts; ST-17
FR-PUB-02	UC-29	—	Current Publication	Planned
FR-LIB-01	UC-22	SCR-09; /knowledge/published	Published Version, Audience	app.test.ts; knowledgeLibrary.test.ts; ST-20; AT-03
FR-LIB-02	UC-22	SCR-10; /knowledge/published/{id}	Published Version	app.test.ts; knowledgeDocumentDetail.test.ts; ST-20
FR-LIB-03	UC-23	—	Knowledge Chunk	Planned
FR-AI-01	UC-24	SCR-06; /ai/answers; Arden-AI /v1/answers	Index Generation, Citation	contracts.test.ts; AT-07

FR	Use case	Screen / API / job	Main entities	Tests
FR-AI-02	UC-25	SCR-10 deep link	Citation	citation-page.test.ts
FR-AI-03	UC-24	SCR-06	Knowledge Chunk	Planned
FR-AI-04	UC-27	SCR-10	—	Planned
FR-AI-05	UC-26	SCR-06	AI Answer	Planned
FR-AI-06	UC-28	SCR-06	AI Question, Answer	Planned
FR-DASH-01	UC-30	SCR-19	Aggregates	Planned
FR-DASH-02	UC-30	SCR-19	Aggregates	Planned

Test references: automated test files are in the Arden repository (apps/api, packages/ui, packages/domain), including apps/api/src/answers for Ask Arden. ST-xx and AT-xx are the system and acceptance scenarios of Report 5; scenarios ST-10, ST-15, ST-17 and ST-18 must be updated to the approve/reject and grant-scope rules described here.